Reconstructing ancient genomes from gene counts: a robust likelihood framework with sampling bias correction
Supporting Information

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SI Appendix

The Appendix consists of two parts: Section A shows the mathematical and algorithmic details for Methods, Section B details the supporting experiments and data for Results.

Contents

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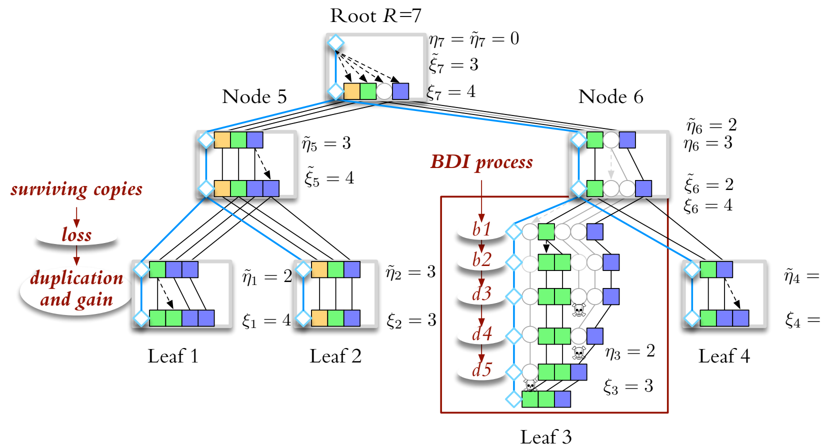

Fig. S1. Gene copy evolution on a 4-leaf phylogeny. Squares denote conserved copies with lines connecting homologs among them. The blue diamonds mark a "virtual" copy outside the genome that is the progenitor of *xenolog* copies. The random variables ξ and η count all copies at a node and founding copies inherited from the parent, respectively. On the edge to Leaf 3, the birth-death process is illustrated with 5 events at random time intervals: gain (b1, originating from outside) and duplication (b2, originating from the first inherited copy), and losses (d3, d4, d5). The full birth-death history includes the copies that are extinct in all descendants (white circles, shown along the path to Leaf 3). The *surviving copies* $\tilde{\xi}$ and $\tilde{\eta}$ include only the copies that have homologs in at least one descendant. Black edges show their reconciled history. An equivalent probability distribution is generated by a two-step manufacturing of surviving copies: children inherit the founding surviving copies by diferential loss; gained and inherited founding copies generate duplicates. (Modified from (1).)

A. Mathematical details

A.1. Likelihood recurrences for copy evolution. The likelihood and posterior calculations are factorized by conditioning on the surviving edge end node copies (see Figure S1) which are necessarily bounded by the sum of copies at the leaves in the subtree of v. The calculations use the survival parameters from Theorem 2.

Definition 1 (Survival parameters)**.** *Define the (per-copy)* survival parameters $\tilde{p}_u, \tilde{q}_u$ *and the (per-copy)* extinction probability ϵ_u*:* $\epsilon_u = 0$ *for all leaves* u*, and for every non-leaf* u *with children* $uv, uw \in T$*,* $\epsilon_u = \tilde{p}_v \tilde{p}_w$*. At every node* u*, define* $\tilde{p}_u$ *and* $\tilde{q}_u$ *so that*

$$
\begin{aligned}
\operatorname{logit} \tilde{q}_u &= \ln \frac{\tilde{q}_u}{1-\tilde{q}_u} = \operatorname{logit} q_u + \ln(1-\epsilon_u) \\
1 - \tilde{p}_u &= (1-p_u)\big(1 - \epsilon_u(1-\tilde{q}_u)\big).
\end{aligned}
$$

At every node u *with* $\lambda_u = 0$*, define* $\tilde{r}_u = r_u(1-\epsilon_u)$*.*

Let Ξ be a given profile. Define the (inner) log-likelihoods of the partial profiles conditioned on $\tilde{\xi}_u$ or $\tilde{\eta}_u$:

$$
\tilde{\mathbf{C}}_u(n) = \ln \mathbb{P}\big\{\Xi_u \bigm| \tilde{\xi}_u = n\big\} \qquad \text{and} \qquad \tilde{\mathbf{K}}_u(s) = \ln \mathbb{P}\big\{\Xi_u \bigm| \tilde{\eta}_u = s\big\},
$$

along with the complementary outer log-priors

$$
\tilde{\mathbf{B}}_u(n) = \ln \mathbb{P}\big\{\Xi - \Xi_u ; \tilde{\xi}_u = n\big\} \qquad \text{and} \qquad \tilde{\mathbf{J}}_u(s) = \ln \mathbb{P}\big\{\Xi - \Xi_u ; \tilde{\eta}_u = s\big\}.
$$

By design, the log-likelihood and the posterior counts can be calculated using $(\tilde{\mathbf{C}}, \tilde{\mathbf{B}})$ or $(\tilde{\mathbf{K}}, \tilde{\mathbf{J}})$ pairs at every node u by finite formulas:

$$
\begin{aligned}
\ln L(\Xi) &= \ln \sum_{n=0}^{\Omega(\Xi_u)} \exp\big(\tilde{\mathbf{C}}_u(n) + \tilde{\mathbf{B}}_u(n)\big) = \ln \sum_{s=0}^{\Omega(\Xi_u)} \exp\big(\tilde{\mathbf{K}}_u(s) + \tilde{\mathbf{J}}_u(s)\big) \\
\mathbb{P}\big\{\tilde{\xi}_u = n \bigm| \Xi\big\} &= \exp\big(\tilde{\mathbf{C}}_u(n) + \tilde{\mathbf{B}}_u(n) - \ln L(\Xi)\big) \quad \text{for } 0 \le n \le \Omega(\Xi_u) \\
\mathbb{P}\big\{\tilde{\eta}_u = s \bigm| \Xi\big\} &= \exp\big(\tilde{\mathbf{K}}_u(s) + \tilde{\mathbf{J}}_u(s) - \ln L(\Xi)\big) \quad \text{for } 0 \le s \le \Omega(\Xi_u)
\end{aligned}
$$

where $\Omega(\Xi_u) = \sum_{w \in \mathcal{L}_u} \xi_w$ is the sum of copies at the leaves in the subtree rooted at u.

Theorem 10 shows the postorder recurrences for inner likelihoods and preorder recurrences for outer priors.

Theorem 10 (Survival log-likelihood recurrences)**.** *Let* $\Omega_u = \Omega(\Xi_u)$ *denote the size of the profile in the subtree of node* u*. We have the following recurrences*

(a) At every node u*: for all* $\Omega_u < s$*,* $\tilde{\mathbf{K}}_u(s) = -\infty$ *and for all* $\Omega_u < n$*,* $\tilde{\mathbf{C}}_u(n) = -\infty$*.*

(b) *At every node u, for all $0 \le s \le \Omega_u$,*

$$\tilde{\mathbf{K}}_u(s) = \ln \sum_{n=s}^{\Omega_u} \exp\bigl(\tilde{\mathbf{K}}'_u(n,s)\bigr) \quad \textit{with}$$

$$\tilde{\mathbf{K}}'_u(n,s) = \tilde{\mathbf{C}}_u(n) + \begin{cases} \ln\binom{\kappa_u+n-1}{n-s} + \bigl(\kappa_u+s\bigr)\ln(1-\tilde{q}_u) + (n-s)\ln(\tilde{q}_u) & \textit{if } 0<q_u \\ (n-s)\ln(\tilde{r}_v) - \tilde{r}_u - \ln((n-s)!) & \textit{if } 0=q_u \end{cases}$$

(c) *If u is a leaf, then $\tilde{\mathbf{C}}_u(n) = -\infty$ when $n \ne n_u$ and $\tilde{\mathbf{C}}_u(n) = 0$ when $n = n_u$. If u is an inner node with children $uv, uw \in T$, then for all $0 \le n \le \Omega_u = \Omega_v + \Omega_w$,*

$$\tilde{\mathbf{C}}_u(n) = \ln \sum_{s=0}^{\min\{n,\Omega_v\}} \exp\bigl(\tilde{\mathbf{K}}_v(s) + \tilde{\mathbf{L}}_w(n,n-s)\bigr) \times \binom{n}{s}(1-p_v^-)^s(p_v^-)^{n-s}$$

where $p_v^- = \frac{\tilde{p}_v - \tilde{p}_v\tilde{p}_w}{1-\tilde{p}_v\tilde{p}_w}$, or $\operatorname{logit} p_v^- = \operatorname{logit}\tilde{p}_v + \ln(1-\tilde{p}_w)$, and $\tilde{\mathbf{L}}_w(n,n) = \tilde{\mathbf{K}}_w(n)$, and, for all $0 \le t < n$,

$$\tilde{\mathbf{L}}_w(n,t) = \ln\Bigl((1-\tilde{p}_w) \times \exp\bigl(\tilde{\mathbf{L}}_w(n,t+1)\bigr) + \tilde{p}_w \times \exp\bigl(\tilde{\mathbf{L}}_w(n-1,t)\bigr)\Bigr).$$

(d) *At the root, $\mathbf{J}_R(0) = 0$ and $\mathbf{J}_R(s) = -\infty$ for $0 < s$.*

(e) *At any node u, for all $0 \le n \le \Omega_u$,*

$$\tilde{\mathbf{B}}_u(n) = \ln \sum_{s=0}^{n} \exp\bigl(\tilde{\mathbf{B}}'_u(n,s)\bigr) \quad \textit{with}$$

$$\tilde{\mathbf{B}}'_u(n,s) = \mathbf{J}_u(s) + \begin{cases} \ln\binom{\kappa_u+n-1}{n-s} + (\kappa_u+s)\ln(1-\tilde{q}_u) + (n-s)\ln(\tilde{q}_u) & \textit{if } 0<q_u \\ (n-s)\ln(\tilde{r}_v) - \tilde{r}_u - \ln((n-s)!) & \textit{if } 0=q_u \end{cases}$$

(f) *At every non-root node v with parent u and sibling w (i.e, $uv, uw \in T$), for all $0 \le s \le \Omega_v$,*

$$\tilde{\mathbf{J}}_v(s) = \ln \sum_{n=s}^{\Omega_u} \exp(\tilde{\mathbf{J}}'_v(n,s)) \quad \textit{with}$$

$$\tilde{\mathbf{J}}'_v(n,s) = \tilde{\mathbf{B}}_u(n) + \tilde{\mathbf{L}}_w(n,n-s) + \ln\binom{n}{s} + s\ln(1-p_v^-) + (n-s)\ln(p_v^-)$$

Proof. Follows from Theorem 2.

(a) At every node u, we have surely $\tilde{\eta}_u \le \tilde{\xi}_u \le \Omega_u$ since surviving copies at node u have at least one descendant among the extant copies in the subtree of u and $\tilde{\xi}_u$ includes copy births on top of inherited copies $\tilde{\eta}_u$.

(b) At every node u, for all $0 \le s \le \Omega_u$, define

$$\tilde{\mathbf{K}}'_u(n,s) = \ln \mathbb{P}\bigl\{\Xi_u; \tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s\bigr\}.$$

We have, by summing disjoint events,

$$\tilde{\mathbf{K}}_u(s) = \ln \mathbb{P}\bigl\{\Xi_u \bigm| \tilde{\eta}_u = s\bigr\} = \ln \sum_{n=s}^{\Omega_u} \mathbb{P}\bigl\{\Xi_u; \tilde{u}_u = n \bigm| \tilde{\eta}_u = s\bigr\} = \ln \sum_{n=s}^{\Omega_u} \exp\bigl(\tilde{\mathbf{K}}'_u(n,s)\bigr).$$

By Theorem 2,

$$\tilde{\mathbf{K}}'_u(n,s) = \ln \mathbb{P}\bigl\{\Xi_u; \tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s\bigr\} = \ln\Bigl(\mathbb{P}\bigl\{\Xi_u \bigm| \tilde{\xi}_u = n\bigr\} \times \mathbb{P}\bigl\{\tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s\bigr\}\Bigr) = \tilde{\mathbf{C}}_u(n) + \ln \mathbb{P}\bigl\{\tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s\bigr\}$$

(c) If u is a leaf, then

$$\exp\bigl(\tilde{\mathbf{C}}_u(n)\bigr) = \mathbb{P}\bigl\{\Xi_u \bigm| \tilde{\xi}_u = n\bigr\} = \mathbb{P}\bigl\{\tilde{\xi}_u = n_u \bigm| \tilde{\xi}_u = n\bigr\} = \begin{cases} 1 & \text{if } n = n_u \\ 0 & \text{if } n \ne n_u \end{cases}$$

Let u be an inner node with children $uv, uw \in T$. Define the random variable $\tilde{\psi}_u$ as the surviving copies that are inherited by both children (from among $\tilde{\xi}_u$ surviving copies at u). For all $0 \le n \le \Omega_u = \Omega_v + \Omega_w$,

$$
\begin{aligned}
\tilde{\mathbf{C}}_u(n) &= \ln \mathbb{P}\bigl\{\Xi_u \bigm| \tilde{\xi}_u = n\bigr\} = \ln \sum_{s=0}^{\min\{n,\Omega_v\}} \sum_{j=0}^{s} \mathbb{P}\bigl\{\Xi_v, \Xi_w; \tilde{\eta}_v = s, \tilde{\eta}_w = n-s+j, \tilde{\psi}_u = j \bigm| \tilde{\xi}_u = n\bigr\} \\
&= \ln\Biggl(\sum_{s=0}^{\min\{n,\Omega_v\}} \mathbb{P}\bigl\{\Xi_v \bigm| \tilde{\eta}_v = s\bigr\} \times \binom{n}{s} \underbrace{\left(\frac{1-\tilde{p}_v}{1-\tilde{p}_v\tilde{p}_w}\right)^{n-s}}_{=(1-p_v^-)^{n-s}} \underbrace{\left(\frac{\tilde{p}_v(1-\tilde{p}_w)}{1-\tilde{p}_v\tilde{p}_w}\right)^{s}}_{=(p_v^-)^{s}} \\
&\qquad \times \underbrace{\sum_{j=0}^{s} \underbrace{\mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_w = n-s+j\bigr\}}_{=\exp\left(\tilde{\mathbf{K}}_w(n-s+j)\right)} \binom{s}{j} (1-\tilde{p}_w)^j (\tilde{p}_w)^{s-j}}_{=\exp\left(\tilde{\mathbf{L}}_w(n,n-s)\right)} \Biggr)
\end{aligned}
$$

For $1 - p_v^- = \frac{1-\tilde{p}_v}{1-\tilde{p}_v\tilde{p}_w}$ and $p_v^- = \frac{\tilde{p}_v(1-\tilde{p}_w)}{1-\tilde{p}_v\tilde{p}_w}$, we have

$$
\operatorname{logit} p_v^- = \ln \frac{\tilde{p}_v(1-\tilde{p}_w)}{1-\tilde{p}_v} = \operatorname{logit} \tilde{p}_v + \ln(1-\tilde{p}_w).
$$

Let

$$
\begin{aligned}
\tilde{\mathbf{L}}_w(n,t) &= \ln \mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_v = n-t; \tilde{\xi}_u = n\bigr\} = \ln \sum_{j=0}^{n-t} \mathbb{P}\bigl\{\Xi_w; \tilde{\eta}_w = t+j \bigm| \tilde{\eta}_v = n-t; \tilde{\xi}_u = n\bigr\} \\
&= \ln \sum_{j=0}^{n-t} \mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_w = t+j\bigr\} \times \mathbb{P}\bigl\{\tilde{\eta}_w = t+j \bigm| \tilde{\eta}_v = n-t; \tilde{\xi}_u = n\bigr\} \\
&= \ln \sum_{j=0}^{n-t} \mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_w = t+j\bigr\} \times \binom{n-t}{j} (1-\tilde{p}_w)^j (\tilde{p}_w)^{n-t-j};
\end{aligned}
$$

with non-zero terms only when $t + j \le \Omega_w$ so $j \le \Omega_w - t$. So,

$$
\tilde{\mathbf{L}}_w(n,n) = \ln \mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_v = 0; \tilde{\xi}_u = n\bigr\} = \ln \mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_w = n\bigr\} = \tilde{\mathbf{K}}_w(n);
$$

and, for $t < n$

$$
\begin{aligned}
\tilde{\mathbf{L}}_w(n,t) &= \ln \sum_{j=0}^{n-t} \mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_w = t+j\bigr\} \times \left(\binom{n-t-1}{j} + \binom{n-t-1}{j-1}\right) (1-\tilde{p}_w)^j (\tilde{p}_w)^{n-t-j} \\
&= \ln\Biggl((1-\tilde{p}_w) \sum_{i=0}^{n-(t+1)} \left(\mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_w = (t+1)+i\bigr\} \times \binom{n-(t+1)}{i} (1-\tilde{p}_w)^i (\tilde{p}_w)^{n-(t+1)-i} \right) \\
&\qquad + \tilde{p}_w \sum_{j=0}^{(n-1)-t} \left(\mathbb{P}\bigl\{\Xi_w \bigm| \tilde{\eta}_w = t+j\bigr\} \times \binom{(n-1)-t}{j} (1-\tilde{p}_w)^j (\tilde{p}_w)^{(n-1)-t-j} \right) \Biggr) \\
&= \ln\Bigl((1-\tilde{p}_w) \exp\bigl(\tilde{\mathbf{L}}_w(n,t+1)\bigr) + \tilde{p}_w \exp\bigl(\tilde{\mathbf{L}}_w(n-1,t)\bigr) \Bigr)
\end{aligned}
$$

(d) At the root,

$$
\mathbf{J}_R(0) = \ln \mathbb{P}\bigl\{\tilde{\eta}_R = 0\bigr\} = \ln 1 = 0
$$

and for $0 < s$, we have $\mathbf{J}_R(s) = \ln 0 = -\infty$.

(e) Let u be any node. Since $\tilde{\xi}_u$ depends of $\tilde{\eta}_u$,

$$
\begin{aligned}
\tilde{\mathbf{B}}'_u(n,s) &= \ln \mathbb{P}\bigl\{\Xi - \Xi_u; \tilde{\xi}_u = n, \tilde{\eta}_u = s\bigr\} = \ln\Bigl(\mathbb{P}\bigl\{\Xi - \Xi_u; \tilde{\eta}_u = s\bigr\} \times \mathbb{P}\bigl\{\tilde{\xi}_u = n \bigm| \Xi - \Xi_u; \tilde{\eta}_u = s\bigr\} \Bigr) \\
&= \tilde{\mathbf{J}}_u(s) + \ln \mathbb{P}\bigl\{\tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s\bigr\}
\end{aligned}
$$

where, by Theorem 2,

$$\ln \mathbb{P}\big\{\tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s\big\} = \begin{cases} \ln \binom{\kappa_u+n-1}{n-s} + (\kappa_u + s)\ln(1-\tilde{q}_u) + (n-s)\ln(\tilde{q}_u) & \text{if } 0 < q_u \\ (n-s)\ln(\tilde{r}_v) - \tilde{r}_u - \ln((n-s)!) & \text{if } 0 = q_u \end{cases}$$

By summing across disjoint events,.

$$\tilde{\mathbf{B}}_u(n) = \ln \mathbb{P}\big\{\Xi - \Xi_u; \tilde{\xi}_u = n\big\} = \ln \sum_{s=0}^{n} \mathbb{P}\big\{\Xi - \Xi_u; \tilde{\xi}_u = n, \tilde{\eta}_u = s\big\} = \ln \sum_{s=0}^{n} \exp\big(\tilde{\mathbf{B}}'_u(n,s)\big).$$

(f) Let v be a non-root node with parent u and sibling w (i.e, $uv, uw \in T$). Define $\tilde{\mathbf{J}}'_v(n,s) = \ln \mathbb{P}\big\{\Xi - \Xi_v; \tilde{\xi}_u = n, \tilde{\eta}_v = s\big\}$. Then

$$\tilde{\mathbf{J}}_v(s) = \ln \sum_{n=s}^{\Omega_u} \exp(\tilde{\mathbf{J}}'_v(n,s)).$$

We have

$$\begin{aligned} \tilde{\mathbf{J}}'_v(n,s) &= \ln \mathbb{P}\big\{\Xi - \Xi_v; \tilde{\xi}_u = n, \tilde{\eta}_v = s\big\} = \ln \mathbb{P}\big\{\Xi - \Xi_u, \Xi_w; \tilde{\xi}_u = n, \tilde{\eta}_v = s\big\} \\ &= \ln\Big(\mathbb{P}\big\{\Xi - \Xi_u; \tilde{\xi}_u = n\big\} \times \mathbb{P}\big\{\tilde{\eta}_v = s \bigm| \tilde{\xi}_u = n\big\} \times \mathbb{P}\big\{\Xi_w \bigm| \tilde{\xi}_u = n, \tilde{\eta}_v = s\big\}\Big) \\ &= \tilde{\mathbf{B}}_u(n) + \ln \mathbb{P}\big\{\tilde{\eta}_v = s \bigm| \tilde{\xi}_u = n\big\} + \tilde{\mathbf{L}}_w(n, n-s). \end{aligned}$$

By Theorem 2,

$$\ln \mathbb{P}\big\{\tilde{\eta}_v = s \bigm| \tilde{\xi}_u = n\big\} = \ln \binom{n}{s} + s\ln(1-p_v^-) + (n-s)\ln(p_v^-).$$

□

A.2. Gradient of the log-likelihood. We arrive to the gradient in three steps. First, we calculate the gradient with respect to the survival paramaters (Theorem 11). Second, the survival gradient is transformed into the gradient with respect to model parameters (Theorem 6) in a prefix traversal of the tree. Third, we compute the gradient with respect to the duplication and gan rates logit λ and $\ln \gamma$ using the chain rule (Theorem 7).

Recall Eq. 4: for any model parameter θ, the partial derivatives of the corrected log-likelihood are calculated as

$$\frac{\partial}{\partial\theta} \ln L^* = \sum_{f=1}^{F} \frac{\partial}{\partial\theta} \ln L(\Xi^f) + F\frac{L(0)}{1-L(0)} \frac{\partial}{\partial\theta} \ln L(0). \qquad \textbf{[5]}$$

Theorem 11 bypasses the calculation of gradients for observed and unobserved profiles and computes instead the linear combination of expectations implied by Eq. (5).

Theorem 11 (Gradient of the corrected survival log-likelihood)**.** *Let $\mathcal{S} = \big\{\Xi^1, \ldots, \Xi^F\big\}$ be a sample of observed profiles with some observation threshold $\Omega_{\min}$ and let*

$$\Phi = \ln L^* = \sum_{f=1}^{F} \ln \mathbb{P}\big\{\Xi^f \bigm| \Omega_{\min} \leq \Omega(\Xi^f)\big\}$$

denote its corrected log-likelihood. Define the following expectations across the sample and across the unobserved profiles at every node $1 \leq u \leq R$

$$\begin{aligned} \tilde{N}_{u,i} &= \sum_{f=1}^{F} \mathbb{P}\big\{i < \tilde{\xi}_u \bigm| \Xi^f\big\} + F\frac{L(0)}{1-L(0)} \text{‘}\mathbb{P}\big\{i < \tilde{\xi}_u \bigm| \Omega(\Xi) < \Omega_{\min}\big\} \\ \tilde{S}_{u,i} &= \sum_{f=1}^{F} \mathbb{P}\big\{i < \tilde{\eta}_u \bigm| \Xi^f\big\} + F\frac{L(0)}{1-L(0)} \text{‘}\mathbb{P}\big\{i < \tilde{\eta}_u \bigm| \Omega(\Xi) < \Omega_{\min}\big\}, \end{aligned}$$

$\tilde{N}_u = \sum_{i=0}^{M_u} \tilde{N}_{u,i}$ and $\tilde{S}_u = \sum_{i=0}^{M_u} \tilde{S}_{u,i}$, where $L(0) = \mathbb{P}\big\{\Omega(\Xi) < \Omega_{\min}\big\}$ is the probability of no-observation and M_u denotes the maximum copy sum at leaves descending from u across all families. Set $F^* = F/(1 - L(0))$.

At every node $1 \le u \le R$, if $0 < \tilde{q}_u$, then

$$\Phi^{(\operatorname{logit} \tilde{q}_u)} = \frac{\partial}{\partial \operatorname{logit} \tilde{q}_u} \ln L^* = (1 - \tilde{q}_u)\tilde{D}_u^+ - \tilde{q}_u\bigl(\tilde{S}_u + F^*\kappa_u\bigr)$$

$$\Phi^{(\ln \kappa_u)} = \frac{\partial}{\partial \ln \kappa_u} \ln L^* = \left(\sum_{i=0}^{M_u} \tilde{D}_{u,i}^+ \frac{\kappa_u}{\kappa_u + i}\right) - F^*\kappa_u\bigl(-\ln(1 - \tilde{q}_u)\bigr),$$

with the expected births

$$\tilde{D}_{u,i}^+ = \tilde{N}_{u,i} - \tilde{S}_{u,i} \qquad \tilde{D}_u^+ = \tilde{N}_u - \tilde{S}_u = \sum_{0 \le i} \tilde{D}_{u,i}^+;$$

and, if $q_u = 0$, then

$$\Phi^{(\ln \tilde{r}_u)} = \frac{\partial}{\partial \ln \tilde{r}_u} \ln L^* = \tilde{D}_u^+ - F^*\tilde{r}_u.$$

At the root, $\Phi^{(\operatorname{logit} \tilde{p}_v)} = 0$. On every edge $uv \in T$ and sibling $uw \in T$,

$$\Phi^{(\operatorname{logit} \tilde{p}_v)} = \frac{\partial}{\partial \operatorname{logit} \tilde{p}_v} \ln L^* = (1 - p_v^-)\tilde{D}_v^- - p_v^-\tilde{S}_v$$

with the expected deaths

$$\tilde{D}_v^- = \tilde{N}_u - \tilde{S}_v$$

and $\operatorname{logit} p_v^- = \operatorname{logit} \tilde{p}_v + \ln(1 - \tilde{p}_w)$.

Proof of Theorem 11. First consider a single family with profile Ξ. By definition,

$$L(\Xi) = \sum_{s=0}^{\Omega_u} \exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{K}}_u(s)\bigr).$$

For $0 < \tilde{q}_u$, we have, by Theorem 10(b)

$$\begin{aligned}
\frac{\partial \ln L(\Xi)}{\partial \operatorname{logit} \tilde{q}_u} &= \frac{\tilde{q}_u(1 - \tilde{q}_u)}{L(\Xi)} \times \frac{\partial L(\Xi)}{\partial \tilde{q}_u} \\
&= \frac{\tilde{q}_u(1 - \tilde{q}_u)}{L(\Xi)} \times \frac{\partial}{\partial \tilde{q}_u} \sum_{s=0}^{\Omega_u} \sum_{n=s}^{\Omega_u} \exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{K}}'_u(n,s)\bigr) \\
&= \frac{\tilde{q}_u(1 - \tilde{q}_u)}{L(\Xi)} \sum_{s=0}^{\Omega_u} \sum_{n=s}^{\Omega_u} \left(\exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{C}}_u(n)\bigr)\binom{\kappa_u + n - 1}{n - s} \times \frac{\partial}{\partial \tilde{q}_u}\Bigl((1 - \tilde{q}_u)^s(\tilde{q}_u)^{n-s}\Bigr)\right) \\
&= \frac{\tilde{q}_u(1 - \tilde{q}_u)}{L(\Xi)} \sum_{0 \le s} \sum_{s \le n} \exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{K}}'_u(n,s)\bigr) \times \left(\frac{n - s}{\tilde{q}_u} - \frac{s}{1 - \tilde{q}_u}\right) \\
&= (1 - \tilde{q}_u) \times \mathbb{E}\bigl[\tilde{\xi}_u - \eta_u \bigm| \Xi\bigr] - \tilde{q}_u \times \mathbb{E}\bigl[\eta_u \bigm| \Xi\bigr].
\end{aligned}$$

For the partial derivative by κ_u, we have

$$\frac{\partial}{\partial \kappa}\binom{\kappa + n - 1}{n - s} = \frac{\partial}{\partial \kappa} \frac{\prod_{i=s}^{n-1}(\kappa + i)}{(n - s)!} = \binom{\kappa + n - 1}{n - s} \times \sum_{i=s}^{n-1} \frac{1}{\kappa + i};$$

so,

$$\frac{\partial}{\partial \kappa}\binom{\kappa + n - 1}{n - s}(1 - \tilde{q})^{\kappa + s}(\tilde{q})^{n-s} = \binom{\kappa + n - 1}{n - s}(1 - \tilde{q})^{\kappa + s}(\tilde{q})^{n-s} \times \left(\ln(1 - \tilde{q}) + \sum_{i=s}^{n-1} \frac{1}{\kappa + i}\right)$$

Thus,

$$\begin{aligned}
\frac{\partial \ln L(\Xi)}{\partial \ln \kappa_u} &= \frac{\kappa_u}{L(\Xi)} \sum_{s=0}^{\Omega_u} \sum_{n=s}^{\Omega_u} \left(\exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{K}}'_u(n,s)\bigr) \times \left(\ln(1 - \tilde{q}_u) + \sum_{i=s}^{n-1} \frac{1}{\kappa_u + i}\right)\right) \\
&= \kappa_u \ln(1 - \tilde{q}_u) + \frac{1}{L(\Xi)} \sum_{i=0}^{\Omega_u - 1} \frac{\kappa_u}{\kappa_u + i} \sum_{s=0}^{i} \sum_{n=i+1}^{\Omega_u} \exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{K}}'_u(n,s)\bigr) \\
&= \kappa_u \ln(1 - \tilde{q}_u) + \sum_{0 \le i} \frac{\kappa_u}{\kappa_u + i}\Bigl(\mathbb{P}\bigl\{i < \tilde{\xi}_u \bigm| \Xi\bigr\} - \mathbb{P}\bigl\{i < \eta_u \bigm| \Xi\bigr\}\Bigr)
\end{aligned}$$

If $0 = \tilde{q}_u$, then

$$\begin{aligned}
\frac{\partial \ln L(\Xi)}{\partial \ln \tilde{r}_u} &= \frac{\tilde{r}_u}{L(\Xi)} \times \sum_{s=0}^{\Omega_u} \sum_{n=s}^{\Omega_u} \exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{C}}_u(n)\bigr) \times \frac{\partial}{\partial \tilde{r}_u}\left(e^{-\tilde{r}_u} \frac{(\tilde{r}_u)^{n-s}}{(n-s)!}\right) \\
&= \frac{\tilde{r}_u}{L(\Xi)} \times \sum_{s=0}^{\Omega_u} \sum_{n=s}^{\Omega_u} \exp\bigl(\tilde{\mathbf{J}}_u(s) + \tilde{\mathbf{C}}_u(n)\bigr) \times e^{-\tilde{r}_u} \frac{(\tilde{r}_u)^{n-s}}{(n-s)!}\left(-1 + \frac{n-s}{\tilde{r}_u}\right) \\
&= -\tilde{r}_u + \mathbb{E}\bigl[\tilde{\xi}_u - \tilde{\eta}_u \bigm| \Xi\bigr].
\end{aligned}$$

For the partial derivative by $\operatorname{logit} \tilde{p}_v$ on edge uv, we have $\frac{\partial \operatorname{logit} p_v^-}{\partial \operatorname{logit} \tilde{p}_v} = 1$; thus,

$$\begin{aligned}
\frac{\partial \ln L(\Xi)}{\partial \operatorname{logit} \tilde{p}_v} &= \frac{1}{L(\Xi)} \sum_{s=0}^{\Omega_v} \sum_{n=s}^{\Omega_u} \frac{\partial}{\partial \operatorname{logit} \tilde{p}_v} \exp\bigl(\tilde{\mathbf{J}}_v'(n,s) + \tilde{\mathbf{K}}_v(s)\bigr) \\
&= \frac{1}{L(\Xi)} \sum_{s=0}^{\Omega_v} \sum_{n=s}^{\Omega_u} \left(\exp\bigl(\tilde{\mathbf{B}}_u(n) + \tilde{\mathbf{L}}_w(n, n-s) + \tilde{\mathbf{K}}_v(s)\bigr) \times \frac{\partial}{\partial \operatorname{logit} p_v^-} \binom{n}{s} (1-p_v^-)^s (p_v^-)^{n-s}\right) \\
&= \frac{1}{L(\Xi)} \sum_{s=0}^{\Omega_v} \sum_{n=s}^{\Omega_u} \left(\exp\bigl(\tilde{\mathbf{B}}_u(n) + \tilde{\mathbf{L}}_w(n, n-s) + \tilde{\mathbf{K}}_v(s)\bigr) \times \binom{n}{s} (1-p_v^-)^s (p_v^-)^{n-s} \times \bigl(-p_v^- s + (n-s)(1-p_v^-)\bigr)\right) \\
&= (1-p_v^-) \times \mathbb{E}\bigl[\tilde{\xi}_u - \tilde{\eta}_v \bigm| \Xi\bigr] - (p_v^-) \times \mathbb{E}\bigl[\tilde{\eta}_v \bigm| \Xi\bigr].
\end{aligned}$$

The Theorem now follows from Eq. (5) by the linearity of expectations: for any observation bias $\Omega_{\min}$,

$$\ln L(0) = \ln\left(\sum_{\Xi \in \mathcal{S}_0} L(\Xi)\right) = \ln\left(\sum_{\Xi \in \mathcal{S}_0} \exp\bigl(\ln L(\Xi)\bigr)\right),$$

where $\mathcal{S}_0$ is the set of profiles with less than $\Omega_{\min}$ copies. Thus, the partial derivatives are

$$\frac{\partial}{\partial \theta} \ln L(0) = \sum_{\Xi \in \mathcal{S}_0} \underbrace{\exp\bigl(\ln L(\Xi) - \ln L(0)\bigr)}_{=\mathbb{P}\left\{\Xi \middle| \Omega(\Xi) < \Omega_{\min}\right\}} \times \frac{\partial}{\partial \theta} \ln L(\Xi).$$

□

The posterior expectations can be written as sums

$$\mathbb{E}\bigl[\xi_u \bigm| \Xi\bigr] = \sum_{0 \le n} \mathbb{P}\bigl\{n < \xi_u \bigm| \Xi\bigr\} \quad \mathbb{E}\bigl[\eta_u \bigm| \Xi\bigr] = \sum_{0 \le s} \mathbb{P}\bigl\{s < \eta_u \bigm| \Xi\bigr\}.$$

At every node u, the expected number of copy births equals

$$\mathbb{E}\bigl[\xi_u \bigm| \Xi\bigr] - \mathbb{E}\bigl[\eta_u \bigm| \Xi\bigr] = \sum_{0 \le i} \Bigl(\mathbb{P}\bigl\{i < \xi_u \bigm| \Xi\bigr\} - \mathbb{P}\bigl\{i < \eta_u \bigm| \Xi\bigr\}\Bigr) \qquad \text{with}$$

$$\mathbb{P}\bigl\{i < \xi_u \bigm| \Xi\bigr\} - \mathbb{P}\bigl\{i < \eta_u \bigm| \Xi\bigr\} = \sum_{i+1 \le n} \sum_{s=0}^{i} \mathbb{P}\bigl\{\xi_u = n, \eta_u = s \bigm| \Xi\bigr\}, \tag{6}$$

On every edge uv, the expected number of copy deaths equals

$$\mathbb{E}\bigl[\xi_u \bigm| \Xi\bigr] - \mathbb{E}\bigl[\eta_v \bigm| \Xi\bigr] = \sum_{0 \le i} \Bigl(\mathbb{P}\bigl\{i < \xi_u \bigm| \Xi\bigr\} - \mathbb{P}\bigl\{i < \eta_v \bigm| \Xi\bigr\}\Bigr) \qquad \text{with}$$

$$\mathbb{P}\bigl\{i < \xi_u \bigm| \Xi\bigr\} - \mathbb{P}\bigl\{i < \eta_v \bigm| \Xi\bigr\} = \sum_{i+1 \le n} \sum_{s=0}^{i} \mathbb{P}\bigl\{\xi_u = n, \eta_v = s \bigm| \Xi\bigr\}. \tag{7}$$

Theorem 12 (Exact gradient (Theorem 6)). *(a) At the root $v = R$, if $q_v \neq 0$, then*

$$\nabla \Phi(\operatorname{logit} q_v, \operatorname{logit} \epsilon_v) = \begin{pmatrix} \Phi^{(\operatorname{logit} \tilde{q}_v)} \\ -\epsilon_v \Phi^{(\operatorname{logit} \tilde{q}_v)} \end{pmatrix};$$

and if $q_v = 0$, then

$$\nabla\Phi(\operatorname{logit} r_v, \operatorname{logit}\epsilon_v) = \begin{pmatrix} \Phi^{(\operatorname{logit}\tilde r_v)} \\ -\epsilon_v \Phi^{(\operatorname{logit}\tilde r_v)} \end{pmatrix}.$$

(b) Let v be any nonroot node with parent u. If $q_v \neq 0$, then

$$\nabla^T\Phi(\operatorname{logit} p_v, \operatorname{logit} q_v, \operatorname{logit}\epsilon_v) = \nabla^T\Phi(\operatorname{logit}\tilde p_v, \operatorname{logit}\tilde q_v, \operatorname{logit}\epsilon_u) \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1-p_v^- & 0 \end{pmatrix} \times \begin{pmatrix} \frac{p_v}{\tilde p_v} & -q_v\frac{\epsilon_v(1-\tilde q_v)}{\tilde p_v} & \frac{\epsilon_v(1-\tilde q_v)}{\tilde p_v} \\ 0 & 1 & -\epsilon_v \end{pmatrix}$$

If $q_v = 0$, then

$$\nabla^T\Phi(\operatorname{logit} p_v, \operatorname{logit}\epsilon_v, \ln r_v) = \nabla^T\Phi(\operatorname{logit}\tilde p_v, \operatorname{logit}\epsilon_u, \ln\tilde r_v) \times \begin{pmatrix} 1 & 0 \\ 1-p_v^- & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} \frac{p_v}{\tilde p_v} & \frac{\epsilon_v}{\tilde p_v} & 0 \\ 0 & -\epsilon_v & 1 \end{pmatrix}$$

Proof of Theorem 6/Theorem 12. The proof follows the techniques for the gradient by $\Phi^{(p_u)}$, $\Phi^{(q_u)}$ and $\Phi^{(\kappa_u)}$ or $\Phi^{(r_u)}$ described by Theorems 12 and 14 in (1), via the chain rule:

$$\frac{\partial\Phi}{\partial\operatorname{logit}\theta} = \theta(1-\theta)\frac{\partial\Phi}{\partial\theta} \quad\text{and}\quad \frac{\partial\Phi}{\partial\ln\theta} = \theta\frac{\partial\Phi}{\partial\theta}.$$

and the conversion

$$\frac{\partial\Phi}{\partial\operatorname{logit}\theta} = (1-\theta)\frac{\partial\Phi}{\partial\ln\theta}.$$

We use the rules that $\operatorname{logit}(1-x) = -\operatorname{logit} x$ and if $\ln x = \ln y + f(z)$ with $\frac{\partial z}{\partial y} = 0$ then

$$\frac{\partial\operatorname{logit} x}{\partial\operatorname{logit} y} = \frac{\partial\ln x}{\partial\ln y} \times \frac{\partial\operatorname{logit} x}{\partial\ln x} \times \frac{\partial\ln y}{\partial\operatorname{logit} y} = \frac{1-y}{1-x}.$$

Let v be any node, including the root. If $q_v \neq 0$, then $\operatorname{logit}\tilde q_v = \operatorname{logit} q_v + \ln(1-\epsilon_v)$, thus

$$\frac{\partial\operatorname{logit}\tilde q_v}{\partial\operatorname{logit} q_v} = 1 \qquad \frac{\partial\operatorname{logit}\tilde q_v}{\partial\operatorname{logit}\epsilon_v} = \frac{\partial\operatorname{logit}\tilde q_v}{\partial\ln(1-\epsilon_v)} \times \frac{\partial\ln(1-\epsilon_v)}{\partial\operatorname{logit}(1-\epsilon_v)} \times \frac{\partial\operatorname{logit}(1-\epsilon_v)}{\partial\operatorname{logit}\epsilon_v} = -\epsilon_v. \tag{8}$$

If $q_v = 0$, then $\ln\tilde r_v = \ln r_v + \ln(1-\epsilon_v)$, so

$$\frac{\partial\ln\tilde r_R}{\partial\ln r_R} = 1 \qquad \frac{\partial\ln\tilde r_R}{\partial\operatorname{logit}\epsilon_R} = \frac{\partial\ln\tilde r_R}{\partial\ln(1-\epsilon_v)} \times \frac{\ln(1-\epsilon_v)}{\partial\operatorname{logit}(1-\epsilon_v)} \times \frac{\operatorname{logit}(1-\epsilon_v)}{\operatorname{logit}\epsilon_v} = -\epsilon_v. \tag{9}$$

Let θ_v be a model parameter at node v such as $\operatorname{logit} p_v$, and either $\ln r_v$ (if $q_v = 0$) or $\operatorname{logit} q_v$. Since θ_v influences the survival parameters at all nodes along the path from v to the root but not at any other node, we unfold the dependencies via the extinction probabilities by descending from the root. Now, survival parameters from the parent u to the root depend from θ_v and $\operatorname{logit}\epsilon_v$ only through the parent's extinction probability ϵ_u.

Since $\ln\epsilon_u = \ln\tilde p_v + \ln\tilde p_w$, we have $\frac{\partial\operatorname{logit}\epsilon_u}{\partial\operatorname{logit}\tilde p_v} = \frac{1-\tilde p_v}{1-\epsilon_u} == \frac{1-\tilde p_v}{1-\tilde p_v\tilde p_w} = 1-p_v^-$.

If $0 \neq q_v$, then we have

$$\tilde p_v = \frac{p_v(1-\epsilon_v) + (1-q_v)\epsilon_v}{1-q_v\epsilon_v} \quad\text{and}\quad \ln(1-\tilde p_v) = \ln(1-p_v) + \ln\bigl(1-\epsilon_v(1-\tilde q_v)\bigr)$$

Consequently,

$$\frac{\partial\operatorname{logit}\tilde p_v}{\partial\operatorname{logit} p_v} = \frac{\partial\operatorname{logit}(1-\tilde p_v)}{\partial\operatorname{logit}(1-\operatorname{logit} p_v)} = \frac{p_v}{\tilde p_v}$$

$$\frac{\partial\operatorname{logit}\tilde p_v}{\partial\operatorname{logit} q_v} = \frac{q_v(1-q_v)}{\tilde p_v(1-\tilde p_v)} \times \frac{\partial\tilde p_v}{\partial q_v} = \frac{q_v(1-q_v)}{\tilde p_v(1-\tilde p_v)} \times \frac{-(1-p_v)\epsilon_v(1-\epsilon_v)}{(1-q_v\epsilon_v)^2} = \underbrace{\frac{1-q_v}{1-q_v\epsilon_v}}_{=1-\tilde q_v} \times \frac{1}{\tilde p_v} \times \underbrace{\frac{(1-p_v)(1-\epsilon_v)}{(1-q_v\epsilon_v)(1-\tilde p_v)}}_{=1} \times (-q_v\epsilon_v)$$

$$= -q_v\frac{\epsilon_v(1-\tilde q_v)}{\tilde p_v}$$

$$\frac{\partial\operatorname{logit}\tilde p_v}{\partial\operatorname{logit}\epsilon_v} = \frac{\epsilon_v(1-\epsilon_v)}{\tilde p_v(1-\tilde p_v)} \times \frac{\partial\tilde p_v}{\partial\epsilon_v} = \frac{\epsilon_v(1-\epsilon_v)}{\tilde p_v(1-\tilde p_v)} \times \frac{(1-p_v)(1-q_v)}{(1-q_v\epsilon_v)^2} = \frac{\epsilon_v}{\tilde p_v} \times \underbrace{\frac{(1-p_v)(1-\epsilon_v)}{(1-q_v\epsilon_v)(1-\tilde p_v)}}_{=1} \times \underbrace{\frac{1-q_v}{1-q_v\epsilon_v}}_{=1-\tilde q_v} = \frac{\epsilon_v(1-\tilde q_v)}{\tilde p_v}.$$

Using the chain rule,

$$
\begin{aligned}
\Phi^{(\operatorname{logit} p_v)} = \frac{\partial \Phi}{\partial \operatorname{logit} p_v} &= \Big(\Phi^{(\operatorname{logit} \tilde{p}_v)} + (1-p_v^-)\Phi^{(\operatorname{logit} \epsilon_u)}\Big) \times \frac{\partial \operatorname{logit} \tilde{p}_v}{\partial \operatorname{logit} p_v} \\
&= \Big(\Phi^{(\operatorname{logit} \tilde{p}_v)} + (1-p_v^-)\Phi^{(\operatorname{logit} \epsilon_u)}\Big) \times \frac{p_v}{\tilde{p}_v} \\
\Phi^{(\operatorname{logit} q_v)} = \frac{\partial \Phi}{\partial \operatorname{logit} q_v} &= \Big(\Phi^{(\operatorname{logit} \tilde{p}_v)} + (1-p_v^-)\Phi^{(\operatorname{logit} \epsilon_u)}\Big) \times \frac{\partial \operatorname{logit} \tilde{p}_v}{\partial \operatorname{logit} q_v} + \Phi^{(\operatorname{logit} \tilde{q}_v)} \times \frac{\partial \operatorname{logit} \tilde{q}_v}{\partial \operatorname{logit} q_v} \\
&= \Big(\Phi^{(\operatorname{logit} \tilde{p}_v)} + (1-p_v^-)\Phi^{(\operatorname{logit} \epsilon_u)}\Big) \times \frac{-q_v\epsilon_v(1-\tilde{q}_v)}{\tilde{p}_v} + \Phi^{(\operatorname{logit} \tilde{q}_v)} \\
\Phi^{(\operatorname{logit} \epsilon_v)} = \frac{\partial \Phi}{\partial \operatorname{logit} \epsilon_v} &= \Big(\Phi^{(\operatorname{logit} \tilde{p}_v)} + (1-p_v^-)\Phi^{(\operatorname{logit} \epsilon_u)}\Big) \times \frac{\partial \operatorname{logit} \tilde{p}_v}{\partial \operatorname{logit} \epsilon_v} + \Phi^{(\operatorname{logit} \tilde{q}_v)} \times \frac{\partial \operatorname{logit} \tilde{q}_v}{\partial \operatorname{logit} \epsilon_v} \\
&= \Big(\Phi^{(\operatorname{logit} \tilde{p}_v)} + (1-p_v^-)\Phi^{(\operatorname{logit} \epsilon_u)}\Big) \times \frac{\epsilon_v(1-\tilde{q}_v)}{\tilde{p}_v} + \Phi^{(\operatorname{logit} \tilde{q}_v)} \times (-\epsilon_v)
\end{aligned}
$$

as the theorem claims in matrix form with the Jacobians.

If $q_v = 0$, then we have $\ln(1-\tilde{p}_v) = \ln(1-p_v) + \ln(1-\epsilon_v)$, thus

$$
\frac{\partial \operatorname{logit} \tilde{p}_v}{\partial \operatorname{logit} p_v} = \frac{\partial \operatorname{logit}(1-\tilde{p}_v)}{\partial \operatorname{logit}(1-p_v)} = \frac{p_v}{\tilde{p}_v} \qquad\qquad \frac{\partial \operatorname{logit} \tilde{p}_v}{\partial \operatorname{logit} \epsilon_v} = \frac{\epsilon_v}{\tilde{p}_v}.
$$

Hence,

$$
\nabla^T \Phi(\operatorname{logit} p_v, \operatorname{logit} \epsilon_v, \ln r_v) = \begin{pmatrix} \Phi^{(\operatorname{logit} \tilde{p}_v)} + (1-p_v^-)\Phi^{(\operatorname{logit} \epsilon_u)} \\ \Phi^{(\ln \tilde{r}_v)} \end{pmatrix} \times \begin{pmatrix} \frac{p_v}{\tilde{p}_v} & \frac{\epsilon_v}{\tilde{p}_v} & 0 \\ 0 & -\epsilon_v & 1 \end{pmatrix}
$$

□

The next step is to first convert into gradients by the log-gain rate $\ln \gamma$ (Theorem 7 below) and then by logit λ.

Theorem 13 (Gradient by relative gain and duplication rates (Theorem **??**)). *For every node $1 \le v \le R$, if $q_v \neq 0$, then*

$$
\nabla^T \Phi(\operatorname{logit} p_v, \operatorname{logit} q_v, \ln \gamma_v) = \nabla^T \Phi(\operatorname{logit} p_v, \operatorname{logit} q_v, \ln \kappa_v) \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1-p_v & -(1-q_v) & 1 \end{pmatrix}; \tag{10a}
$$

and if $q_v = 0$, then

$$
\nabla^T \Phi(\operatorname{logit} p_v, \ln \gamma_v) = \nabla^T \Phi(\operatorname{logit} p_v, \ln r_v) \times \begin{pmatrix} 1 & 0 \\ 1-p_v & 1 \end{pmatrix}. \tag{10b}
$$

For a duplication rate $\lambda_v = q_v/p_v \le 1$, we have

$$
\nabla^T \Phi(\operatorname{logit} p_v, \operatorname{logit} \lambda_v) = \begin{pmatrix} \Phi^{(\operatorname{logit} p_v)} \\ \Phi^{(\operatorname{logit} \lambda_v)} \end{pmatrix} = \nabla^T \Phi(\operatorname{logit} p_v, \operatorname{logit} q_v) \times \begin{pmatrix} 1 & 0 \\ \frac{1-p_v}{1-q_v} & \frac{1-\lambda_v}{1-q_v} \end{pmatrix} \tag{11}
$$

Proof of Theorem 7/Theorem 13. If $q_v \neq 0$, then we have $q_v\kappa_v = p_v\gamma_v$; if $q_v = 0$, then we have $r_v = p_v\gamma_v$. The formulas of Eq. (10) show the corresponding Jacobian matrix. Eq. (11) shows the Jacobian matrix corresponding to $q_v = p_v\lambda_v$. □

A.3. Unobserved profiles.

Proof of Theorem 3. At every node,

$$
\mathbb{P}\{\Omega_R \le M\} = \sum_{m=0}^{M} \mathbb{P}\big\{\Omega_u = m; \Omega_u^* \le M-m\big\}
$$

We have

$$
L(0) = \mathbb{P}\{\Omega_R \le M\} = \sum_{m=0}^{M}\sum_{n=m}^{M} \underbrace{\mathbb{P}\Big\{\Omega_u = m \,\Big|\, \tilde{\xi}_u = n\Big\}}_{=\exp(\tilde{\mathbf{C}}_{u,m}(n))} \times \sum_{s=0}^{n} \underbrace{\mathbb{P}\big\{\tilde{\xi}_u = n, \tilde{\eta}_u = s, \Omega_u^* \le M-m\big\}}_{=\exp(\tilde{\mathbf{B}}'_{u,m}(n,s))}
$$

(a) By definition each of the surviving copies counted in $\tilde{\xi}_u$ and $\tilde{\eta}_v$ have a surviving descendant copy at the leaves of the subtreee rooted at u. Thus surely $\tilde{\eta}_u \le \tilde{\xi}_u \le \Omega_u$.

(b) Recurrence for $\tilde{\mathbf{K}}'_{u,m}(n,s)$:

$$\begin{aligned}\tilde{\mathbf{K}}'_{u,m}(n,s) &= \ln \mathbb{P}\{\Omega_u = m; \tilde{\xi}_u = n \mid \tilde{\eta}_u = s\} = \ln \mathbb{P}\{\Omega_u = m \mid \tilde{\xi}_u = n\} + \ln \mathbb{P}\{\tilde{\xi}_u = n \mid \tilde{\eta}_u = s\} \\ &= \tilde{\mathbf{C}}_{u,m}(n) + \ln \underbrace{\mathbb{P}\{\tilde{\xi}_u = n \mid \tilde{\eta}_u = s\}}_{\text{Poisson}(\tilde{r}_u) \text{ or Pólya}(\kappa_u, \tilde{q}_u)}\end{aligned}$$

(c) Recurrence for $\tilde{\mathbf{C}}_{u,m}$

$$\tilde{\mathbf{C}}_{u,m}(n) = \ln \mathbb{P}\Big\{\Omega_u = m \;\Big|\; \tilde{\xi}_u = n\Big\}.$$

If u is a leaf, then

$$\tilde{\mathbf{C}}_{u,m}(n) = \begin{cases} \ln 0 & \text{if } n \neq m \\ \ln 1 & \text{if } n = m \end{cases}$$

Suppose u is an inner node with two child edges $uv, uw \in T$. Define the random variable $\tilde{\psi}_u$ as the surviving copies that are inherited by both children (from among $\tilde{\xi}_u$ surviving copies at u). Necessarily, $\tilde{\xi}_u = \tilde{\eta}_v + \tilde{\eta}_w - \tilde{\psi}_u$. We have

$$\begin{aligned}\tilde{\mathbf{C}}_{u,m}(n) &= \ln \sum_{s=0}^{n} \sum_{k=n-s}^{m-s} \sum_{j=0}^{s} \mathbb{P}\{\Omega_v = m-k; \tilde{\eta}_v = s; \Omega_w = k; \tilde{\eta}_w = n-s+j; \tilde{\psi}_u = j \mid \tilde{\xi}_u = n\} \\ &= \ln \Bigg(\sum_{s=0}^{n} \sum_{k=n-s}^{m-s} \underbrace{\mathbb{P}\{\Omega_v = m-k \mid \tilde{\eta}_v = s\}}_{=\exp(\tilde{\mathbf{K}}_{v,m-k}(s))} \times \binom{n}{s} \left(\frac{1-\tilde{p}_v}{1-\tilde{p}_v\tilde{p}_w}\right)^{s} \left(\frac{\tilde{p}_v(1-\tilde{p}_w)}{1-\tilde{p}_v\tilde{p}_w}\right)^{n-s} \\ &\qquad \underbrace{\times \sum_{j=0}^{s} \underbrace{\mathbb{P}\{\Omega_w = k \mid \tilde{\eta}_w = n-s+j\}}_{=\exp(\tilde{\mathbf{K}}_{w,k}(n-s+j))} \times \binom{s}{j} (1-\tilde{p}_w)^j (\tilde{p}_w)^{s-j}}_{=\exp(\tilde{\mathbf{L}}_{w,k}(n,n-s))} \Bigg)\end{aligned}$$

For $1 - p_v^- = \frac{1-\tilde{p}_v}{1-\tilde{p}_v\tilde{p}_w}$ and $p_v^- = \frac{\tilde{p}_v(1-\tilde{p}_w)}{1-\tilde{p}_v\tilde{p}_w}$, we have

$$\operatorname{logit} p_v^- = \ln \frac{\tilde{p}_v(1-\tilde{p}_w)}{1-\tilde{p}_v} = \operatorname{logit} \tilde{p}_v + \ln(1-\tilde{p}_w).$$

Let

$$\begin{aligned}\tilde{\mathbf{L}}_{w,k}(n,t) &= \ln \mathbb{P}\{\Omega_w = k \mid \tilde{\eta}_v = n-t; \tilde{\xi}_u = n\} = \ln \sum_{j=0}^{n-t} \mathbb{P}\{\Omega_w = k; \tilde{\eta}_w = t+j \mid \tilde{\eta}_v = n-t; \tilde{\xi}_u = n\} \\ &= \ln \sum_{j=0}^{n-t} \mathbb{P}\{\Omega_w = k \mid \tilde{\eta}_w = t+j\} \times \mathbb{P}\{\tilde{\eta}_w = t+j \mid \tilde{\eta}_v = n-t; \tilde{\xi}_u = n\} \\ &= \ln \sum_{j=0}^{n-t} \mathbb{P}\{\Omega_w = k \mid \tilde{\eta}_w = t+j\} \times \binom{n-t}{j} (1-\tilde{p}_w)^j (\tilde{p}_w)^{n-t-j};\end{aligned}$$

with non-zero terms only for $t + j \le k$ or $j \le k - t$. So,

$$\tilde{\mathbf{L}}_{w,k}(n,n) = \ln \mathbb{P}\{\Omega_w = k \mid \tilde{\eta}_v = 0; \tilde{\xi}_u = n\} = \ln \mathbb{P}\{\Omega_w = k \mid \tilde{\eta}_w = n\} = \tilde{\mathbf{K}}_{w,k}(n);$$

and, for $t < n$

$$\begin{aligned}
\tilde{\mathbf{L}}_{w,k}(n,t) &= \ln \sum_{j=0}^{n-t} \mathbb{P}\big\{\Omega_w = k \bigm| \tilde{\eta}_v = t+j\big\} \times \left(\binom{n-t-1}{j} + \binom{n-t-1}{j-1}\right)(1-\tilde{p}_w)^j(\tilde{p}_w)^{n-t-j} \\
&= \ln\Bigg((1-\tilde{p}_w)\sum_{i=0}^{n-(t+1)}\Big(\mathbb{P}\big\{\Omega_w = k \bigm| \tilde{\eta}_v = (t+1)+i\big\} \times \binom{n-(t+1)}{i}(1-\tilde{p}_w)^i(\tilde{p}_w)^{n-(t+1)-i}\Big) \\
&\quad + \tilde{p}_w \sum_{j=0}^{(n-1)-t}\Big(\mathbb{P}\big\{\Omega_w = k \bigm| \tilde{\eta}_v = t+j\big\} \times \binom{(n-1)-t}{j}(1-\tilde{p}_w)^j(\tilde{p}_w)^{(n-1)-t-j}\Big)\Bigg) \\
&= \ln\Big((1-\tilde{p}_w)\exp\big(\tilde{\mathbf{L}}_{w,k}(n,t+1)\big) + \tilde{p}_w \exp\big(\tilde{\mathbf{L}}_{w,k}(n-1,t)\big)\Big)
\end{aligned}$$

(d) Recurrence for $\tilde{\mathbf{B}}'_{u,k}$:

$$\begin{aligned}
\tilde{\mathbf{B}}'_{u,k}(n,s) &= \ln \mathbb{P}\big\{\tilde{\xi}_u = n, \tilde{\eta}_u = s, \Omega^*_u = k\big\} = \ln \mathbb{P}\big\{\tilde{\eta}_u = s, \Omega^*_u = k\big\} + \ln \mathbb{P}\big\{\tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s, \Omega^*_u = k\big\} \\
&= \tilde{\mathbf{J}}_{u,k}(s) + \ln \underbrace{\mathbb{P}\big\{\tilde{\xi}_u = n \bigm| \tilde{\eta}_u = s\big\}}_{\text{Poisson}(\tilde{r}_u)\text{ or Pólya}(\kappa_u,\tilde{q}_u)} \qquad \textit{since } \tilde{\xi}_u \textit{ depends of } \tilde{\eta}_u
\end{aligned}$$

(e) For the root, we have $\Omega^*_R = 0$ thus

$$\tilde{\mathbf{J}}_{R,k}(s) = \ln \mathbb{P}\big\{\Omega^*_R = k; \tilde{\eta}_R = s\big\} = \begin{cases} \ln 1 & \text{if } k = 0 \text{ and } 0 = s \\ \ln 0 & \text{if } 0 < k \text{ or } 0 < s \end{cases}$$

For uniformity, we also set $\tilde{\mathbf{J}}'_{R,0}(0,0) = 0$.

(f) Recurrence for $\tilde{\mathbf{J}}'_{v,k}$ on edge uv with sibling edge uw: summation by $\Omega_w = j$

$$\begin{aligned}
\tilde{\mathbf{J}}'_{v,k}(n,s) &= \ln \mathbb{P}\big\{\tilde{\xi}_u = n, \tilde{\eta}_v = s, \Omega^*_v = k\big\} \\
&= \ln \sum_{j=0}^{k} \mathbb{P}\big\{\tilde{\xi}_u = n, \tilde{\eta}_v = s, \Omega^*_v = k, \Omega_w = j\big\} \\
&= \ln \sum_{j=0}^{k} \mathbb{P}\big\{\tilde{\xi}_u = n, \tilde{\eta}_v = s, \Omega^*_u = k-j, \Omega_w = j\big\} \\
&= \ln \sum_{j=0}^{k} \mathbb{P}\big\{\tilde{\xi}_u = n, \Omega^*_u = k-j\big\} \times \mathbb{P}\big\{\Omega_u = j \bigm| \tilde{\eta}_v = s; \tilde{\xi}_u = n\big\} \times \binom{n}{s}(1-p^-_v)^s(p^-_v)^{n-s} \\
&= \ln \sum_{j=n-s}^{k} \exp\big(\tilde{\mathbf{B}}_{u,k-j}(n) + \tilde{\mathbf{L}}_{w,j}(n,n-s)\big) \times \binom{n}{s}(1-p^-_v)^s(p^-_v)^{n-s}
\end{aligned}$$

□

Theorem 14 (Posteriors for unobserved profiles (Theorem 4))**.** *We have the posteriors at nodes u and edges uv*

$$\begin{aligned}
\ln \mathbb{P}\big\{\tilde{\eta}_u = s, \tilde{\xi}_u = n \bigm| \Omega_R \le M\big\} &= \ln \sum_{m=n}^{M}\sum_{k=0}^{M-m} \exp\big(\mathbf{B}'_{u,k}(n,s) + \mathbf{C}_{u,m}(n) - \ln L(0)\big), \\
\ln \mathbb{P}\big\{\tilde{\xi}_u = n, \tilde{\eta}_v = s \bigm| \Omega_R \le M\big\} &= \ln \sum_{m=s}^{M}\sum_{k=0}^{M-m} \exp\big(\mathbf{J}'_{v,k}(n,s) + \mathbf{K}_{v,m}(s) - \ln L(0)\big), \\
\ln \mathbb{P}\big\{\tilde{\eta}_v = s \bigm| \Omega_R \le M\big\} &= \ln \sum_{m=s}^{M}\sum_{k=0}^{M-m} \exp\big(\mathbf{J}_{v,k}(s) + \mathbf{K}_{v,m}(s) - \ln L(0)\big),
\end{aligned}$$

Proof of Theorem 4/Theorem 14. We have $\tilde{\mathbf{B}}'_{u,k}(n,s) = \ln \mathbb{P}\{\tilde{\xi}_u = n, \tilde{\eta}_u = s, \Omega^*_u = k\}$ and $\tilde{\mathbf{J}}'_{v,k}(n,s) = \ln \mathbb{P}\{\tilde{\xi}_u = n, \tilde{\eta}_v = s, \Omega^*_v = k\}$. By summing over possible values of Ω^*_u and Ω_u, we have

$$\mathbb{P}\{\tilde{\eta}_u = s, \tilde{\xi}_u = n; \Omega_R \le M\} = \sum_{m=n}^{M} \sum_{k=0}^{M-m} \underbrace{\mathbb{P}\{\tilde{\xi}_u = n, \tilde{\eta}_u = s, \Omega^*_u = k\}}_{=\exp\left(\tilde{\mathbf{B}}'_{u,k}(n,s)\right)} \times \underbrace{\mathbb{P}\{\Omega_u = m \mid \tilde{\xi}_u = n\}}_{=\exp\left(\tilde{\mathbf{C}}_{u,m}(n)\right)}.$$

$$\mathbb{P}\{\tilde{\xi}_u = n, \tilde{\eta}_v = s; \Omega_R \le M\} = \sum_{m=s}^{M} \sum_{k=0}^{M-m} \underbrace{\mathbb{P}\{\tilde{\xi}_u = n, \tilde{\eta}_v = s, \Omega^*_v = k\}}_{=\exp\left(\tilde{\mathbf{J}}'_{v,k}(n,s)\right)} \times \underbrace{\mathbb{P}\{\Omega_v = m \mid \tilde{\eta}_v = s\}}_{=\exp\left(\tilde{\mathbf{K}}_{v,m}(s)\right)}$$

$$\mathbb{P}\{\tilde{\eta}_v = s; \Omega_R \le M\} = \sum_{m=s}^{M} \sum_{k=0}^{M-m} \underbrace{\mathbb{P}\{\tilde{\eta}_v = s, \Omega^*_v = k\}}_{=\exp\left(\tilde{\mathbf{J}}_{v,k}(s)\right)} \times \underbrace{\mathbb{P}\{\Omega_v = m \mid \tilde{\eta}_v = s\}}_{=\exp\left(\tilde{\mathbf{K}}_{v,m}(s)\right)}$$

□

Theorem 15 (Gradient for unobserved profiles (Theorem 5))**.** *Let $\Phi = \ln L(0) = \ln \mathbb{P}\{\Omega_R \le M\}$. At every node $1 \le u \le R$, if $0 < q_u$, then*

$$\begin{aligned}
\Phi^{(\operatorname{logit} \tilde{q}_u)} &= (1-\tilde{q}_u)\exp(\delta^+_u) - \tilde{q}_u \exp\bigl(\mathsf{ladd}(\sigma_u, \ln(\kappa_u))\bigr) \\
\Phi^{(\ln \kappa_u)} &= \sum_{0\le i} \frac{\kappa_u}{\kappa_u + i} \exp(\delta^+_{u,i}) - \kappa_u\bigl(-\ln(1-\tilde{q}_u)\bigr),
\end{aligned}$$

and, if $q_u = 0$, then

$$\Phi^{(\ln \tilde{r}_u)} = \exp(\delta^+_u) - \tilde{r}_u.$$

On every edge $uv \in T$,

$$\Phi^{(\operatorname{logit} p_v)} = (1-p^-_v)\exp\bigl(\delta^-_v\bigr) - p^-_v \exp\bigl(\sigma_v\bigr).$$

Proof of Theorem 5/Theorem 15. With the set of unobserved profiles $\mathcal{S}_0 = \{\Xi \mid \Omega(\Xi) \le M\}$,

$$\ln L(0) = \ln \sum_{\Xi \in \mathcal{S}_0} L(\Xi).$$

For any parameter θ,

$$\frac{\partial \ln L(0)}{\partial \theta} = \sum_{\Xi \in \mathcal{S}_0} \frac{L(\Xi)}{L(0)} \times \frac{\partial}{\partial \theta} \ln L(\Xi).$$

Similarly, for any random variable X,

$$\mathbb{E}\bigl[X \mid \Omega(\Xi) \le M\bigr] = \sum_{\Xi \in \mathcal{S}_0} \frac{L(\Xi)}{L(0)} \times \mathbb{E}\bigl[X \mid \Xi\bigr]$$

From the proof of Theorem 11,

$$\begin{aligned}
\frac{\partial \ln L(0)}{\partial \operatorname{logit} \tilde{q}_u} &= \sum_{\Xi \in \mathcal{S}_0} \frac{L(\Xi)}{L(0)} \times \frac{\partial \ln L(\Xi)}{\tilde{q}_u} \\
&= \sum_{\Xi \in \mathcal{S}_0} \frac{L(\Xi)}{L(0)} \Bigl((1-\tilde{q}_u) \times \mathbb{E}\bigl[\tilde{\xi}_u - \tilde{\eta}_u \mid \Xi\bigr] - \tilde{q}_u \times \mathbb{E}\bigl[\tilde{\eta}_u \mid \Xi\bigr]\Bigr). \\
&= (1-\tilde{q}_u) \sum_{\Xi \in \mathcal{S}_0} \frac{L(\Xi)}{L(0)} \mathbb{E}\bigl[\tilde{\xi}_u - \tilde{\eta}_u \mid \Xi\bigr] - \tilde{q}_u \sum_{\Xi \in \mathcal{S}_0} \frac{L(\Xi)}{L(0)} \mathbb{E}\bigl[\tilde{\eta}_u \mid \Xi\bigr] \\
&= (1-\tilde{q}_u) \times \mathbb{E}\bigl[\tilde{\xi}_u - \tilde{\eta}_u \mid \Omega(\Xi) \le M\bigr] - \tilde{q}_u \times \mathbb{E}\bigl[\tilde{\eta}_u \mid \Omega(\Xi) \le M\bigr].
\end{aligned}$$

The formulas for $\frac{\partial \ln L(0)}{\partial \operatorname{logit} \tilde{p}_u}$, $\frac{\partial \ln L(0)}{\partial \ln \kappa_u}$ or $\frac{\partial \ln L(0)}{\partial \ln \tilde{r}_u}$ are inferred analogously. □

A.4. Numerical procedures. The numerical computations are carried out using log-likelihoods. The necessary floating-point functions include the usual $\mathsf{log}(x) = \ln x$ and $\mathsf{exp}(x) = e^x$, and, for maximum floating-point precision, $\mathsf{log1p}(x) = \ln(1+x)$ and $\mathsf{expm1}(x) = e^x - 1$. For optimization purposes, the model is parametrized using logit and logarithmic scales. In particular, our fundamental model parameters for numerical optimization are

$$\operatorname{logit} p_u = \ln \frac{p_u}{1-p_u} \quad \text{and} \quad \operatorname{logit} \lambda_u = \ln \frac{\lambda_u}{1-\lambda_u}$$

and the logarithmic gain $\ln \gamma_u$.

Multiplication on the logarithmic scale is cheap since it corresponds to addition. Addition, however, needs two floating-point function calls: define $\mathsf{ladd}(x, y) = \ln(e^x + e^y)$

$$\mathsf{ladd}(x,y) = \begin{cases} y + \ln(1+e^{-(y-x)}) = y + \mathsf{log1p}(\mathsf{exp}(x-y)) & \text{if } x \le y \\ x + \ln(1+e^{-(x-y)}) = x + \mathsf{log1p}(\mathsf{exp}(y-x)) & \text{if } y < x \end{cases}$$

The logit scale represents values between 0 and 1 (corresponding to $-\infty$ and ∞ inclusively) and offers equal floating-point precision for a parameter and its complement: if $x = \operatorname{logit} \theta$, then

$$\mathsf{linv}(x) = \big(\mathsf{linv}_0(x), \mathsf{linv}_1(x)\big) = \big(\ln\theta, \ln(1-\theta)\big) = \begin{cases} \big(-\mathsf{log1p}(\exp(-x)), -x - \mathsf{log1p}(\exp(-x))\big) & \text{if } 0 \le x \\ \big(x - \mathsf{log1p}(\exp(x)), -\mathsf{log1p}(\exp(x))\big) & \text{if } x < 0 \end{cases}$$

In order to compute $\ln(-\ln(1-q))$ from $x = \operatorname{logit} q$, we take advantage of $-\ln(1-q) \sim q$ as $q \to 0$:

$$\mathsf{loglogc}(x) = \ln(-\ln(1-q)) = \begin{cases} \mathsf{log}(-\mathsf{linv}_1(x)) & \text{if } \mathsf{log}(-\mathsf{linv}_1(x)) \ne -\infty \\ \mathsf{linv}_0(x) & \text{otherwise (very small } q) \end{cases}$$

We calculate the product of two values represented on logit scale by $\frac{e^x}{1+e^x}$ and $\frac{e^y}{1+e^y}$ using three floating-point function calls as

$$\begin{aligned} \mathsf{limul}(x,y) &= \operatorname{logit} \frac{e^{x+y}}{(1+e^x)(1+e^y)} = \ln \frac{1}{e^{-x} + e^{-y} + e^{-x-y}} \\ &= \begin{cases} \infty & \text{if } x = y = \infty \\ (x+y) - \mathsf{log1p}\big(\exp(x) + \exp(y)\big) & \text{if } x < 0 \text{ and } y < 0 \\ x - \mathsf{log1p}\big(\exp(x-y) + \exp(-y)\big) & \text{if } x \le y, x < \infty \text{ and } 0 \le y \\ y - \mathsf{log1p}\big(\exp(y-x) + \exp(-x)\big) & \text{if } y < x \text{ and } 0 \le x \end{cases} \end{aligned}$$

The complement can be obtained on the logarithmic scale for a parameter $0 \le \theta = e^x \le 1$ close to the boundaries via

$$\mathsf{lcomp}(x) = \ln(1-e^x) = \begin{cases} \mathsf{log}(-\mathsf{expm1}(x)) & \text{if } \ln(1/2) < x \\ \mathsf{log1p}\big(-\mathsf{exp}(x)\big) & \text{if } x \le \ln(1/2) \end{cases}$$

which leads to the conversion from logarithmic to logit scale $\operatorname{logit}\theta = \ln\theta - \mathsf{lcomp}(\ln\theta)$.

Our theorems are suitable for numerical computations, provided that the probability $L(0)$ of unobserved profiles is bounded away from 1. Theorems 6/12 and 7/13 keep the gradient well-scaled because all entries in the Jacobian matrices are bounded by 1 in absolute value. In our implementation we impose bounded gain rates $\gamma \le 1$ and use the logit-scaled rate as the optimizable parameter.

The numerical algorithms work with log-scaled probabilities and expectations, using the following corollary of Theorem 11.

Corollary 16 (Corrected gradient with log-scaled expectations)**.** *Define the logarithms of $\tilde{S}_u, \tilde{D}_u^+, \tilde{D}_u^-$ for all nodes u:*

$$\tilde{\Gamma}_u = \ln \tilde{S}_u.$$

$$\tilde{\Delta}_{u,i}^- = \ln \tilde{D}_{u,i}^- \qquad \tilde{\Delta}_u^- = \ln \tilde{D}_u^- = \ln\left(\sum_{0\le i} \exp(\tilde{\Delta}_{v,i}^-)\right)$$

$$\tilde{\Delta}_{u,i}^+ = \ln \tilde{D}_{u,i}^+ \qquad \tilde{\Delta}_u^+ = \ln \tilde{D}_u^+ = \ln\left(\sum_{0\le i} \exp(\tilde{\Delta}_{u,i}^+)\right)$$

At every node $1 \le u \le R$, if $0 < \tilde{q}_u$, then

$$\Phi^{(\operatorname{logit}\tilde{q}_u)} = \frac{\partial}{\partial \operatorname{logit} \tilde{q}_u} \ln L^* = (1-\tilde{q}_u)\exp\big(\tilde{\Delta}_u^+\big) - \tilde{q}_u \exp\big(\mathsf{ladd}(\tilde{\Gamma}_u, \ln F^* + \ln \kappa_u)\big)$$

$$\Phi^{(\ln \kappa_u)} = \frac{\partial}{\partial \ln \kappa_u} \ln L^* = \left(\sum_{i=0}^{M_u} \frac{\kappa_u}{\kappa_u + i} \exp\big(\tilde{\Delta}_{u,i}^+\big)\right) - F^* \kappa_u \big(-\ln(1-\tilde{q}_u)\big),$$

and, if $q_u = 0$, *then*

$$\Phi^{(\ln \tilde{r}_u)} = \frac{\partial}{\partial \ln \tilde{r}_u} \ln L^* = \exp\bigl(\tilde{\Delta}_u^+\bigr) - F^* \tilde{r}_u.$$

At the root, $\Phi^{(\operatorname{logit} \tilde{p}_v)} = 0$. *On every edge* $uv \in T$ *and sibling* $uw \in T$,

$$\Phi^{(\operatorname{logit} \tilde{p}_v)} = \frac{\partial}{\partial \operatorname{logit} \tilde{p}_v} \ln L^* = (1 - p_v^-) \exp\bigl(\tilde{\Delta}_v^-\bigr) - p_v^- \exp\bigl(\tilde{\Gamma}_v\bigr)$$

A.5. Algorithms. Figure S2 shows the numerical computation of the survival parameters. Figure S3 is the main exact gradient calculation algorithm. Figures S4–S9 show the helper procedures. Figures S10–S13 show the algorithm that computes the posterior statistics for unobserved profiles. We prove here our results on computational efficiency.

A subtle but important calculation for quadratic running time is for the tail differences $\delta_{u,i}^+, \delta_{u,i}^-$ in Theorem 5/15 and Eq. (6)–Eq. (7).

Theorem 17. *Let* $t_j = \sum_{s=0}^{j} \sum_{n=j+1}^{M} p_{n,s}$ *for some distribution* $p_{n,s}$ *over* $0 \le sn \le M$. *Define* $t_{i,j} = \sum_{s=0}^{i} \sum_{n=j+1}^{M} p_{n,s}$ *for* $i \le j$ *so that* $t_j = t_{j,j}$. *Let* $z_{s,j} = \sum_{n=j+1}^{M} p_{n,s}$. *Then*

$$\begin{aligned}
z_{s,M} &= 0 \\
z_{s,j} &= z_{s,j+1} + p_{j+1,s} && \{s \le j < M\} \\
t_{0,j} &= z_{0,j} \\
t_{i,j} &= t_{i-1,j} + z_{i,j} && \{0 < i \le j\}
\end{aligned}$$

Consequently, $\{t_0, t_1, \ldots, t_M\}$ *can be computed with* $M(M+1)$ *additions.*

Proof. We have, by separating the first term in the summation,

$$z_{s,j} = \sum_{n=j+1}^{M} p_{n,s} = p_{j+1,s} + \sum_{n=j+2}^{M} p_{n,s} = p_{j+1,s} + z_{s,j+1}.$$

By separating the last term,

$$t_{i,j} = \sum_{s=0}^{i} \sum_{n=j+1}^{M} p_{n,s} = \sum_{s=0}^{i-1} \sum_{n=j+1}^{M} p_{n,s} + \sum_{n=j+1}^{M} p_{n,i} = t_{i-1,j} + z_{i,j}.$$

The recurrences are evaluated $M(M+1)/2$ times for $z_{s,j}$ with $0 \le s \le j < M$, and $M(M+1)/2$ times for $t_{i,j}$ with $0 < i \le j \le M$. □

Proof of Theorem 9 (Running time). We characterize the running time by counting the number of the typical ladd operations. We assume that $\mathsf{ladd}(-\infty, x) = x$ and $\mathsf{ladd}(x, -\infty) = x$ are computed for free. Let Ξ be a profile and define the size at every node u:

$$n_u = \Omega_u = \sum_{w \in \mathcal{L}_u} n_w$$

with the leaf set $\mathcal{L}_u$ in the subtree of u.

First, consider the number of operations to calculate $\tilde{\mathbf{L}}$ or $\tilde{\mathbf{C}}$ by Algorithm Linner from Figure S7. We have, for a non-leaf node u with children $uv, uw \in T$ and $n_u = n_v + n_w$,

$$a(u) = \sum_{n=1}^{n_v + n_w} \sum_{t=\max\{0, n-n_v\}}^{\min\{n_w, n\}-1} 1 = n_v n_w.$$

Second, in order to calculate $\tilde{\mathbf{K}}$ at a node u, the algorithm makes

$$b(u) = \sum_{s=0}^{n_u} \sum_{n=s+1}^{n_u} 1 = n_u(n_u + 1)/2$$

ladd operations. Algorithm Louter uses $a(u)$ operations to compute $\tilde{\mathbf{J}}$ and $b(u)$ operations to compute $\tilde{\mathbf{B}}$ (supposing that $\mathbf{L}$ is computed only once).

Let $t(n) = n(n+1)/2$ denote the triangular numbers, so that $b(u) = t(n_u)$. For any x, y, we have

$$t(x + y) = xy + t(x) = t(y) \qquad \textbf{[12]}$$

so $a(u) = b(u) - b(v) - b(w)$ at every inner node u with children $uv, uw \in T$. Summing across the inner nodes of the tree, we have the telescoping sum

$$A = \sum_{u=L+1}^{R} a(u) = b(R) - \sum_{w=1}^{L} b(w) = \frac{1}{2}\left(\sum_{w=1}^{L} n_w\right)^2 - \frac{1}{2}\sum_{w=1}^{L}(n_w)^2 \leq \frac{(n_R)^2}{2}.$$

Let $\delta(u)$ denote the depth of each node u. Summing at any level d,

$$N_d = \sum_{u:\delta(u)=d} n_u = \sum_{u:\delta(u)=d} \sum_{w\in\mathcal{L}_u} n_w \leq n_R$$

because the leaf sets $\mathcal{L}_u$ are distinct across nodes at the same level, and $n_R = \sum_{w=1}^{L} n_w$ at the root R. Now,

$$\sum_{u:\delta(u)=d} b(u) = \sum_{u:\delta(u)=d} t(n_u) \leq t(N_d) \leq t(n_R) = b(R).$$

Hence, for a tree of height h,

$$B = \sum_{u=1}^{R-1} b(u) = \sum_{d=1}^{h} \sum_{u:\delta(u)=d} b(u) \leq hb(R) = ht(n_R)$$

The total number of ladd operations for computing one profile's likelihoods is thus bounded as

$$T_{\mathrm{L}}(\Xi) = \underbrace{(1+2+2)A}_{\text{for } \tilde{\mathbf{C}}, \tilde{\mathbf{L}}, \tilde{\mathbf{J}}} + \underbrace{(1+1)B}_{\text{for } \tilde{\mathbf{K}}, \tilde{\mathbf{B}}} + \underbrace{4n_R}_{\tilde{\mathbf{K}} \text{ at root}} \leq 2ht(n_R) + 2(n_R)^2 + 4n_R.$$

The posterior calculations for a profile include nodePosteriors (Figure S9) and the updates of sample-wide expectations in samplePosteriors (Figure S5). By Theorem 17, nodePosteriors uses $2(n_v)(n_v+1) = 4b(v)$ operations to calculate birth and death posteriors at node v. The updates in samplePosteriors at node v use 1 ladd operation for $\tilde{\Gamma}_v$ and

$$c(v) = \sum_{i=0}^{n_v-1} 1 = n_v$$

operations for updating $\tilde{\Delta}_v^-$ or $\tilde{\Delta}_v^+$. Summing across the tree,

$$C = \sum_{u=1}^{R} c(u) = \sum_{u=1}^{R} n_u = \sum_{w=1}^{L} n_w \delta(w) \leq h \sum_{w=1}^{L} n_w = hn_R.$$

with the depth $\delta(w)$ of each leaf w. The total number of ladd operations for computing one profile's posterior statistics is thus bounded as

$$T_{\mathrm{P}}(\Xi) = \underbrace{4B}_{\text{nodePosteriors}} + \underbrace{2C}_{\text{samplePosteriors}} + \underbrace{(2L-1)}_{\text{for } \tilde{\Gamma}} \leq 4ht(n_R) + 2hn_R + (2L-1).$$

Across the entire sample, we have a bound on the total number of ladd operations as

$$\begin{aligned} T(\Xi^1, \ldots, \Xi^F) &= \sum_{f=1}^{F}\big(T_{\mathrm{L}}(\Xi^f) + T_{\mathrm{P}}(\Xi^f)\big) \leq \sum_{f=1}^{F}\left(6h\frac{\Omega(\Xi^f)((\Omega(\Xi^f)+1))}{2} + 2\Omega(\Xi^f)^2\right) + (2h+4)\sum_{f=1}^{F}\Omega(\Xi^f) + (2L-1)F \\ &= (3h+2)L^2\sum_{f=1}^{F}\left(\frac{\Omega(\Xi^f)}{L}\right)^2 + (5h+4)L\sum_{f=1}^{F}\frac{\Omega(\Xi^f)}{L} + (2L-1)F \sim 3FhL^2\bar{\omega}^2 \end{aligned}$$

with $\bar{\omega}$ the sample-wide quadratic average of the mean copy number. The total runtime for computing the uncorrected likelihood and posteriors across the sample is thus $\Theta(FhL^2\bar{\omega}^2)$. The running time for further computations is asymptotically negligible with respect to this term. By Theorem 8, the calculations for unobserved profiles take $\Theta(LM^4) = o(FL^2)$ time when $\Omega_{\min} = o(FL^{1/4})$. The corrections to the posteriors and likelihood in samplePosteriors are done only once, in $\Theta(L\Omega_{\max})$ time with the largest profile $\Omega_{\max} = \max\big\{\Omega(\Xi^f) \bigm| 1 \leq f \leq F\big\}$. The gradient calculations are done only once, also in $\Theta(L\Omega_{\max})$ time.

□

Proof of Theorem 8 (Running time for unobserved profiles). We characterize the running time by counting the number of the typical ladd operations, denoted by $a(M)$, in function of the maximum unobserved profile size M. Since the tree is binary, we have $R = 2L - 1$ nodes over L leaves. We assume that $\mathsf{ladd}(-\infty, x) = x$ is computed for free. For the number of operations by Algorithm Uinner on Figure S11 we have

$$
\begin{aligned}
a(M) &= \sum_{m=0}^{M} \left(m + \sum_{u=1}^{R-1} \sum_{n=0}^{m} \sum_{s=0}^{n-1} 1 + \sum_{u=1}^{R-1} \sum_{n=0}^{m} \sum_{s=0}^{n-1} 1 \right) \\
&\quad + \sum_{m=0}^{M} \sum_{u=L+1}^{R} \sum_{n=0}^{m} \left(n + \sum_{s=0}^{n} \sum_{k=n-s+1}^{m-s} 1 \right) \\
&= \sum_{m=0}^{M} \left(m + 2 \sum_{u=1}^{R-1} \sum_{n=0}^{m} n \right) \\
&\quad + \sum_{m=0}^{M} \sum_{u=L+1}^{R} \sum_{n=0}^{m} \bigl(n + (m-n)(n+1) \bigr) \\
&= \frac{M(M+1)}{2} + 2(R-1)\frac{M(M+1)(M+2)}{6} + (R-L)\left(\frac{M(M+1)(M+1)(M+2)(M+3)}{24} + \frac{M(M+1)(M+2)}{6} \right) \\
&\sim (L-1)\binom{M+3}{4} \sim L\frac{M^4}{24}
\end{aligned}
$$

The same number of operations is needed for Algorithm Uouter on Figure S12. Algorithm unobservedNodePosteriors on Figure S13 needs a cubic number of ladd operations in M as it has three nested iterations with the variables s, n, m. Consequently, the posteriors are computed using $\sim L\frac{M^4}{12}$ log-scale additions. The actual running time is proportional to $a(M)$; thus, it is $\Theta(LM^4)$. $\square$

Procedure computeSurvivalParameters()
// *computes the survival parameters* $\mathsf{logite}[v] = \operatorname{logit} \epsilon_v$
// $\mathsf{logitp}[v] = \operatorname{logit} \tilde{p}_v, \mathsf{logitq}[v] = \operatorname{logit} \tilde{q}_v, \mathsf{logr}[v] = \ln \tilde{r}_v$
for $u \leftarrow 1, 2, \ldots, R$ **do** // *nodes in postorder*
  if $(u \le L)$ **then** // *leaf*
    $\mathsf{logite}[u] \leftarrow -\infty$ // $\epsilon_u = 0$
  else
    $\mathsf{logite}[u] \leftarrow +\infty$
    for every child $uv \in T$ **do** $\mathsf{logite}[u] \leftarrow \mathsf{limul}(\mathsf{logite}[u], \mathsf{logitp}[v])$
  $\mathsf{log1e} \leftarrow \mathsf{linv}_1(\mathsf{logite}[u])$
  $\mathsf{logitq}[u] \leftarrow \operatorname{logit}(q_u) + \mathsf{log1e}$ // $\operatorname{logit}(\tilde{q}_u) = \operatorname{logit}(q_u) \log(1 - \epsilon_u)$
  if $(\mathsf{logitq}[u] = -\infty)$ **then**
    $\mathsf{logr}[u] \leftarrow \log(r_u) + \mathsf{log1e}$
  else
    $\mathsf{logk} \leftarrow \log(\kappa_u); \mathsf{logq} \leftarrow \mathsf{inv}_0(\mathsf{logitq}[u]); \mathsf{logr}[u] \leftarrow \mathsf{logk} + \mathsf{logq}$
  // *set* $1 - \tilde{p}_u = (1 - p_u)(1 - \epsilon_u(1 - \tilde{q}_u))$
  $z \leftarrow \mathsf{limul}(\mathsf{logite}[u], -\mathsf{logitq}[u])$
  $\mathsf{logitp}[u] \leftarrow -\mathsf{limul}(-\operatorname{logit}(p_v), -z)$

Fig. S2. Algorithm computeSurvivalParameters computes the survival parameters from Definition 1.

Function $\mathsf{gradient}\big(\mathcal{S} = \{\Xi^f\}_{f=1}^{F}, \Omega_{\min}\big)$
$\Big(\mathsf{LL}, F^*, \Big\{(\tilde{\Gamma}_u, \tilde{\Delta}_u^-[0..M_u], \tilde{\Delta}_u^+[0..M_u])\Big\}_{u=1}^{R}\Big) \leftarrow \mathsf{samplePosteriors}\big(\mathcal{S}, \Omega_{\min}\big)$
// *intermediate logarithmic terms:* $\frac{\partial \Phi}{\partial \operatorname{logit} p} = \exp(\mathsf{ldp0}) - \exp(\mathsf{ldp1})$
// *similarly for* $\frac{\partial \Phi}{\partial \operatorname{logit} q} = \exp(\mathsf{ldq0}) - \exp(\mathsf{ldq1})$*, and gain terms* $\mathsf{ldg0}, \mathsf{ldg1}$
$v \leftarrow R$; **while** $(0 < v)$ **do** // *in preorder*
  $i \leftarrow M_v; \Delta^- \leftarrow \tilde{\Delta}_{v,i}^-; \Delta^+ \leftarrow \tilde{\Delta}_{v,i}^+$
  while $(0 < i)$ **do**
    $i \leftarrow i - 1;\ \Delta^- \leftarrow \mathsf{ladd}(\Delta^-, \tilde{\Delta}_{v,i}^-)\ ;\ \Delta^+ \leftarrow \mathsf{ladd}(\Delta^+, \tilde{\Delta}_{v,i}^+)$
  if $(\operatorname{logit} \tilde{q}_v = -\infty)$ **then**
    $(\mathsf{ldq0}[v], \mathsf{ldq1}[v]) \leftarrow (-\infty, -\infty)$
    $\mathsf{ldg0}[v] \leftarrow \Delta^+; \mathsf{ldg1}[v] \leftarrow \ln(F^*) + \ln(\tilde{r}_v)$
  else
    $\mathsf{ldq0}[v] \leftarrow \ln(1 - \tilde{q}_v) + \Delta^+; \mathsf{ldq1}[v] \leftarrow \ln(\tilde{q}_v) + \mathsf{ladd}\big(\tilde{\Gamma}_v, \ln(\kappa_v) + \ln(F^*)\big)$
    $i \leftarrow 0; \mathsf{ldg0}[v] \leftarrow \tilde{\Delta}_{v,0}^+$
    while $(i < M_v)$ **do**
      $i \leftarrow i + 1; x \leftarrow \ln(\kappa_v) - \ln(i)$
      if $(x < 0)$ **then** $y \leftarrow x - \mathsf{log1p}\big(\exp(x)\big)$
      else $y \leftarrow -\mathsf{log1p}\big(\exp(-x)\big)$
      $\mathsf{ldg0}[v] \leftarrow \mathsf{ladd}\big(\mathsf{ldg0}[v], y + \tilde{\Delta}_{v,i}^+\big)$
    $\mathsf{ldg1}[v] \leftarrow \ln(\kappa_v) + \ln(F^*) + \mathsf{loglogc}(\operatorname{logit} \tilde{q}_v)$
  if $(v = R)$ **then** $(\mathsf{ldp0}[v], \mathsf{ldp1}[v]) \leftarrow (-\infty, -\infty)$
  else
    determine parent $uv \in$ and sibling $w \neq v$ with $uw \in T$
    $z \leftarrow \operatorname{logit} \tilde{p}_v + \ln(1 - \tilde{p}_w); (z_0, z_1) \leftarrow \mathsf{linv}(z)$
    $\mathsf{ldp0}[v] \leftarrow z_1 + \Delta^-; \mathsf{ldp1}[v] \leftarrow z_0 + \tilde{\Gamma}_v$
  $v \leftarrow v - 1$
// *we have the gradient by survival parameters; apply Theorem 13*
$\mathsf{transformGradient}\big(\mathsf{ldp0}, \mathsf{ldp1}, \mathsf{ldq0}, \mathsf{ldq1}, \mathsf{ldg0}, \mathsf{ldg1}\big)$
for $v \leftarrow 1, 2, \ldots, R$ **do** // *convert from logarithmic terms*
  $\mathsf{dp}[v] \leftarrow \mathsf{ldiff}\big(\mathsf{ldp0}[v], \mathsf{ldp1}[v]\big)$
  $\mathsf{dq}[v] \leftarrow \mathsf{ldiff}\big(\mathsf{ldq0}[v], \mathsf{ldq1}[v]\big)$
  $\mathsf{dg}[v] \leftarrow \mathsf{ldiff}\big(\mathsf{ldg0}[v], \mathsf{ldg1}[v]\big)$
return $\big(\mathsf{LL}, \mathsf{dp}[1..R], \mathsf{dg}[1..R], \mathsf{dg}[1..R]\big)$

Function $\mathsf{ldiff}(a, b)$ // *calculates* $e^a - e^b$
$c \leftarrow a - b$; **if** $(c < 0)$ **then** $y \leftarrow -\exp\Big(b + \mathsf{log1p}\big(-\exp(c)\big)\Big)$
else $y \leftarrow \exp\Big(a + \mathsf{log1p}\big(-\exp(-c)\big)\Big)$
return y

Fig. S3. Algorithm gradient computes the survival gradient by Theorem 11 and transforms it to model parameter gradient.

Procedure $\mathsf{transformGradient}\big(\mathsf{ldp0}, \mathsf{ldp1}, \mathsf{ldq0}, \mathsf{ldq1}, \mathsf{ldg0}, \mathsf{ldg1}\big)$

$v \leftarrow R$; **while** $(0 < v)$ **do** *// in preorder*

// compute gradient by model parameters from Theorem 12

if $(v = R)$ **then**

if $(\operatorname{logit} \tilde{q}_v = -\infty)$ **then**

$\mathsf{lde0}[v] \leftarrow \mathsf{ldg1} + \ln(\epsilon_v)$; $\mathsf{lde1}[v] \leftarrow \mathsf{ldg0} + \ln(\epsilon_v)$

else

$\mathsf{lde0}[v] \leftarrow \mathsf{ldq1} + \ln(\epsilon_v)$; $\mathsf{lde1}[v] \leftarrow \mathsf{ldq0} + \ln(\epsilon_v)$

else

determine parent $uv \in$ and sibling $w \neq v$ with $uw \in T$

$z \leftarrow \operatorname{logit} \tilde{p}_v + \ln(1 - \tilde{p}_w)$; $(z_0, z_1) \leftarrow \mathsf{linv}(z)$

$\mathsf{ldpe0} \leftarrow \mathsf{ladd}\big(\mathsf{ldp0}, z_1 + \mathsf{lde0}[u]\big)$; $\mathsf{ldpe1} \leftarrow \mathsf{ladd}\big(\mathsf{ldp1}, z_1 + \mathsf{lde1}[u]\big)$

$x \leftarrow \ln \tilde{p}_v - \ln p_v$; $\mathsf{ldp0} \leftarrow \mathsf{ldpe0} + x$; $\mathsf{ldp1} \leftarrow \mathsf{ldpe1} + x$

if $(\operatorname{logit} \tilde{q}_v = -\infty)$ **then**

$y \leftarrow \ln \epsilon_v - \ln \tilde{p}_v$

$\mathsf{lde0}[v] \leftarrow \mathsf{ladd}\big(\mathsf{ldpe0} + y, \mathsf{ldg1} + \ln(\epsilon_v)\big)$

$\mathsf{lde1}[v] \leftarrow \mathsf{ladd}\big(\mathsf{ldpe1} + y, \mathsf{ldg0} + \ln(\epsilon_v)\big)$

else

$y \leftarrow \ln(1 - \tilde{q}_v) + \ln \epsilon_v - \ln \tilde{p}_v$

$\mathsf{lde0}[v] \leftarrow \mathsf{ladd}\big(\mathsf{ldpe0} + y, \mathsf{ldq1} + \ln(\epsilon_v)\big)$

$\mathsf{lde1}[v] \leftarrow \mathsf{ladd}\big(\mathsf{ldpe1} + y, \mathsf{ldq0} + \ln(\epsilon_v)\big)$

$\mathsf{ldq0} \leftarrow \mathsf{ladd}\big(\mathsf{ldq0}, \mathsf{ldpe1} + \ln(q_v)\big)$; $\mathsf{ldq1} \leftarrow \mathsf{ladd}\big(\mathsf{ldq1}, \mathsf{ldpe0} + \ln(q_v)\big)$

if $(\operatorname{logit} \tilde{q}_v = -\infty)$ **then**

// employ Theorem 13

$\mathsf{ldp0} \leftarrow \mathsf{ladd}\big(\mathsf{ldp0}, \mathsf{ldg0} + \ln(1 - p_v)\big)$; $\mathsf{ldp1} \leftarrow \mathsf{ladd}\big(\mathsf{ldp1}, \mathsf{ldg1} + \ln(1 - p_v)\big)$

else

// employ Theorem 13

$\mathsf{ldp0} \leftarrow \mathsf{ladd}\big(\mathsf{ldp0}, \mathsf{ldq0} + \ln(1 - p_v)\big)$; $\mathsf{ldp1} \leftarrow \mathsf{ladd}\big(\mathsf{ldp1}, \mathsf{ldq1} + \ln(1 - p_v)\big)$

$\mathsf{ldq0} \leftarrow \mathsf{ladd}\big(\mathsf{ldq0}, \mathsf{ldg1} + \ln(1 - q_v)\big)$; $\mathsf{ldq1} \leftarrow \mathsf{ladd}\big(\mathsf{ldq1}, \mathsf{ldg0} + \ln(1 - q_v)\big)$

$\mathsf{ldg0} \leftarrow \mathsf{ldg0} + \ln(1 - \gamma_v)$; $\mathsf{ldg1} \leftarrow \mathsf{ldg1} + \ln(1 - \gamma_v)$ *// by* $\operatorname{logit} \gamma_v$

// employ Eq. (11)

$y \leftarrow \ln(1 - p_v) - \ln(1 - q_v)$; $z \leftarrow \ln(1 - \lambda_v) - \ln(1 - q_v)$

$\mathsf{ldp0} \leftarrow \mathsf{ladd}\big(\mathsf{ldp0}, \mathsf{ldg0} + y\big)$; $\mathsf{ldp1} \leftarrow \mathsf{ladd}\big(\mathsf{ldp1}, \mathsf{ldg1} + y\big)$

$\mathsf{ldq0} \leftarrow z + \mathsf{ldq0}$; $\mathsf{ldq1} \leftarrow z + \mathsf{ldq1}$

$v \leftarrow v - 1$

Fig. S4. Algorithm transformGradient transforms the survival gradient to model parameter gradient by p, λ, γ using Theorems 12 and 13

Function samplePosteriors$\big(\mathcal{S} = \{\Xi^f\}_{f=1}^F, \Omega_{\min}\big)$

for $(u \leftarrow 1, 2, \ldots, R)$ **do**

  $\tilde{\Gamma}_u \leftarrow -\infty; M_u \leftarrow 0$

for $(f \leftarrow 1, 2, \ldots, F)$ **do** — *// determine max copy sum M_u*

  $u \leftarrow 1$; **while** $(u \leq L)$ **do** $m_u \leftarrow \xi_{f,u}; M_u \leftarrow \max\big\{M_u, m_u\big\}; u \leftarrow u+1$

  while $(u \leq R)$ **do**

    $m_u \leftarrow 0$; **for** all children $uv \in T$ **do** $m_u \leftarrow m_u + m_v$

    $M_u \leftarrow \max\big\{M_u, m_u\big\}; u \leftarrow u+1$

for $(u \leftarrow 1, 2, \ldots, R)$ **do**

  initialize $\tilde{\Delta}_u^-[0..M_u]$ and $\tilde{\Delta}_u^+[0..M_u]$ arrays with $-\infty$

$\mathsf{LL} \leftarrow 0$ — *// sample log-likelihood*

for $(f \leftarrow 1, 2, \ldots, F)$ **do**

  $\Big(\ell, \big\{(s_u, \delta_u^-[0..n_u], \delta_u^+[0..n_u])\big\}_{u=1}^R\Big) \leftarrow \mathsf{profilePosteriors}(\Xi^f)$

  $\mathsf{LL} \leftarrow \mathsf{LL} + \ell$

  for $(u \leftarrow 1, 2, \ldots, R)$ **do**

    $\tilde{\Gamma}_u \leftarrow \mathsf{ladd}\big(\tilde{\Gamma}_u, s_u\big)$

    for $(i \leftarrow 0, 1, \ldots, n_u)$ **do**

      $\tilde{\Delta}_u^-[i] \leftarrow \mathsf{ladd}\big(\tilde{\Delta}_u^-[i], \delta_u^-[i]\big)$

      $\tilde{\Delta}_u^+[i] \leftarrow \mathsf{ladd}\big(\tilde{\Delta}_u^+[i], \delta_u^+[i]\big)$

$\Big(\mathsf{LL0}, \big\{(\sigma_u, \delta_u^-[0..m_u], \delta_u^+[0..m_u])\big\}_{u=1}^R\Big) \leftarrow \mathsf{smallProfilePosteriors}(\Omega_{\min} - 1)$

$\mathsf{cLL0} \leftarrow \mathsf{lcomp}(\mathsf{LL0})$ — *// so $\mathsf{cLL0} = \ln\big(1 - \exp(\mathsf{LL0})\big)$*

$F^* \leftarrow F \times \exp(-\mathsf{cLL0})$ — *// corrected sample size*

$\mathsf{LL} \leftarrow \mathsf{LL} - F \times \mathsf{cLL0}$ — *// corrected log-likelihood*

$z \leftarrow \ln(F) + \mathsf{LL0} - \mathsf{cLL0}$ — *// factor $e^z = F\frac{L(0)}{1-L(0)}$ for unobserved E-statistics*

for $(u \leftarrow 1, 2. \ldots, R)$ **do**

  $\tilde{\Gamma}_u \leftarrow \mathsf{ladd}\big(\tilde{\Gamma}_u, z + \sigma_u\big)$

  for $(i \leftarrow 0, 1, \ldots, m_u)$ **do**

    $\tilde{\Delta}_u^-[i] \leftarrow \mathsf{ladd}\big(\tilde{\Delta}_u^-[i], z + \delta_u^-[i]\big)$

    $\tilde{\Delta}_u^+[i] \leftarrow \mathsf{ladd}\big(\tilde{\Delta}_u^+[i], z + \delta_u^+[i]\big)$

return $\Big(\mathsf{LL}, F^*, \big\{(\tilde{\Gamma}_u, \tilde{\Delta}_u^-[0..M_u], \tilde{\Delta}_u^+[0..M_u])\big\}_{u=1}^R\Big)$

Fig. S5. Algorithm samplePosteriors computes the posteriors in Theorem 11

Function profilePosteriors$(\Xi = \{\xi_1 = n_1, \ldots, \xi_L = n_L\})$

// statistics for one profile Ξ

initialize data structure for n_u and $\tilde{\mathbf{K}}_u, \tilde{\mathbf{K}}'_u, \tilde{\mathbf{C}}_u, \tilde{\mathbf{J}}_u, \tilde{\mathbf{B}}_u, \tilde{\mathbf{B}}'_u$ at $1 \leq u \leq R$

$u \leftarrow 1$

while $(u \leq R)$ **do** — *// in postfix order*

  if $(u \leq L)$ **then** $n_u \leftarrow \xi_u$

  else $n_u \leftarrow 0$; **for** $(uv \in T)$ **do** $n_u \leftarrow n_u + n_v$ — *// copy sum in $\mathcal{L}_u$*

  $\mathsf{Linner}(\{n_v\}_{v=1}^R, u); u \leftarrow u+1$ — *// inner log-likelihoods*

while $(0 < u)$ **do** — *// in prefix order*

  $u \leftarrow u-1; \mathsf{Louter}(\{n_v\}_{v=1}^R, u)$ — *// outer log-priors*

while $(u < R)$ **do**

  $(\mathsf{LL}, \tilde{\Gamma}_u, \tilde{\Delta}_u^-[0..n_u], \tilde{\Delta}_u^+[0..n_u]) \leftarrow \mathsf{nodePosteriors}(\Xi, u)$

  $u \leftarrow u+1$

return $\Big(\mathsf{LL}, \{(\tilde{\Gamma}_u, \tilde{\Delta}_u^-[0..n_u], \tilde{\Delta}_u^+[0..n_u])\}_{u=1}^R\Big)$

Fig. S6. Algorithm profilePosteriors does the posterior computations for a single profile

```
Procedure Linner({n_v}_{v=1}^R, u)
  // calculates C̃_u, K̃_u, M_u
  if (u ≤ L) then
      C̃_u[n_u] ← 0; for n ← 0, 1, ..., n_u − 1 do C_u[n] ← −∞
  else
      L̃[0..n_w] ← K̃_w[0..n_w]
      z ← logit p̃_v + ln(1 − p̃_w)                          // z = logit p_v^−
      (z_0, z_1) ← linv(z)                                  // z_0 = ln p_v^−, z_1 = ln(1 − p_v^−)
      for n ← 0, 1, ..., n_u do
          if (n_w < n) then t ← n_w + 1; x ← −∞ else t ← n; x ← L̃[t] ← K̃_w[t]
          s ← n − t
          while (0 < t ∧ s < n_v) do                        // update L̃
              t ← t − 1; s ← s + 1
              x ← x + ln(1 − p̃_w); y ← L̃[t] + ln(p̃_w)
              x ← L̃[t] ← ladd(x, y)
          C̃_u[n] ← K̃_u[s] + L̃[t] + ln binom(n, s) + s z_1 + t z_0
          s ← s − 1; t ← t + 1
          while (0 ≤ s ∧ t ≤ n_w) do
              C̃_u[n] ← ladd(C̃_u[n], K̃_u[s] + L̃[t] + ln binom(n, s) + s z_1 + t z_0)
              s ← s − 1; t ← t + 1
  if (u = R) then s_max ← 0 else s_max ← n_u              // maximum surviving copies
  for s ← 0, 1, ..., s_max do
      t ← 0; n ← s
      if (logit q_u = −∞) then y ← −r̃_u
      else y ← (s + κ_u) ln(1 − q̃_u)
      K̃_u[s] ← C̃_u[n] + y
      while (n < n_u) do
          n ← n + 1; t ← t + 1
          if (logit q_u = −∞) then y ← −r̃_u + t ln(r̃_u) − ln(t!)
          else y ← ln binom(κ_u + n − 1, t) + (s + κ_u) ln(1 − q̃_u) + t ln(q̃_v)
          K̃_u[s] ← ladd(K̃_u[s], C̃_u[n] + y)
```

Fig. S7. Algorithm Linner computes the inner log-likelihood recurrences from Theorem 10

```
Procedure Louter({n_v}_{v=1}^R, v)
  // calculates B̃_v, B̃'_v, J̃_v, J̃'_v
  if (v = R) then
      s_max ← 0; B̃_v[0] ← 0; B̃'_v[0][0] ← 0
  else
      s_max ← n_v
      get parent u with uv ∈ T and sibling w ≠ v with uw ∈ T
      z ← logit p̃_v + ln(1 − p̃_w)                          // z = logit p_v^−
      (z_0, z_1) ← linv(z)                                  // z_0 = ln p_v^−, z_1 = ln(1 − p_v^−)
      L̃[0..n_w] ← K̃_w[0..n_w]
      s ← 0
      while (s ≤ s_max) do
          J̃_v[s] ← −∞
          if (0 < s) then
              t ← n_w; y ← L̃[t]; L̃[t] ← L̃[t] + ln(p̃_w)
              while (0 < t) do
                  t ← t − 1; x ← y + ln(1 − p̃_w); y ← L̃[t]
                  L̃[t] ← ladd(x, y + ln(p̃_w))
          t ← 0; n ← s + t
          while (t ≤ n_w) do
              J̃'_v[n][s] ← B̃_u[n] + L̃[t] + ln binom(n, s) + s z_0 + t z_1; J̃_v[s] ← ladd(J̃_v[n], J̃'_v[n][s])
              t ← t + 1; n ← n + 1
          s ← s + 1
  for n ← 0, 1, ..., n_v do
      t ← n; s ← 0; B̃_v[n] ← −∞
      while (s ≤ s_max ∧ s ≤ n) do
          if (logit q_v = −∞) then y ← −r̃_v + t ln(r̃_v) − ln(t!)
          else y ← ln binom(κ_u + n − 1, t) + (s + κ_u) ln(1 − q̃_u) + t ln(q̃_v)
          B̃'_v[n][s] ← J̃_v[s] + y; B̃_v[n] ← ladd(B̃_v[n], B̃'_v[n][s])
          s ← s + 1; t ← t − 1
```

Fig. S8. Algorithm Louter computes the outer conditional priors from Theorem 10

Function nodePosteriors$(\{n_u\}_{u=1}^{R}, v)$
// *calculates* $\mathsf{LL} = \ln L(\Xi)$, $\tilde{\Gamma}_v = \ln \mathbb{E}\Big[\tilde{\eta}_v \;\Big|\; \Xi\Big]$, $\tilde{\Delta}^{-}_{v,i} = \ln \mathbb{P}\Big\{i < \tilde{\xi}_u, \tilde{\eta}_v \le i \;\Big|\; \Xi\Big\}$ *(parent* u*)*, $\tilde{\Delta}^{+}_{v,i} = \ln \mathbb{P}\Big\{i < \tilde{\xi}_v, \tilde{\eta}_v \le i \;\Big|\; \Xi\Big\}$
$\tilde{\Gamma}_v \leftarrow -\infty; x \leftarrow -\infty$
if $(v = R)$ **then** $s_{\max} \leftarrow 0$ **else** $s_{\max} \leftarrow n_v$
for $(s \leftarrow s_{\max}, s_{\max} - 1, \ldots, 0)$ **do**
  $\tilde{\Gamma}_v \leftarrow \mathsf{ladd}\big(\tilde{\Gamma}_v, x\big)$
  $x \leftarrow \mathsf{ladd}\big(x, \tilde{\mathbf{J}}_v[s] + \tilde{\mathbf{K}}_v[s]\big)$
$\mathsf{LL} \leftarrow x; \tilde{\Gamma}_v \leftarrow \tilde{\Gamma}_v - \mathsf{LL}$
for $(s \leftarrow 0, 1, \ldots, s_{\max})$ **do**
  for $(n \leftarrow 0, 1, \ldots, n_v)$ **do**
    if $(n < s)$ **then** $\Delta^{-}_{n,s} \leftarrow -\infty; \Delta^{+}_{n,s} \leftarrow -\infty$
    else
      $\Delta^{-}_{n,s} \leftarrow \mathbf{J}'_v[n, s] + \mathbf{K}_v[s] - \mathsf{LL}$
      $\Delta^{+}_{n,s} \leftarrow \mathbf{B}'_v[n, s] + \mathbf{C}_v[n] - \mathsf{LL}$
$\tilde{\Delta}^{-}_v[0..n_v] \leftarrow \mathsf{tailDiff}\big(n_v, \{\Delta^{-}_{n,s}\}\big)$
$\tilde{\Delta}^{+}_v[0..n_v] \leftarrow \mathsf{tailDiff}\big(n_v, \{\Delta^{+}_{n,s}\}\big)$
return $\Big(\mathsf{LL}, \tilde{\Gamma}_v, \tilde{\Delta}^{-}_v[0..n_v], \tilde{\Delta}^{+}_v[0..n_v]\Big)$

Function tailDiff$\big(M, \{\ln p_{n,s}\}_{0 \le s \le n \le M}\big)$
// *calculates* $T_j = \ln \sum_{s=0}^{j} \sum_{n=j+1}^{M} p_{n,s}$ *for all* $j = 0, \ldots, M$
$s \leftarrow 0; j \leftarrow M; T_j \leftarrow -\infty; t \leftarrow \ln p_{j,s}$
while $(s < j)$ **do**
  $j \leftarrow j - 1$
  $T_j \leftarrow t$; **if** $(s < j)$ **then** $t \leftarrow \mathsf{ladd}(t, \ln p_{j,s})$
while $(s < M)$ **do**
  $s \leftarrow s + 1$
  // *at the end of each iteration* T_s *has the correct value*
  $j \leftarrow M; t \leftarrow \ln p_{j,s}$
  while $(s < j)$ **do**
    $j \leftarrow j - 1$
    $T_j \leftarrow \mathsf{ladd}(T_j, t)$
    // *now* $t = \ln \sum_{n=j+1}^{M} p_{n,s}$ *and* $T_j = \ln \sum_{i=0}^{s} \Big(\sum_{n=j+1}^{M} p_{n,i}\Big)$
    if $(s < j)$ **then** $t \leftarrow \mathsf{ladd}(t, \ln p_{j,s})$
return $\{T_j : j = 0, \ldots, M\}$

Fig. S9. Algorithm nodePosteriors computes the posteriors at a single node. Crucially, the death and birth statistics are calculated in quadratic time from the posterior transitions Δ^{-}, Δ^{+} by tailDiff as suggested by Theorem 17.

Function unobservedProfilePosteriors(M)
// *computes posteriors for profiles with at most* M *members*
initialize data structure for $\tilde{\mathbf{K}}_{u,m}, \tilde{\mathbf{K}}'_{u,m}, \tilde{\mathbf{C}}_{u,m}, \mathbf{J}, \tilde{m}_u, \tilde{\mathbf{B}}_{u,m}, \tilde{\mathbf{B}}'_{u,m}, \tilde{\mathbf{L}}_{u,m}$ at $1 \le u \le R$ and $0 \le m \le M$
while $(u \le R)$ **do** // *in postfix order*
  Uinner(u, M) // *inner log-likelihoods*
  $u \leftarrow u + 1$
while $(0 < u)$ **do** // *in prefix order*
  $u \leftarrow u - 1$
  Uouter(u, M) // *outer log-priors*
while $(u < R)$ **do**
  $\Big(\mathsf{LL0}, \{(\sigma_u, \delta^{-}_u, \delta^{+}_u)\}_{u=1}^{R}\Big) \leftarrow$ unobservedNodePosteriors(u, M)
  $u \leftarrow u + 1$
return $\Big(\mathsf{LL0}, \{(\sigma_u, \delta^{-}_u[0..M], \delta^{+}_u[0..M])\}_{u=1}^{R}\Big)$

Fig. S10. Algorithm unobservedProfilePosteriors does the posterior computations for unobserved profiles with at most $M = \Omega_{\min} - 1$ members

Procedure $\mathsf{Uinner}(u, M)$
// *calculates* $\tilde{\mathbf{C}}_{u,m}$, $\tilde{\mathbf{K}}_{u,m}$, $\tilde{\mathbf{L}}_{u,m}$ *for profiles with at most* M *copies*
for $(m \leftarrow 0, 1, \ldots, M)$ **do**
  if $(u \leq L)$ **then**
    $\tilde{\mathbf{C}}_{u,m}[m] \leftarrow 0$; **for** $n \leftarrow 0, 1, \ldots, m-1$ **do** $\mathbf{C}_{u,m}[n] \leftarrow -\infty$
  else // *recurrences for* $\tilde{\mathbf{C}}$
    $z \leftarrow \operatorname{logit} \tilde{p}_v + \ln(1 - \tilde{p}_w)$ // $z = \operatorname{logit} p_v^-$
    $(z_0, z_1) \leftarrow \mathsf{linv}(z)$ // $z_0 = \ln p_v^-, z_1 = \ln(1 - p_v^-)$
    for $n \leftarrow 0, 1, \ldots, m$ **do**
      $s \leftarrow n; t \leftarrow 0; \tilde{\mathbf{C}}_{u,m}[n] \leftarrow -\infty$
      while $(0 \leq s)$ **do**
        $x \leftarrow \tilde{\mathbf{K}}_{v,m-t}[s] + \tilde{\mathbf{L}}_{w,t}[n][t];\ k \leftarrow t+1$
        while $(k \leq m - s)$ **do** $x \leftarrow \mathsf{ladd}\big(x, \tilde{\mathbf{K}}_{v,m-k}[s] + \tilde{\mathbf{L}}_{w,k}[n][t]\big); k \leftarrow k+1$
        $\tilde{\mathbf{C}}_{u,m}[n] \leftarrow \mathsf{ladd}\big(\tilde{\mathbf{C}}_{u,m}[n], x + \ln \binom{n}{s} + z_1 s + z_0 t\big)$
        $s \leftarrow s - 1; t \leftarrow t + 1$
  if $(u = R)$ **then** $s \leftarrow 0$ **else** $s \leftarrow m$
  while $0 \leq s$ **do** // *recurrences for* $\tilde{\mathbf{K}}$
    $t \leftarrow 0; n \leftarrow s$
    if $(\operatorname{logit} q_u = -\infty)$ **then** $y \leftarrow -\tilde{r}_u$
    else $y \leftarrow (s + \kappa_u)\ln(1 - \tilde{q}_u)$
    $\tilde{\mathbf{K}}_{u,m}[s] \leftarrow \tilde{\mathbf{C}}_{u,m}[n] + y$
    while $(n < m)$ **do**
      $n \leftarrow n + 1; t \leftarrow t + 1$
      if $(\operatorname{logit} q_u = -\infty)$ **then** $y \leftarrow -\tilde{r}_u + t\ln(\tilde{r}_u) - \ln(t!)$
      else $y \leftarrow \ln \binom{\kappa_u + n - 1}{t} + (s + \kappa_u)\ln(1 - \tilde{q}_u) + t\ln(\tilde{q}_v)$
      $\tilde{\mathbf{K}}_{u,m}[s] \leftarrow \mathsf{ladd}\big(\tilde{\mathbf{K}}_{u,m}[s], \tilde{\mathbf{C}}_{u,m}[n] + y\big)$
    $s \leftarrow s - 1$
  for $n \leftarrow 0, 1, \ldots, m$ **do**
    $t \leftarrow n; x \leftarrow \tilde{\mathbf{L}}_{v,m}[n][t] \leftarrow \tilde{\mathbf{K}}_{v,m}[t]$
    while $(0 < t)$ **do** // *recurrences for* $\tilde{\mathbf{L}}$
      $t \leftarrow t - 1$
      $x \leftarrow x + \ln(1 - \tilde{p}_v); y \leftarrow \tilde{\mathbf{L}}_{v,m}[n-1][t] + \ln(\tilde{p}_v)$
      $x \leftarrow \tilde{\mathbf{L}}_{v,m}[n][t] \leftarrow \mathsf{ladd}(x, y)$

Fig. S11. Algorithm Uinner computes the inner log-likelihoods for unobserved profiles from Theorem 3

Procedure $\mathsf{Uouter}(v, M)$
// *calculates* $\tilde{\mathbf{B}}_{v,m}, \tilde{\mathbf{B}}'_{v,m}$, $\tilde{\mathbf{J}}_{v,m}, \tilde{\mathbf{J}}'_{v,m}$ *for profiles with at most* M *copies*
if $(v = R)$ **then**
  $n_{\max} \leftarrow 0; m \leftarrow 0; \tilde{\mathbf{J}}'_{v,m}[0][0] \leftarrow 0$
  while $(m < M)$ **do** $m \leftarrow m + 1; \tilde{\mathbf{J}}'_{v,m}[0][0] \leftarrow -\infty$
else
  $n_{\max} \leftarrow M$; get parent u with $uv \in T$ and sibling $w \neq v$ with $uw \in T$
  $z \leftarrow \operatorname{logit} \tilde{p}_v + \ln(1 - \tilde{p}_w)$ // $z = \operatorname{logit} p_v^-$
  $(z_0, z_1) \leftarrow \mathsf{linv}(z)$ // $z_0 = \ln p_v^-, z_1 = \ln(1 - p_v^-)$
  for $(m \leftarrow 0, 1, \ldots M)$ **do**
    for $(s \leftarrow 0, 1, \ldots M - m)$ **do**
      $t \leftarrow 0; n \leftarrow s; \tilde{\mathbf{J}}_{v,m}[s] \leftarrow -\infty$
      while $(n \leq M)$ **do**
        $j \leftarrow t; x \leftarrow \tilde{\mathbf{B}}_{u,m-j}[n] + \tilde{\mathbf{L}}_{w,j}[n][t]$
        while $(j < m)$ **do** $j \leftarrow j + 1; x \leftarrow \mathsf{ladd}\big(x, \tilde{\mathbf{B}}_{u,m-j}[n] + \tilde{\mathbf{L}}_{w,j}[n][t]\big)$
        $\tilde{\mathbf{J}}'_{v,m}[n][s] \leftarrow x + \ln \binom{n}{s} + sz_1 + tz_0$
        $\tilde{\mathbf{J}}_{v,m}[s] \leftarrow \mathsf{ladd}\big(\tilde{\mathbf{J}}_{v,m}[s], \tilde{\mathbf{J}}'_{v,m}[n][s]\big)$
        $t \leftarrow t + 1; n \leftarrow n + 1$
for $(m \leftarrow 0, 1, \ldots M)$ **do**
  for $n \leftarrow 0, 1, \ldots, n_{\max}$ **do**
    $t \leftarrow 0; s \leftarrow n$
    if $(\operatorname{logit} q_v = -\infty)$ **then** $\tilde{\mathbf{B}}_{v,m}[n] \leftarrow \tilde{\mathbf{B}}'_{v,m}[n][s] \leftarrow \tilde{\mathbf{J}}_{v,m}[s] - \tilde{r}_v$
    else $\tilde{\mathbf{B}}_{v,m}[n] \leftarrow \tilde{\mathbf{B}}'_{v,m}[n][s] \leftarrow \tilde{\mathbf{J}}_{v,m}[s] + (s + \kappa_u)\ln(1 - \tilde{q}_u)$
    while $(0 < s)$ **do**
      $s \leftarrow s - 1; t \leftarrow t + 1$
      if $(\operatorname{logit} q_v = -\infty)$ **then** $y \leftarrow -\tilde{r}_v + t\ln(\tilde{r}_v) - \ln(t!)$
      else $y \leftarrow \ln \binom{\kappa_u + n - 1}{t} + (s + \kappa_u)\ln(1 - \tilde{q}_u) + t\ln(\tilde{q}_v)$
      $\tilde{\mathbf{B}}'_{v,m}[n][s] \leftarrow \tilde{\mathbf{J}}_{v,m}[s] + y$
      $\tilde{\mathbf{B}}_{v,m}[n] \leftarrow \mathsf{ladd}\big(\tilde{\mathbf{B}}_{v,m}[n], \tilde{\mathbf{B}}'_{v,m}[n][s]\big)$

Fig. S12. Algorithm Uouter computes the outer conditional priors for unobserved profiles from Theorem 3

Function unobservedNodePosteriors(u, M)

// *calculates* $\mathsf{LL0} = \ln \mathbb{P}\Big\{\Omega_R \le M\Big\}$, σ_u, $\delta^-_{u,i}$, $\delta^+_{u,i}$

if $(v = R)$ **then** $s_{\max} \leftarrow 0$ **else** $s_{\max} \leftarrow M$

for $(m \leftarrow 1, 2, \ldots, M)$ **do** // *compute cumulatives*

  for $(s \leftarrow 0, 1, \ldots, \min\{s_{\max}, M - m\})$ **do**

    $\mathbf{J}_{u,m}[s] \leftarrow \mathsf{ladd}\Big(\mathbf{J}_{u,m}[s], \mathbf{J}_{u,m-1}[s]\Big)$

    for $(n \leftarrow s, s+1, \ldots, M)$ **do**

      $\mathbf{J}'_{u,m}[n][s] \leftarrow \mathsf{ladd}\Big(\mathbf{J}'_{u,m}[n][s], \mathbf{J}'_{u,m-1}[n][s]\Big)$

      $\mathbf{B}'_{u,m}[n][s] \leftarrow \mathsf{ladd}\Big(\mathbf{B}'_{u,m}[n][s], \mathbf{B}'_{u,m-1}[n][s]\Big)$

$\mathsf{LL0} \leftarrow -\infty; \sigma_u \leftarrow -\infty$

for $(s \leftarrow s_{\max}, s_{\max} - 1, \ldots, 0)$ **do**

  $m \leftarrow s; y \leftarrow \tilde{\mathbf{J}}_{u,M-m}[s] + \tilde{\mathbf{K}}_{u,m}[s]$

  while $(m < M)$ **do** $m \leftarrow m + 1; y \leftarrow \mathsf{ladd}(y, \tilde{\mathbf{J}}_{u,M-m}[s] + \tilde{\mathbf{K}}_{u,m}[s])$

  $\sigma_u \leftarrow \mathsf{ladd}(\sigma_u, \mathsf{LL0})$ // *expectation by tail sum*

  $\mathsf{LL0} \leftarrow \mathsf{ladd}(\mathsf{LL0}, y)$

$\sigma_u \leftarrow \sigma_u - \mathsf{LL0}$

for $(s \leftarrow 0, 1, \ldots, s_{\max})$ **do**

  for $(n \leftarrow s, s+1, \ldots, M)$ **do**

    $\Delta^-_{n,s} \leftarrow \mathbf{J}'_u[n, s] + \mathbf{K}_u[s] - \mathsf{LL0}$

    $\Delta^+_{n,s} \leftarrow \mathbf{B}'_u[n, s] + \mathbf{C}_u[n] - \mathsf{LL0}$

$\delta^-_u[0..M] \leftarrow \mathsf{tailDiff}\Big(M, \{\Delta^-_{n,s}\}\Big)$

$\delta^+_u[0..M] \leftarrow \mathsf{tailDiff}\Big(M, \{\Delta^+_{n,s}\}\Big)$

return $\Big(\mathsf{LL0}, \sigma_u, \delta^-_u[0..M], \delta^+_u[0..M]\Big)$

Fig. S13. Algorithm unobservedNodePosteriors does the posterior computations at a single node for unobserved profiles from Theorem 4

B. Experiments

B.1. Convergence to optimum. Figures S14, S15, S16, and S17 plot the convergence of the log-likelihood over the datasets E114, D80, P75, and ED194. For each of them, the input table comprises families with at least 4 members within that dataset. Optimization with Count (2) was initialized with a random GLD model (with random seed 2025) and ran until the parameter values did not change anymore (within machine precision). The longest optimization (ED194) costs about 200 vCPU hours (distributed over 9 threads on a MacBook Pro).

The log-linear plots show the L_∞-norm of the gradient (maximum absolute value in the vector) and the change in the log-likelihood value across the BFGS iterations.

One can observe that the optimization navigates a difficult landscape through long ravines (segments of decreasing gradient) and sudden steep slopes. The experiments underline that a small change in the log-likelihood value is not a good condition for stopping the optimization, as the gradient grows by multiple magnitudes when a steep slope is found. A theoretically quadratic convergence to a local maximum would show as a sudden drop only in the last few iterations.

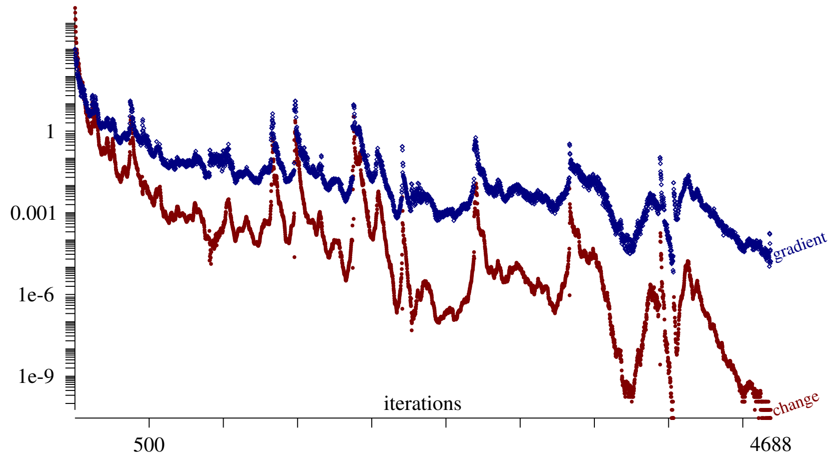

Fig. S14. Convergence of the likelihood optimizaton on the tree of Methanobacteriati (114 genomes, 7335 families with at least 4 copies). The plots show the L_∞-norm of the gradient and the increase of the log-likelihood in each iteration until convergence.

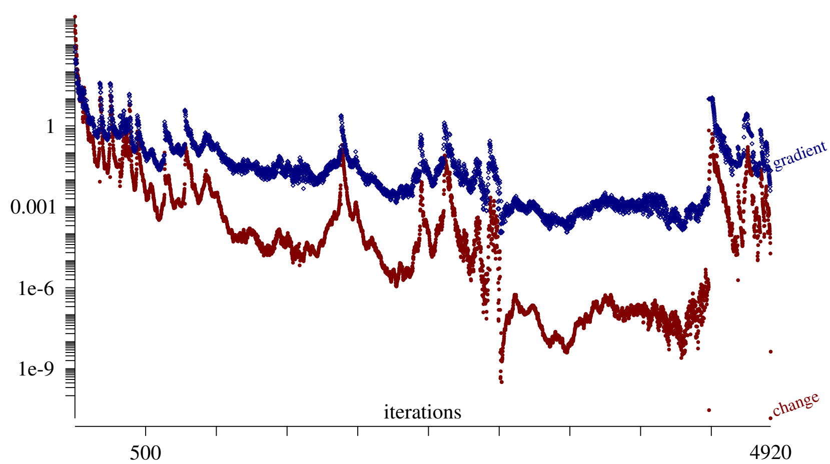

Fig. S15. Convergence of the likelihood optimizaton on the tree of Nanobdellati (80 genomes, 3034 families with at least 4 copies). The plots show the L_∞-norm of the gradient and the increase of the log-likelihood in each iteration until convergence.

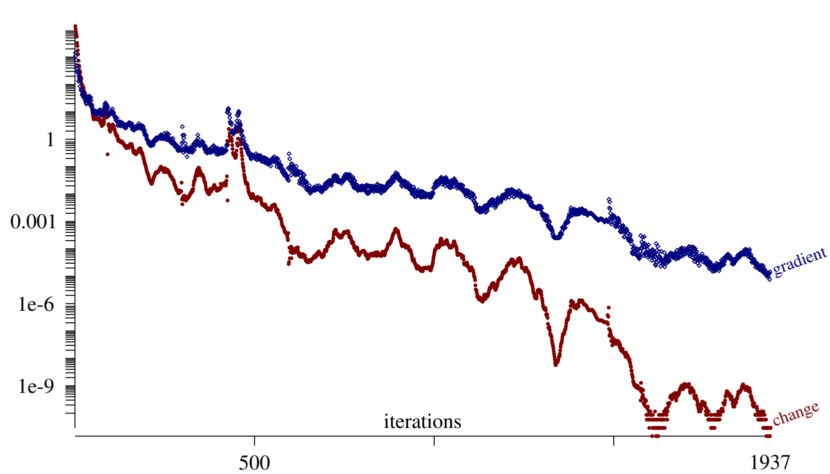

Fig. S16. Convergence of the likelihood optimizaton on the tree of Proteoarchaeota (75 genomes, 5179 families with at least 4 copies). The plots show the L_∞-norm of the gradient and the increase of the log-likelihood in each iteration until convergence.

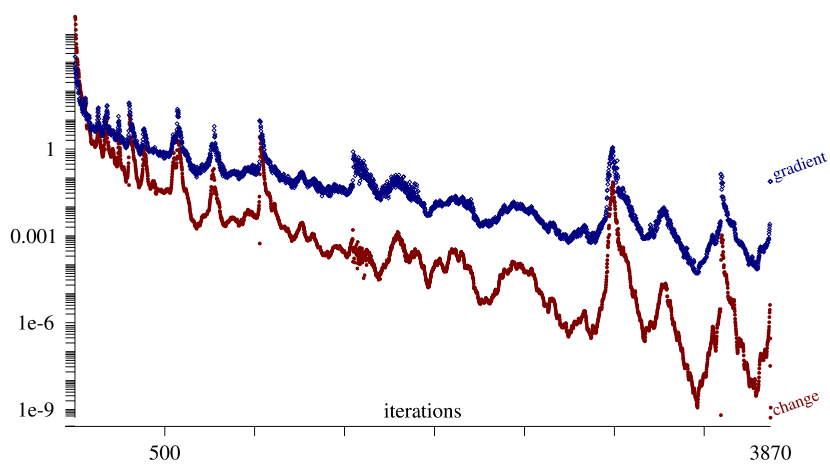

Fig. S17. Convergence of the likelihood optimizaton on the tree of Methanobacterioti and Nanobdellati (194 genomes, 8855 families with at least 4 copies). The plots show the L_∞-norm of the gradient and the increase of the log-likelihood in each iteration until convergence.

B.2. Accuracy of reconstruction. Count (2) implements discrete-event simulation for generating Galton-Watson trees over the gene copies by a given GLD model. In the simulation for one family, every copy is a distinct object with an associated age (increasing from the root) and parent-child relationships. (Future) duplication and loss events are generated at copy births. Gain events (xenolog arrivals) are generated at speciation points. Events are inserted into a priority queue by their associated waiting times drawn randomly by exponential distribution (with rates λ, μ, γ for duplication,loss, and gain). Events are deleted from the queue in their time order and either generate (one or two) new copies born with registered origin or kill one copy. Upon copy gain, the next gain event is inserted into the queue. Moot events over dead copies are ignored. At speciation points, identical child copies are created in the two descendant lineage, with newly generated future events. At each iteration of the simulation, random families are generated until one passes the observation threshold $\Omega_{\min} = 4$; histories over observed (and unobserved) families are recorded as "ground truth."

On each dataset, I simulated the same number of observed families by their optimized GLD model (with different seeds: $s = 2025, 2026, 2027, 2028$ for D80, P75, E114 and ED194), and then optimized a GLD model over the simulated profiles (optimization initialized with seed $17s$). In order to measure true positive rates (TPR) and false positive rates (FPR) by the optimized model, I generated an independent test sample of 10000 random profiles by the original ("true") rate model.

The parsimony reconstructions use a modified Wagner-parsimony penalization π of for the state transitions $\xi_u \to \xi_v$ on each edge uv: if $\xi_u = 0$, then $\pi = g + (\xi_v - 1)$ or else $\pi = \left|\xi_u \to \xi_v\right|$. The gain penalty $g = 3.0001$ is used for comparisons.

A third method considered here is Gloome (3), developed for probabilistic inference from binary presence-absence profiles. Each simulated dataset was converted into absence-presence profiles and used as input to Gloome (Gloome 2.0 web interface with default parameters) along with the true phylogeny for which edge lengths were set to the GLD rate model's edge lengths. The output "Table of nodes" file gives the absence-presence inferences at ancestral nodes that are compared to ground truth for TPR and FPR statistics.

Figure S18 shows the accuracy of ancestral reconstruction over P75, compared to simpler inference methods, and to inference with the true model. At most nodes, the true positive rate for MAP inferences exceeds 98.5% and the false positive rate stays below 0.55%. Figure S19, S20 and S21 show the errors of the same methods on E114, D80 and ED194, respectively.

Figure S22 shows the accuracy of MAP reconstruction on smaller, phylum- and superclass-level datasets.

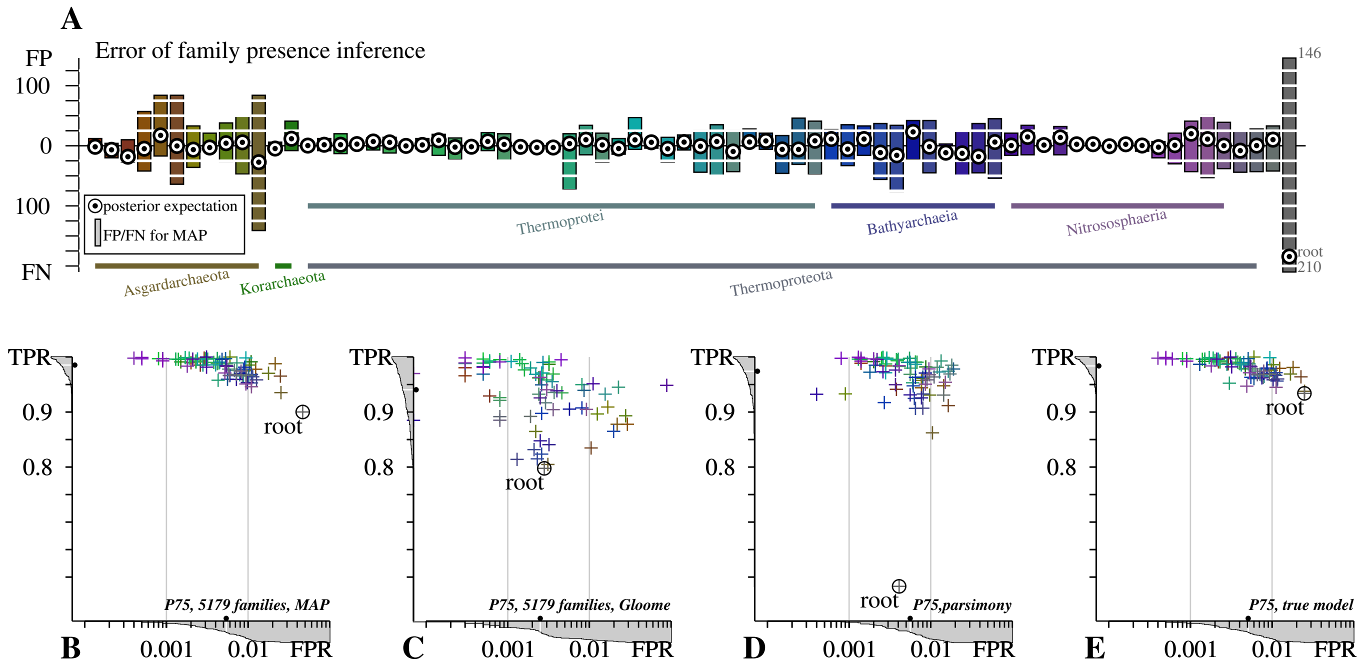

Fig. S18. Error of ancestral reconstructions on Proteoarchaeota (P75). **A** Upward and downward bars show false positives (FP) and false negatives (FN) for the maximum a posteriori (MAP) estimates of family presence, at ancestral nodes enumerated in postorder from left to right. Coloring is by position in the phylogeny as in Figure S23. Circles show the error of the posterior expected family count. **B–E**. Linear-log plot of false positive rate (FPR) and true positive rate (TPR) by the MAP estimate at ancestral nodes (A), by Gloome (B), by numerical parsimony (C), and by posterior inference with the true model (the Bayes classifier). Shaded graphs on the exterior of the axes plot the cumulative distribution function across ancestral nodes, dots mark the median values.

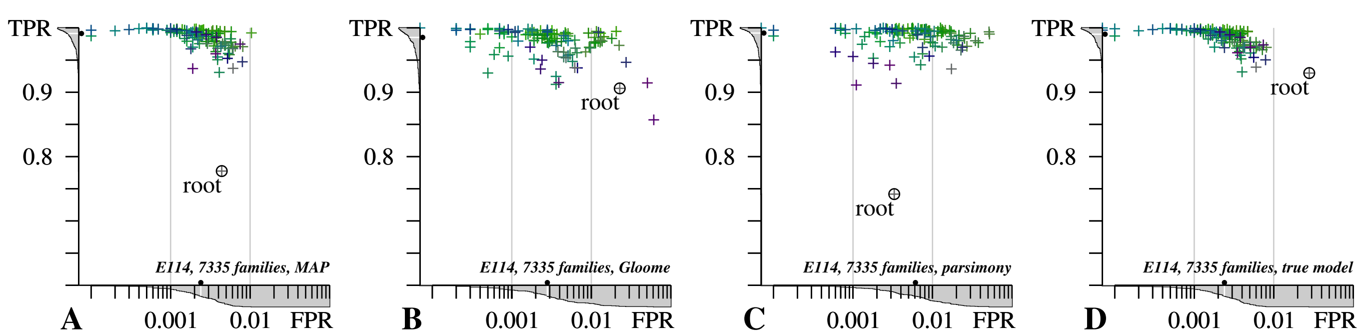

Fig. S19. Error of ancestral reconstructions on Methanobacteriati (E114). **A–D**. Linear-log plot of false positive rate (FPR) and true positive rate (TPR) by the MAP estimate at ancestral nodes (A), by Gloome (B), by numerical parsimony (C), and by posterior inference with the true model (the Bayes classifier). Coloring is by position in the phylogeny as in Figure ??. Shaded graphs on the exterior of the axes plot the cumulative distribution function across ancestral nodes, dots mark the median values.

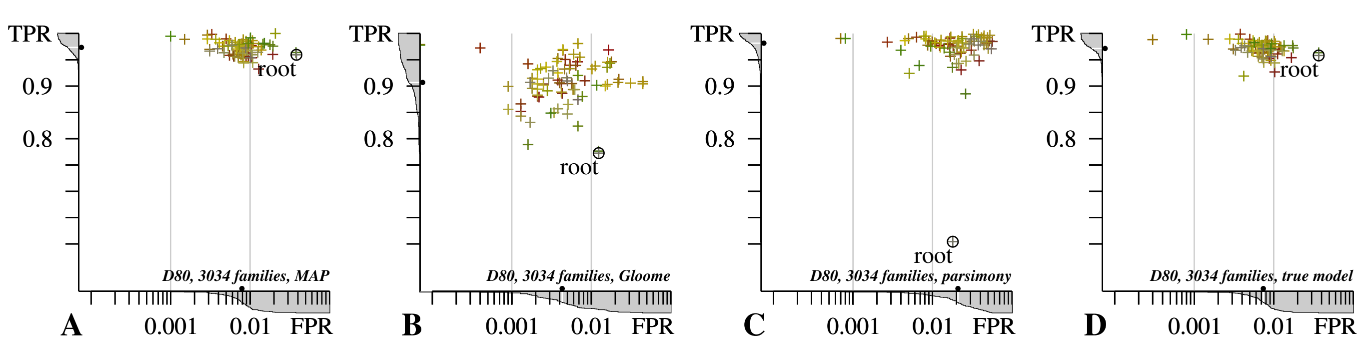

Fig. S20. Error of ancestral reconstructions on Nanobdellati (D80). **A–D**. Linear-log plot of false positive rate (FPR) and true positive rate (TPR) by the MAP estimate at ancestral nodes (A), by Gloome (B), by numerical parsimony (C), and by posterior inference with the true model (the Bayes classifier). Coloring is by position in the phylogeny as in Figure ??. Shaded graphs on the exterior of the axes plot the cumulative distribution function across ancestral nodes, dots mark the median values.

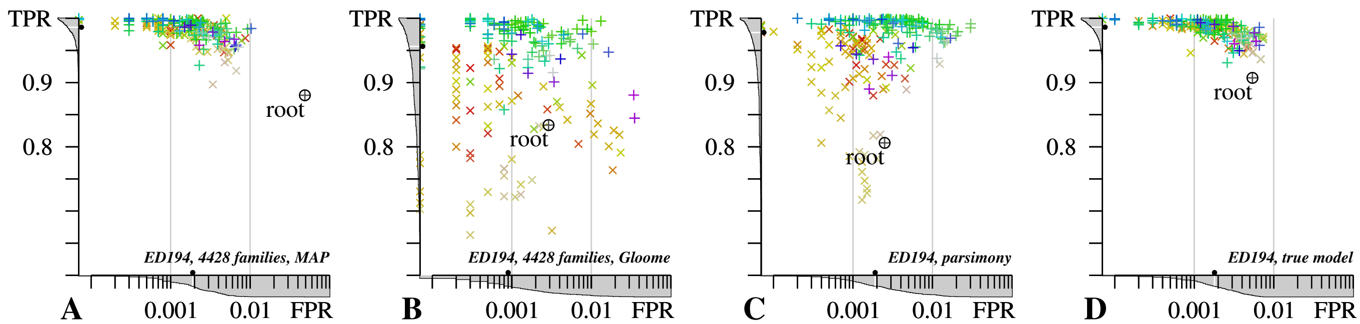

Fig. S21. Error of ancestral reconstructions on ED194. **A–D**. Linear-log plot of false positive rate (FPR) and true positive rate (TPR) by the MAP estimate at ancestral nodes (A), by Gloome (B), by numerical parsimony (C), and by posterior inference with the true model (the Bayes classifier). On **A** and **B**, the models were optimized on half as many families as in the full dataset. Coloring is by position in the phylogeny as in Figure ??. Shaded graphs on the exterior of the axes plot the cumulative distribution function across ancestral nodes, dots mark the median values. Methanobacteriati are plotted with + and Nanobdellati with ×

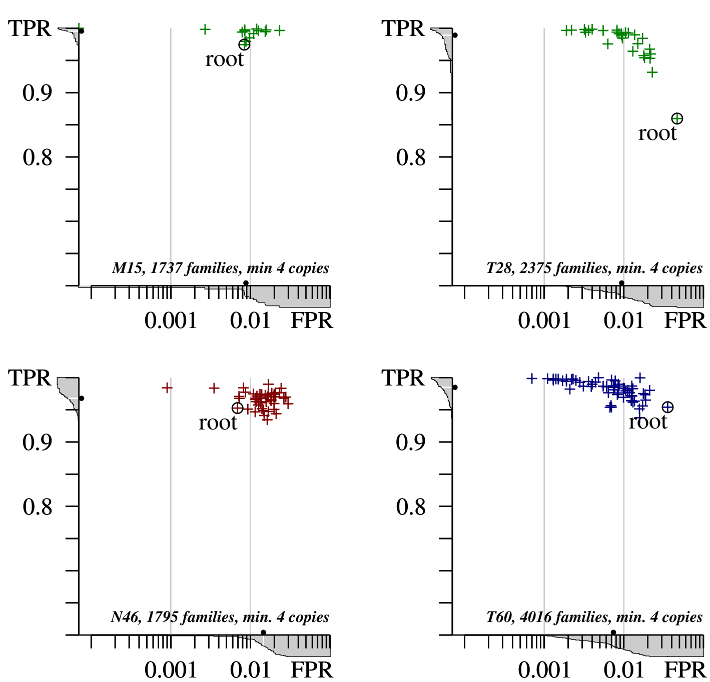

Fig. S22. MAP reconstruction accuracy on phylum- and superclass-level datasets. Linear-log plot of false positive rate (FPR) and true positive rate (TPR) by the MAP estimate at ancestral nodes for M15(Methanomada), T28(Thermoplasmatota), H51(Halobacteriota), N46(Nanobdellota), and T60(Thermoprotei).

B.3. Distribution of family profiles in homolog data and the GLD model. Figure S23 shows the distribution of the number of families at terminal and ancestral taxa, in ancestral reconstruction and in simulations.

Figures S24,S25, and S26 show the distribution of family frequencies (how many terminal taxa have a copy) in simulations with D80, P75 and E114.

Data was collected from 1000 simulations (launched with seeds $s = 2025, 2026, \dots$) over each dataset.

Figure S27 plots the cumulative residuals between the observed and simulated distribution functions. If o_i denotes the observed families in the ortholog dataset that are present in i extant genomes, and m_i denotes the average number of families in i lineages across the model simulations, then

$$F = \sum_i o_i = \sum_i m_i \qquad \mathrm{cdf}_o(k) = \frac{1}{F}\sum_{i=1}^{k} o_i \qquad \mathrm{cdf}_m(k) = \frac{1}{F}\sum_{i=1}^{k} m_i \qquad r(k) = \mathrm{cdf}_o(k) - \mathrm{cdf}_m(k)$$

with the number of families F, cumulative distribution functions cdf, and the residuals $r(k)$. The graphs for $r(k)$ have similar shapes in the four datasets: (1) a left (*transient*) region that passes through the absolute maximum near the sampling bias and extends to a first local minimum near 0, (2) a middle (*adaptive*) region that extends from the first minimum through a second local maximum until $r(k)$ crosses 0, and (3) a right (*stabilizing*) region from 0-crossing that passes through the absolute minimum and then increases to 0. The observed distribution is shifted to the left in the first two regions, and it is shifted to the right in the third region. Table S1 shows that the placement of these notable points is consistent across the four datasets: the second maximum and second minimum provide data-driven cloud-shell and shell-core boundaries (near 1/3 and 2/3 of lineages), while the breakpoints between regions are at 1/6 and 36–48% falling in the center of the data-driven cloud and shell regions.

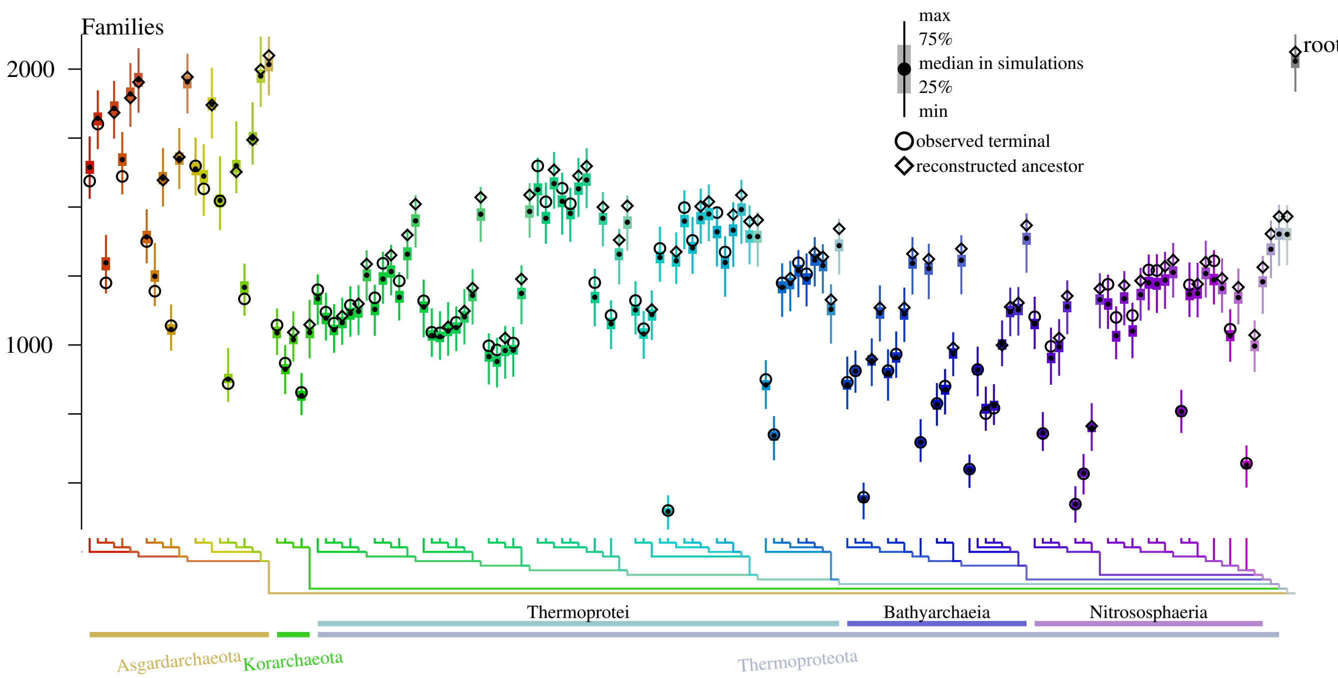

Fig. S23. Posterior expected family counts at ancestral nodes, observed family counts at terminal nodes, and empirical distribution of family counts in 1000 simulated datasets. (Families with at least 4 members are counted.) The small interquartile ranges show that genome size distribution is highly concentrated by the GLD model. Phylogeny nodes are enumerated and colored in postorder, as shown by the tree diagram.

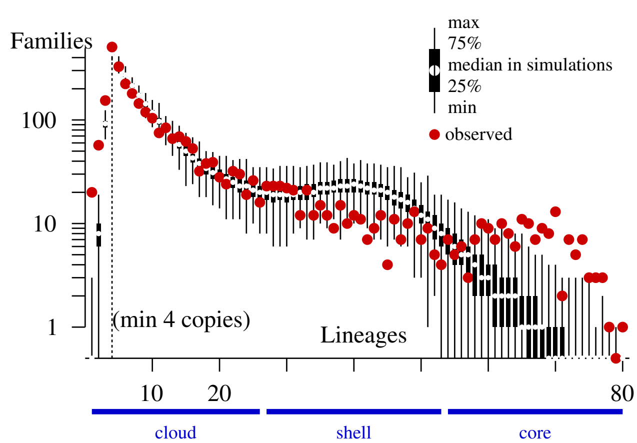

Fig. S24. Distribution of family frequency (extant genomes with a copy) in simulated and in observed profiles over D80; on linear-log scale. Box-and-whisker plot shows the distribution of families with given frequency in simulations; red discs plot the observed counts in ortholog data. The cloud, shell, and core families fall into consecutive thirds of the frequency range.

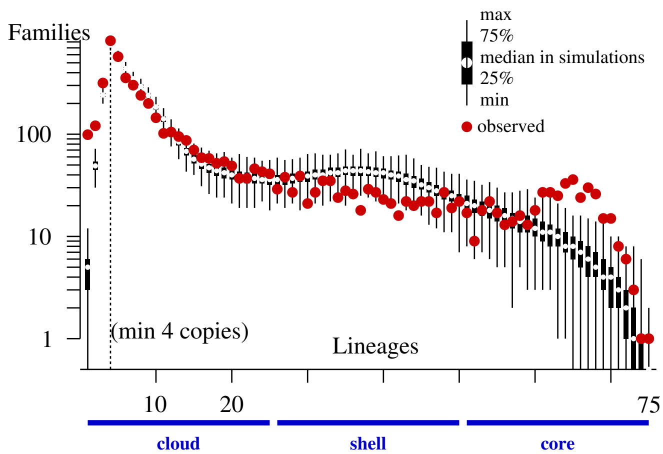

Fig. S25. Distribution of family frequency (extant genomes with a copy) in simulated and in observed profiles over P75; on linear-log scale. Box-and-whisker plot shows the distribution of families with given frequency in simulations; red discs plot the observed counts in ortholog data. The cloud, shell, and core families fall into consecutive thirds of the frequency range.

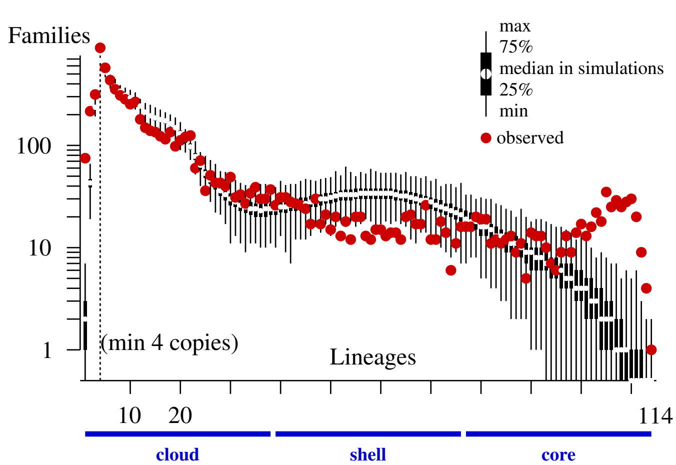

Fig. S26. Distribution of family frequency (extant genomes with a copy) in simulated and in observed profiles over E114; on linear-log scale. Box-and-whisker plot shows the distribution of families with given frequency in simulations; red discs plot the observed counts in ortholog data. The cloud, shell, and core families fall into consecutive thirds of the frequency range.

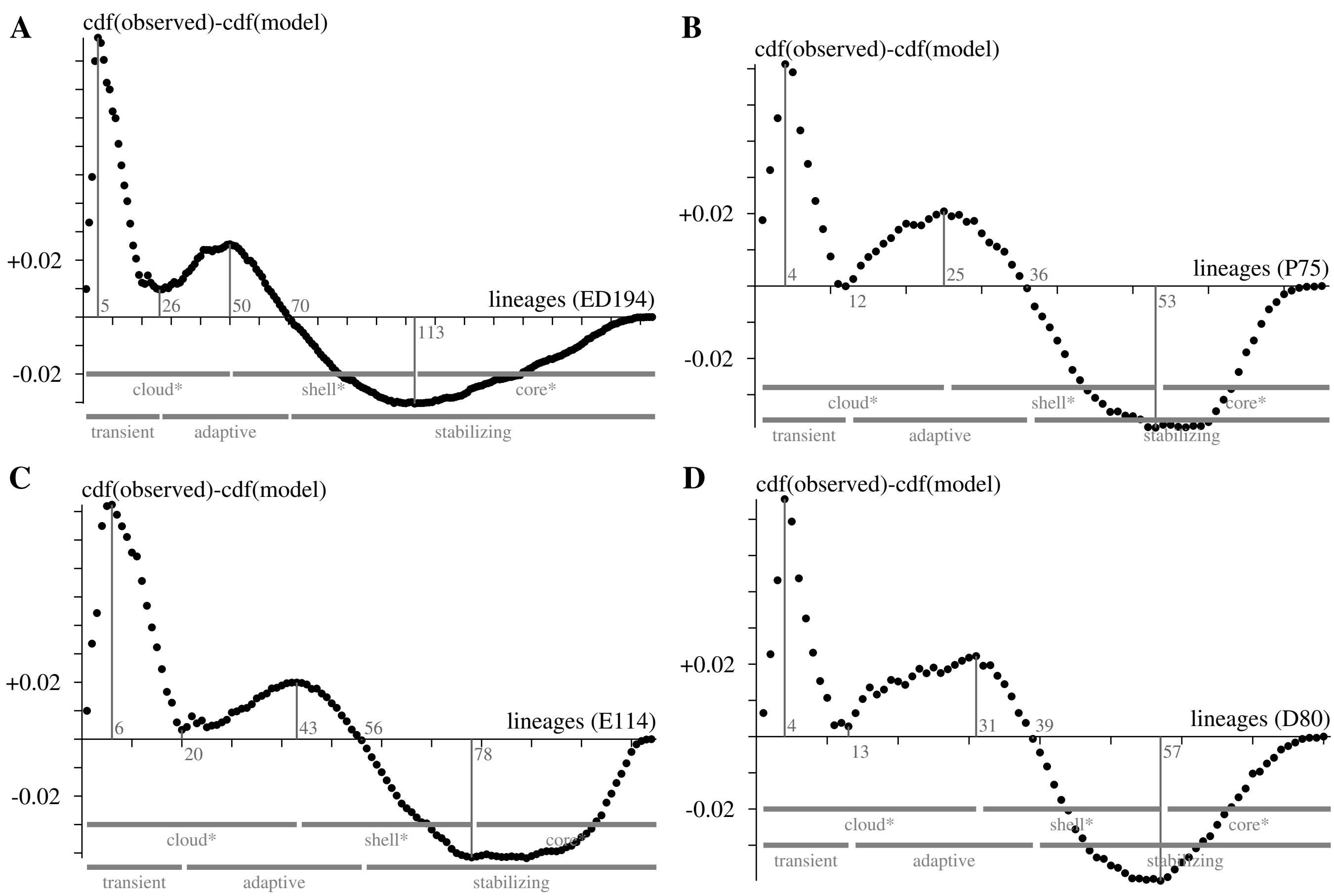

Fig. S27. Cumulative residual plots for family frequency distribution: difference between the cumulative distributions in true datasets (observation) and model simulations (average across 1000 experments). **A–D** show the plots for ED194, P75, E114 and D80, respectively. Main graphical properties are marked along the X axis: an early first maximum is followed by decrease with a first local minimum close to 0 (transient region), then increase to a first local maximum, followed by decrease crossing 0 (adaptive region), and further decrease until a second local minimum and increase until the end (stabilizing region). The second maximum and second minimum define data-driven cloud-shell and shell-core breakpoints (with *).

Table S1. Transitions in the cumulative residual plot. Notable frequency points from Figure S27.

Dataset	genomes	first maximum	first minimum	second maximum	0-crossing	second minimum
ED194	194	5	26 (13%)	50 (26%)	70 (36%)	113 (58%)
P75	75	4	12 (16%)	25 (33%)	36 (48%)	53 (71%)
E114	114	6	20 (18%)	43 (38%)	56 (49%)	78 (68%)
D80	80	4	13 (16%)	31 (39%)	39 (49%)	57 (71%)
average			16%	34%	45%	67%

B.4. Clade-specific distributions. Tables S2, S3 and S4 show the statistics on profile distribution and simulations (over E114, P75, and D80) for every clade of at least 12 taxa, with observation bias $\Omega_{\mathrm{min}} = 4$. The tables show the minimum and maximum observed in 1000 simulated datasets over the three taxon sets. Each simulated dataset was filtered for counting the total and unique profiles with at least 4 copies in a selected clade. P-values for bias were computed by the empirical distribution from the simulations. We can observe that, in general, (1) ortholog data have more families in smaller clades than in the simulation, that (2) ortholog data have fewer families with unique profiles. The first observation holds for smaller clades (exceptions are nodes close to the root), and the second holds for larger clades. But among Nanobdellati, there are many clades with patchy unique profiles so that ortholog profiles seem as random as in the GLD model.

Table S2. Families with at least 4 copies within clades of at least 12 genomes among Methanobacteriati (E114). True datasets tend to have more families (farther from the root) but fewer unique profiles (closer to the root) than what the model predicts.

Clade root	Parent	Taxa	All families			Unique families		
			simulation	observed	$P(o \leq s)$	simulation	observed	$P(s \leq o)$
U129	U133	14	2654–2915	2855	0.036	2497–2732	1936	< 0.001
U133	U134	18	2771–3052	3071	< 0.001	2706–2950	2308	< 0.001
U134	U136	22	2835–3123	3181	< 0.001	2786–3054	2520	< 0.001
U136	U156	24	2885–3168	3259	< 0.001	2840–3102	2632	< 0.001
U147	U149	12	2237–2525	2431	0.168	1699–1942	1353	< 0.001
U149	U155	14	2332–2613	2571	0.015	1912–2190	1595	< 0.001
U155	U156	20	2791–3077	3033	0.006	2437–2697	2184	< 0.001
U156	U163	44	4617–4885	5011	< 0.001	4434–4710	4223	< 0.001
U163/H51	U191	51	4996–5264	5316	< 0.001	4809–5071	4507	< 0.001
U175	U179	13	1376–1604	1520	0.164	1129–1311	931	< 0.001
U179	U186	17	1707–1933	1836	0.249	1373–1561	1212	< 0.001
U186	U190	24	1969–2209	2130	0.172	1658–1872	1593	< 0.001
U190/T28	U191	28	2137–2391	2375	0.002	1867–2085	1940	0.159
U191	U206	79	5961–6181	6196	< 0.001	5804–6048	5462	< 0.001
U205/M15	U206	15	1567–1771	1737	0.030	1003–1176	823	< 0.001
U206/T94	U219	94	6543–6700	6627	0.427	6386–6582	5808	< 0.001
U218	U219	13	1801–2027	1951	0.159	1276–1519	1000	< 0.001
U219	U226	107	7186–7250	7211	0.679	7027–7134	6281	< 0.001
U226/E114	root	114	7335	7335		7192–7243	6411	< 0.001

Table S3. Families with at least 4 copies within clades of at least 12 genomes among Proteoarchaeota (P75). True datasets tend to have more families (farther from the root) but fewer unique profiles (closer to the root) than what the model predicts.

Clade root	Parent	Taxa	All families			Unique families		
			simulation	observed	$P(o \leq s)$	simulation	observed	$P(s \leq o)$
U85/A12	U148	12	2174–2431	2400	0.009	1963–2194	1879	< 0.001
U100	U106	14	1472–1675	1594	0.243	1064–1241	857	< 0.001
U106	U114	20	1966–2195	2163	0.008	1579–1783	1379	< 0.001
U114	U119	28	2340–2639	2599	0.002	1956–2194	1901	< 0.001
U119/T33	U146	33	2653–2915	2902	0.001	2227–2430	2218	< 0.001
U130/B12	U145	12	1178–1376	1240	0.85	1082–1250	1057	< 0.001
U144/N15	U145	15	1470–1668	1625	0.069	1134–1325	996	< 0.001
U145	U146	27	2049–2281	2271	0.003	1842–2095	1984	0.498
U146/T60	U147	60	3857–4053	4016	0.036	3530–3739	3400	< 0.001
U147	U148	63	3938–4137	4080	0.100	3629–3829	3470	< 0.001
U148/P75	root	75	5179	5179		4879–4982	4377	< 0.001

Table S4. Families with at least 4 copies within clades of at least 12 genomes among Nanobdellati (D80). True datasets tend to have more families (farther from the root) but than what the model predicts.

Clade root	Parent	Taxa	All families			Unique families		
			simulation	observed	$P(o \leq s)$	simulation	observed	$P(s \leq o)$
U94	U149	16	894–1075	1044	0.006	790–957	823	0.031
U120	U121	15	899–1048	1060	< 0.001	862–1004	914	0.246
U121/W19	U140	19	967–1128	1178	< 0.001	937–1097	1082	0.994
U133	U134	13	674–825	749	0.386	597–708	552	< 0.001
U134	U135	14	681–795	749	0.42	634–780	609	< 0.001
U135	U136	15	697–851	833	0.002	676–828	720	0.153
U136	U138	16	711–862	850	0.001	694–845	753	0.358
U138	U139	18	726–879	874	0.001	713–864	805	0.799
U139/P19	U140	19	734–887	892	< 0.001	725–875	836	0.947
U140	U144	38	1426–1604	1668	< 0.001	1437–1578	1611	1.0
U144	U145	42	1555–1713	1744	< 0.001	1541–1695	1686	0.997
U145/N46	U146	46	1609–1774	1795	< 0.001	1596–1750	1725	0.966
U146	U147	47	1628–1796	1804	< 0.001	1613–1772	1734	0.931
U147	U148	54	1842–2021	1966	0.062	1829–2001	1893	0.254
U148	U149	55	1864–2047	1982	0.083	1857–2028	1907	0.172
U149	U158	71	2523–2640	2525	0.999	2514–2635	2368	< 0.001
U158/D80	root	80	3034	3034		3012–3033	2830	< 0.001

B.5. Expanding ancestors and cryptic inheritance. Table S5 shows the reconstructed genome sizes at clade ancestors in different phylogenetic contexts. While the root genome is difficult to estimate precisely, a larger contexts suffices for accurate inference, which changes little in even larger contexts. In contrast, ancestral estimates from ortholog datasets may increase more in larger phylogenetic contexts, as shown in Table 1.

Table S6 shows how the ancestral genome estimates in P75 change with phylogenetic context. In general, including a larger outgroup leads to larger estimates.

Figure S28 shows the distribution of arCOG functional categories for cryptic inheritance at Thermoplasmatota (T28): ancestral families detected only in the larger phylogenetic context of THM (T94 dataset). Figure S29 shows the functional categories for cryptic inheritance in Halobacteriota (H51 versus T94) and THM (T94 versus E114). I defined the cryptically inherited genes by the posterior probability of family presence at the ancestral node: less than 1/2 on the small dataset but at least 1/2 in the larger phylogenetic context.

Cryptically inherited genes in Thermplasmatota include sets for the biosynthesis of isoleucine and valine (IlvB, IlvC, IlvD, IlvH), of arginine (ArgB, ArgC, ArgJ, ArgG, ArgH), of histidine (HisA, HisB, HisD, HisG, HisH, HisI) and of aromatic amino acids (TrpA, TrpB, TrpC, TrpD, TrpE, TrpF, TrpG, TyrA, PheA1, PheA2, AroE). Conversely, smaller groups of cryptically inherited genes in Poseidoniia or MBG-D replace the synthesis by amino acid transport for branched-chain amino acids (LivF, LivG, LivH, LivK) and histidine (HisJ, HisM). Cryptically inherited genes for coenzyme metabolism are typically found only in Methanomassiliicoccales among Thermoplasmata, and include the sets for the biosynthesis of cobalamine (13 families), siroheme (6 families), and F430 coenzyme (4 families), as well as the subunits for the alkyl-coenzyme M reductase (MCR) complex (McrA, McrB, McrC, McrD, McrG). Cryptically inherited families in Halobacteriota include the MCR families, as well as the subunits of the tetrahydromethanopterin S-methyltransferase complex (MtrA–G). Cryptically inherited families at the THM ancestor tend to be core families in Acherontia (Thermococci and Theionarchaea), shell or core in Halobacteriota and Thermoplasmatota and largely missing in Methanomada. The distribution of functional classes, with a large contribution of defense mechanisms and energy production, differs from Thermoplasmatota and Halobacteriota cryptically inherited genes.

Table S7 gives the code for functional categories from (4).

Table S5. Reconstructed ancestral genome sizes on simulated data in different phylogenetic contexts (min. 4 copies): posterior expected counts for families (F). M15:Methanomada, T28:Thermoplasmatota, H51:Halobacteriota, T94:THM, N46:Nanobdellota,

dataset:	M15/T28/H51/N46	T94	E114/D80	ED194	truth
Ancestor	F	F	F	F	
M15	1336	1329	1330	1331	1341
T28	1119	1430	1430	1428	1433
H51	1645	1688	1693	1692	1702
T94		1541	1664	1657	1646
N46	1180		1175	1181	1193
D80			1535	1405	1406
Methanobacteriati			1585	1411	1377

Table S6. Reconstructed ancestral genome sizes in different phylogenetic contexts of Proetoarchaeota (minimum 4 copies).

dataset:	B12/N15/T33/A12		T60		P75	
Ancestor	families	copies	families	copies	families	copies
Bathyarchaeia (B12)	1020	1359	1350	1660	1433	1795
Nitrososphaeria (N15)	1017	1187	1178	1414	1281	1572
Thermoprotei (T33)	1098	1322	1255	1538	1421	1797
Thermoproteota (T60)			1200	1457	1465	1895
Asgardarchaeota (A12)	1802	3076			2049	3315
Proteoarchaeota					2062	3346

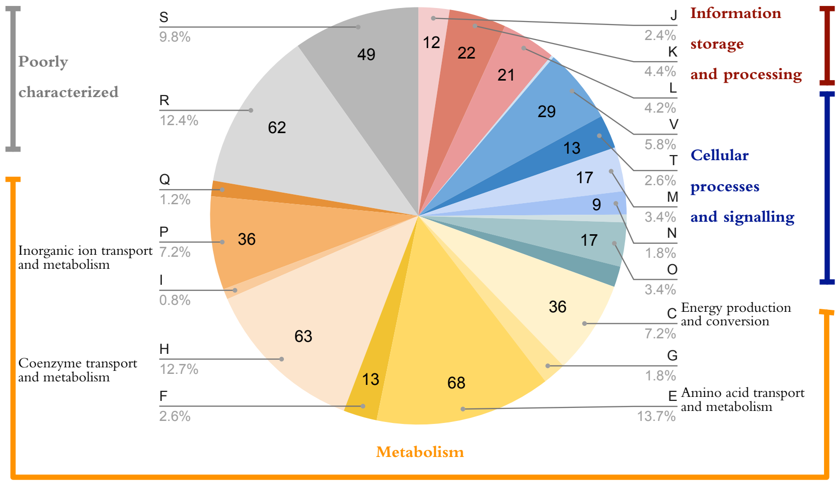

Fig. S28. Functional categories for cryptically inherited genes in Thermoplasmatota.

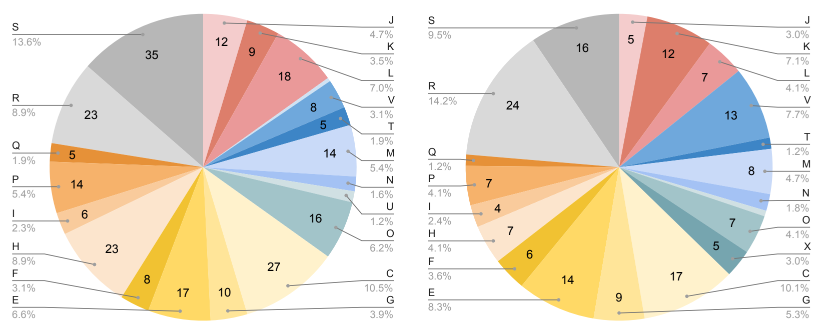

Fig. S29. Functional categories for cryptically inherited genes in Halobacteriota (left) and THM (right) ancestors.

Table S7. ArCOG functional categories from (4).

	Information storage and processing
J	Translation, ribosomal structure and biogenesis
A	RNA processing and modification
K	Transcription
L	Replication, recombination and repair
B	Chromatin structure and dynamics
	Cellular processes and signaling
D	Cell cycle control, cell division, chromosome partitioning
Y	Nuclear structure
V	Defense mechanisms
T	Signal transduction mechanisms
M	Cell wall/membrane/envelope biogenesis
N	Cell motility
Z	Cytoskeleton
W	Extracellular structures
U	Intracellular trafficking, secretion, and vesicular transport
O	Posttranslational modification, protein turnover, chaperones
X	Mobilome: prophages, transposons
	Metabolism
C	Energy production and conversion
G	Carbohydrate transport and metabolism
E	Amino acid transport and metabolism
F	Nucleotide transport and metabolism
H	Coenzyme transport and metabolism
I	Lipid transport and metabolism
P	Inorganic ion transport and metabolism
Q	Secondary metabolites biosynthesis, transport and catabolism
	Poorly characterized
R	General function prediction only
S	Function unknown

B.6. Phylogenetic reconciliation and copy evolution. In order to compile a profile dataset from the supporting data for (5), for (6), and for (7), the ALE reconciliation files (ending with `ml_rec`) or input files (`ale` extension), I parsed the files with ad-hoc Perl scripts. Ancestral copies were counted by summing the reported posterior expectations (using the supplied `branchwise.py` or an equivalent Perl script) when they were not available immediately. From (6), I used the 60 genomes in their reconstruction (and not the 62 genomes in their inferred phylogeny).

Figures S30 compares the inferences of (5) with 30 plants. The GLD and DTL reconstructions are close on the plant dataset; GLD tends to be higher, but ALE infers larger ancestors for bryophytes. The two methods agree on the size of the common ancestor of land plants, and the root's shared inheritance by GLD matches the inferred ancestral size of about 5000 genes at the root. Figure S31 shows that almost all ancestral families inferred with DTL are also inferred by GLD, but GLD infers larger deep ancestors. The reconciliation uses 0–5 transfers to explain family evolution, increasing from shell to core. For some large families, however, ALE infers 20 to 81 transfers (Figure S35).

Figure S32 compares the inferences of (6) with 60 Archaea. On this dataset and phylogeny, the GLD and DTL reconstructions agree on the number of surviving copies at some ancestral nodes, but differ by many hundreds at others. Ancestral estimates tend to be larger by DTL than by GLD when they differ. But DTL infers only around 400 genes (which is less than in the dataset's smallest genome) at the last Archaeal common ancestor (LACA). Only a few genes are present at LACA from main metabolic pathways. Instead, ALE seems to infer more recent gains by horizontal transfer, even in core and shell families.

Figures S33 and S34 show the comparisons over the MCL and COG datasets of (7) over 265 Bacteria.

Figure S35 shows the distribution of transfer events per family on 30 plants and green algae from (5).

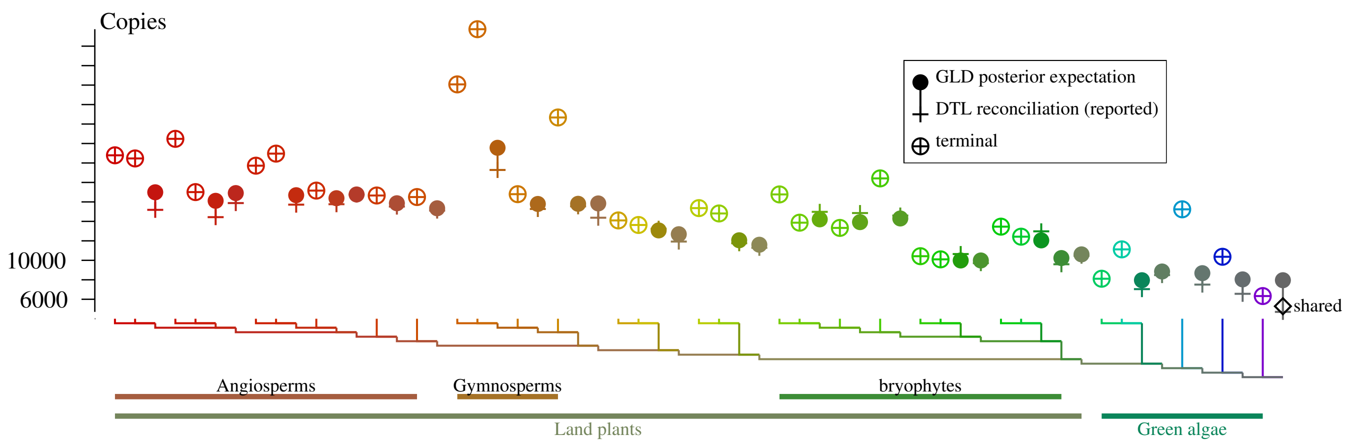

Fig. S30. Total copy numbers in ancestral genomes inferred by ALE reconciliation (DTL model) and by posterior reconstruction (GLD model) on a 30-taxon dataset over land plants and green algae from (5). The nodes are colored and enumerated as shown by the tree diagram along the X axis. The empty diamond at the root marks the shared inheritance by both children.

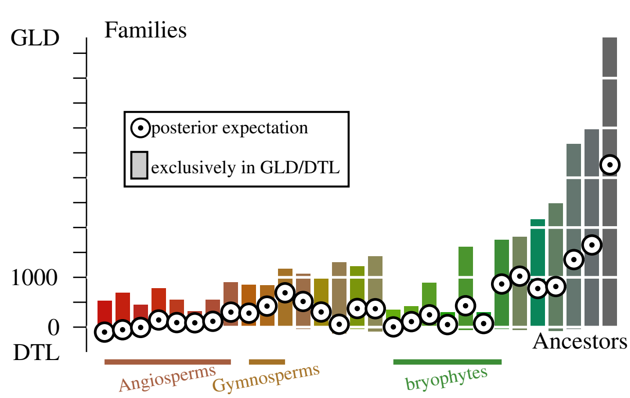

Fig. S31. Differences in ancestral family reconstruction inferred by GLD (MAP estimator) and DTL on a 30-taxon dataset over land plants and green algae from (5). Upward bars from 0 count families where MAP estimation with GLD predicts presence but the DTL expected copy number is less than 0.5 (positives exclusive to GLD). Downward bars from 0 count families where GLD predicts absence but DTL copy number is at least 0.5 (there are very few such families). Ancestral nodes are colored and enumerated as in Figure S30.

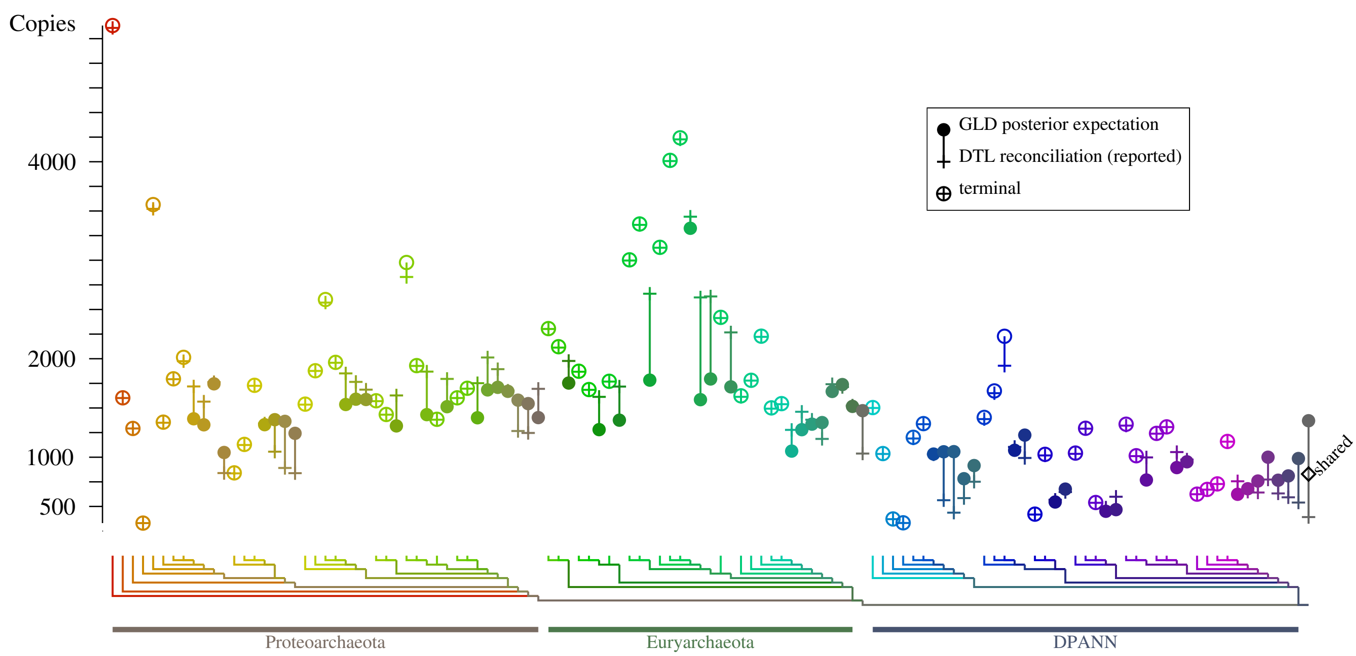

Fig. S32. Total copy numbers in ancestral genomes inferred by ALE reconciliation (DTL model) and by posterior reconstruction (GLD model) on a 60-taxon Archaeal dataset from (6). The nodes are colored and enumerated as shown by the tree diagram along the X axis. The empty diamond at the root marks the shared inheritance by both children.

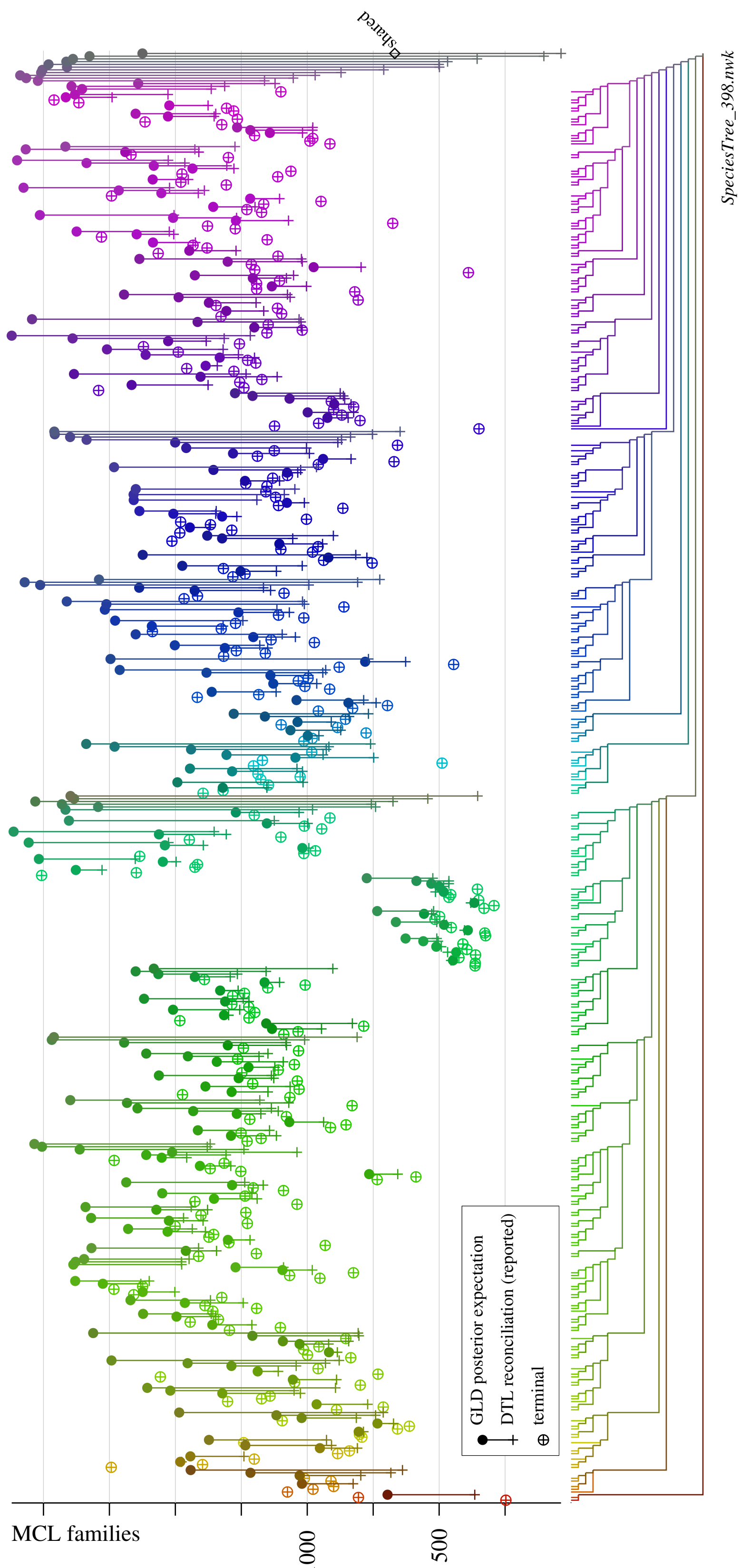

Fig. S33. Number of families at ancestral nodes inferred by ALE reconciliation (DTL model) and by posterior reconstruction (GLD model) on 265 Bacteria over MCL families from (7). The nodes are colored and enumerated as shown by the tree diagram along the X axis. The empty diamond at the root marks shared inheritance.

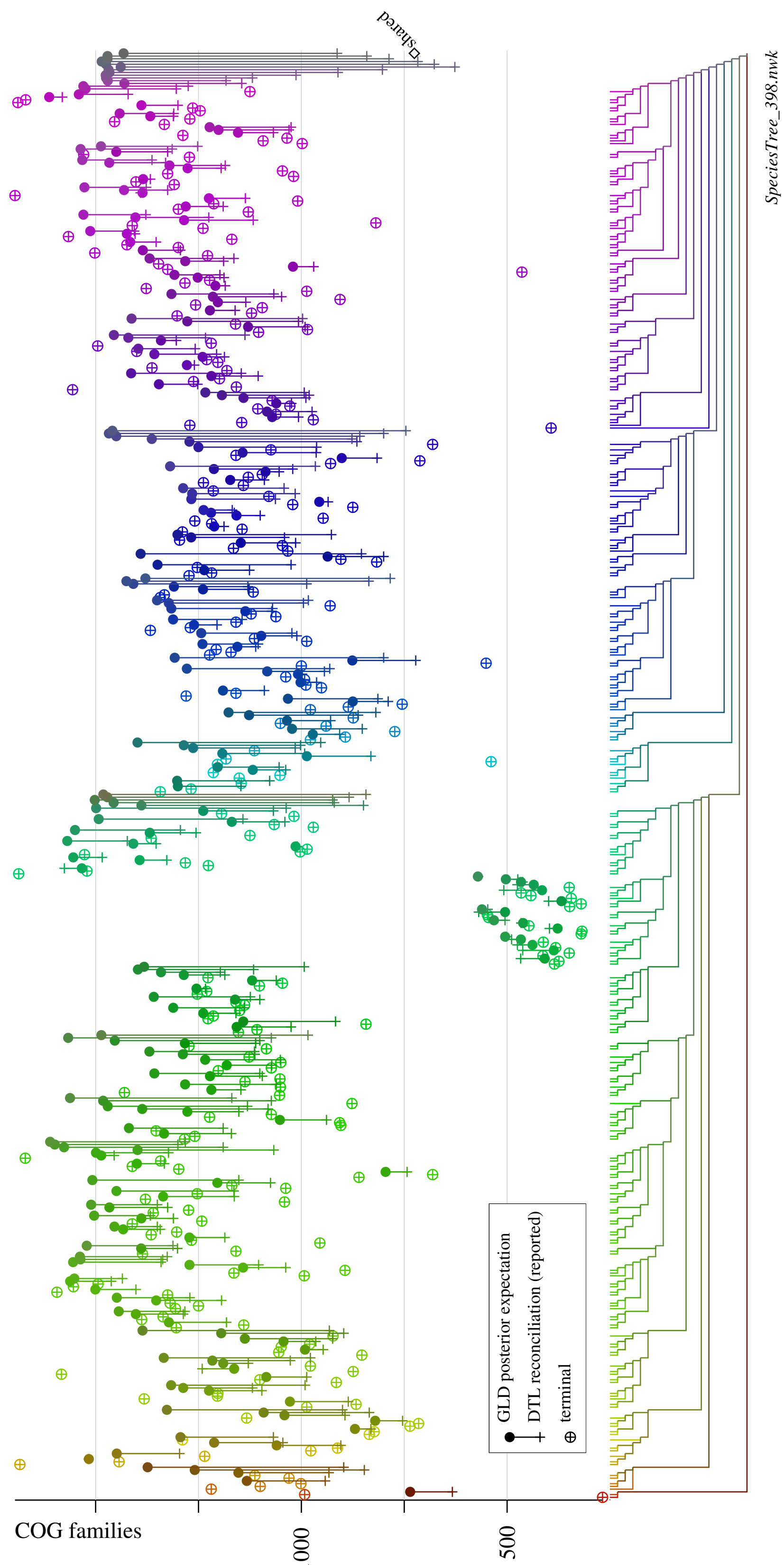

Fig. S34. Number of families at ancestral nodes inferred by ALE reconciliation (DTL model) and by posterior reconstruction (GLD model) on 265 Bacteria over COG families from (7). The nodes are colored and enumerated as shown by the tree diagram along the X axis. The empty diamond at the root marks shared inheritance.

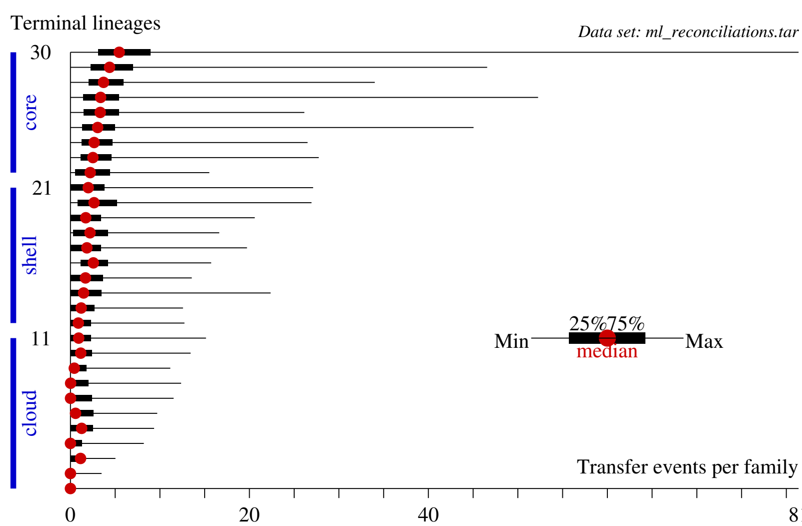

Fig. S35. Distribution of the number of transfer events per family inferred by ALE reconciliation on 30 plants and green algae from (5). The horizontal box-and-whisker plot shows median, interquartile range and extreme values, by family frequency in extant genomes (along the Y axis).

B.7. Large-scale reconciliations. Table S8 shows the results of simulated gene-tree sampling with IQ-TREE3 (8), modelling incongruence in the dataset of (7). The provided species tree (Tree398) over $n = 265$ leaves and two subtrees with $n = 135, 67$ leaves were used as examples for shell and core families with perfect vertical inheritance. The simulated alignment lengths $\ell = 66, 116, 195$ were chosen to match the quartiles for the distribution of informative sites in shell and core across the MCL dataset (namely, $65, 116, 194$). Alignments were generated using AliSim with the `LG+C60` substitution model, which was the study's original choice for core genes in species-tree reconstruction. The random seed is set to the concatenation of tree size and alignment length: for instance,

```
iqtree3 -t SpeciesTree_398.nwk --alisim ali265-195 --length 195 -seed 265195 -m LG+C60
```

Gene trees were sampled by ultrafast bootstrap following the protocol of (7):

```
iqtree3 -s ali265-195.fa -seed 2651951 -bb 1000 -wbt
```

Each edge of a gene tree defines a leaf bipartition (*split*), and the splits exclusive to another tree (true tree or bootstrap tree) measure topological distance, quantifying incongruence. This metric is equivalent to half the Robinson-Foulds distance. When comparing with the true tree, the incongruence is the number of unrecovered internal edges (up to $n - 3$). Robinson-Foulds distances between trees were obtained using `iqtree3 -rf_all` over the bootstrap table and the true (unrooted) tree.

The expected posterior numbers of (transfer or duplication) events per family were calculated by summing the posteriors reported in the ALE reconstruction across nodes. (For (11) they were already tabulated.) Figures S36, S37, and S38 show the distributions of transfer and duplication events per family from the ALE reconciliations of (7) over 265 Bacteria..

Figures S39 and S40 show the distributions of transfer and duplication events from the ALE reconciliations of (11) over 1007 Bacteria.

Figure S41 shows the distribution of transfer events per family from ALE reconciliations of (12) over 700 genomes. I tried only one dataset from the multiple bootstrap replicates (`Constrained_K05`).

The exact provenance of the data is the file indicated on the top right corner of the plots S36–S41.

Table S8. Alignment length and phylogenetic incongruence. The table shows the results over simulated alignments with $n = 67, 135, 265$ sequences and $\ell = 66, 116, 195$ sites, corresponding to typical values of shell and core families in the 265-genome dataset of (7). Data was collected from 1000 bootstrap trees using IQ-TREE. Topological distance (incongruence) is measured by the leaf bipartitions (splits) exclusive to another tree (true tree or another bootstrap tree); this metric is equivalent to half the Robinson-Foulds distance and corresponds to the number of unrecovered internal edges (up to $n - 3$) when comparing to the true tree. PIS: parsimony informative sites.

sequences	alignment length	informative sites		incongruence vs. true	pairwise incongruence
n	ℓ	PIS	PIS/n	min.–max.	average
67	195	187	2.97	25–45 (39–70%)	20.3 (32%)
67	116	105	1.57	31–50 (48–78%)	21.4 (33%)
67	66	62	0.93	44–56 (69–88%)	27.4 (43%)
135	195	191	1.44	56–78 (42–58%)	31.2 (24%)
135	116	115	0.85	73–91 (55–67%)	35.6 (27%)
135	66	65	0.48	94–108 (71–82%)	39.8 (30%)
265	195	194	0.74	82–110 (31–42%)	37.2 (14%)
265	116	116	0.44	152–169 (58–65%)	64.3 (25%)
265	66	66	0.25	207–220 (79–84%)	74.7 (28%)

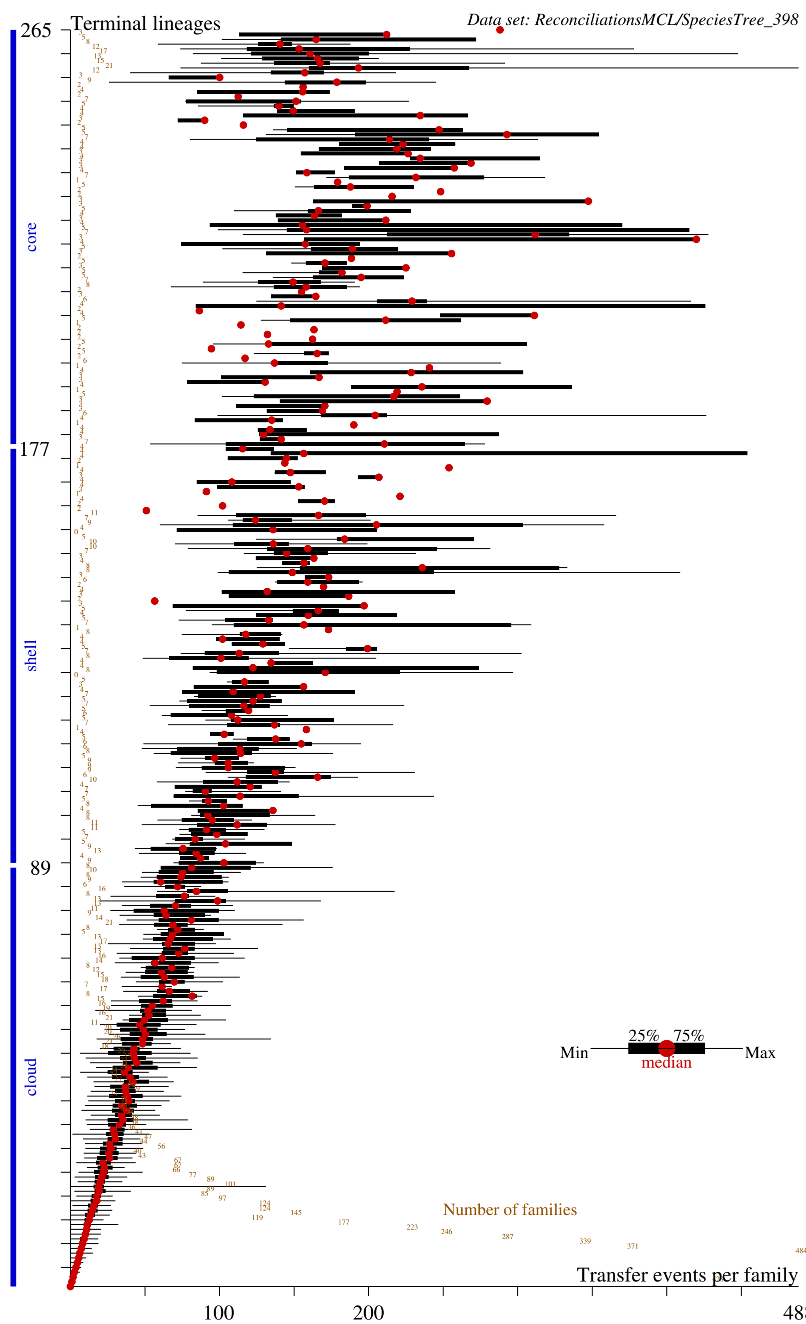

Fig. S36. Number of transfer events per family and number of families by frequency in ALE reconciliations on 265 Bacteria over MCL families from (7). The horizontal box-and-whisker plot shows median, interquartile range and extreme values, by family frequency in extant genomes (along the Y axis).

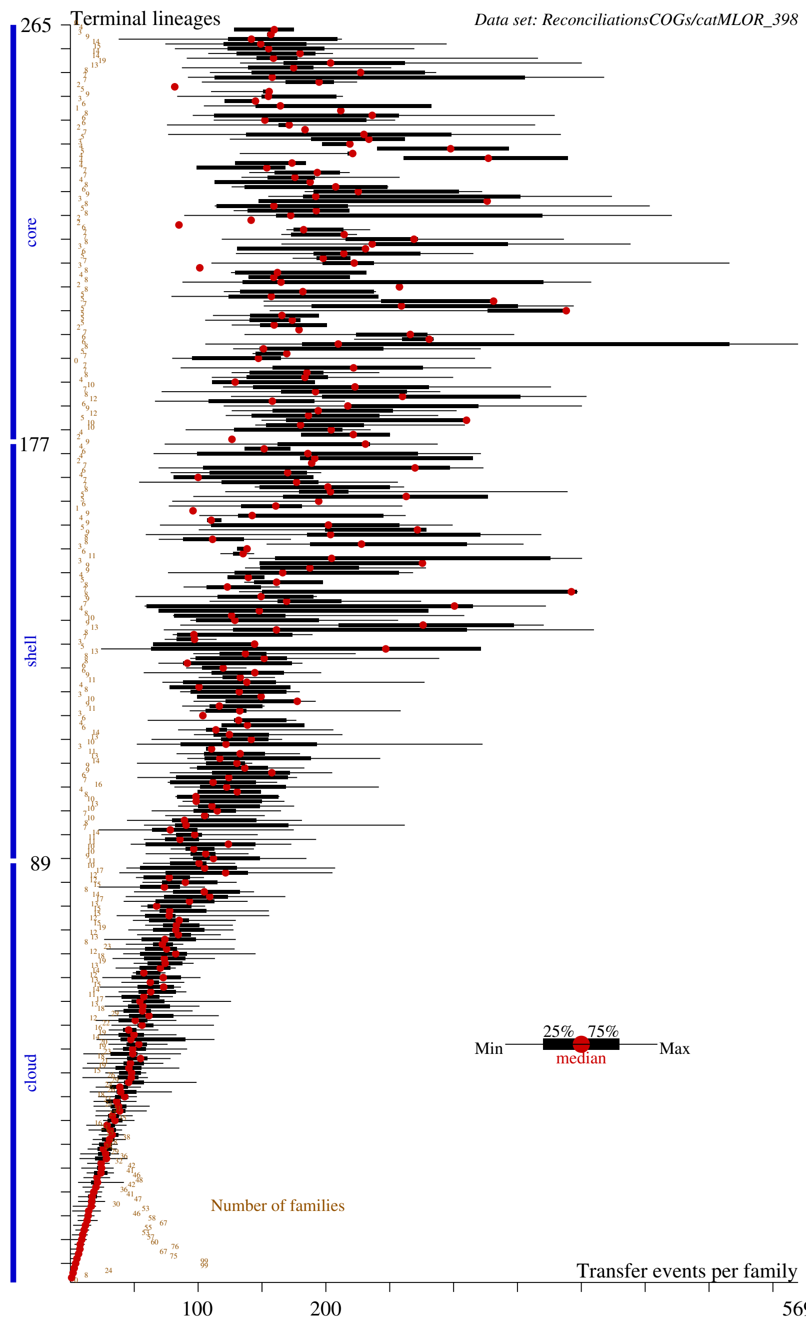

Fig. S37. Number of transfer events per family and number of families by frequency in ALE reconciliations on 265 Bacteria over COG families from (7). The horizontal box-and-whisker plot shows median, interquartile range and extreme values, by family frequency in extant genomes (along the Y axis).

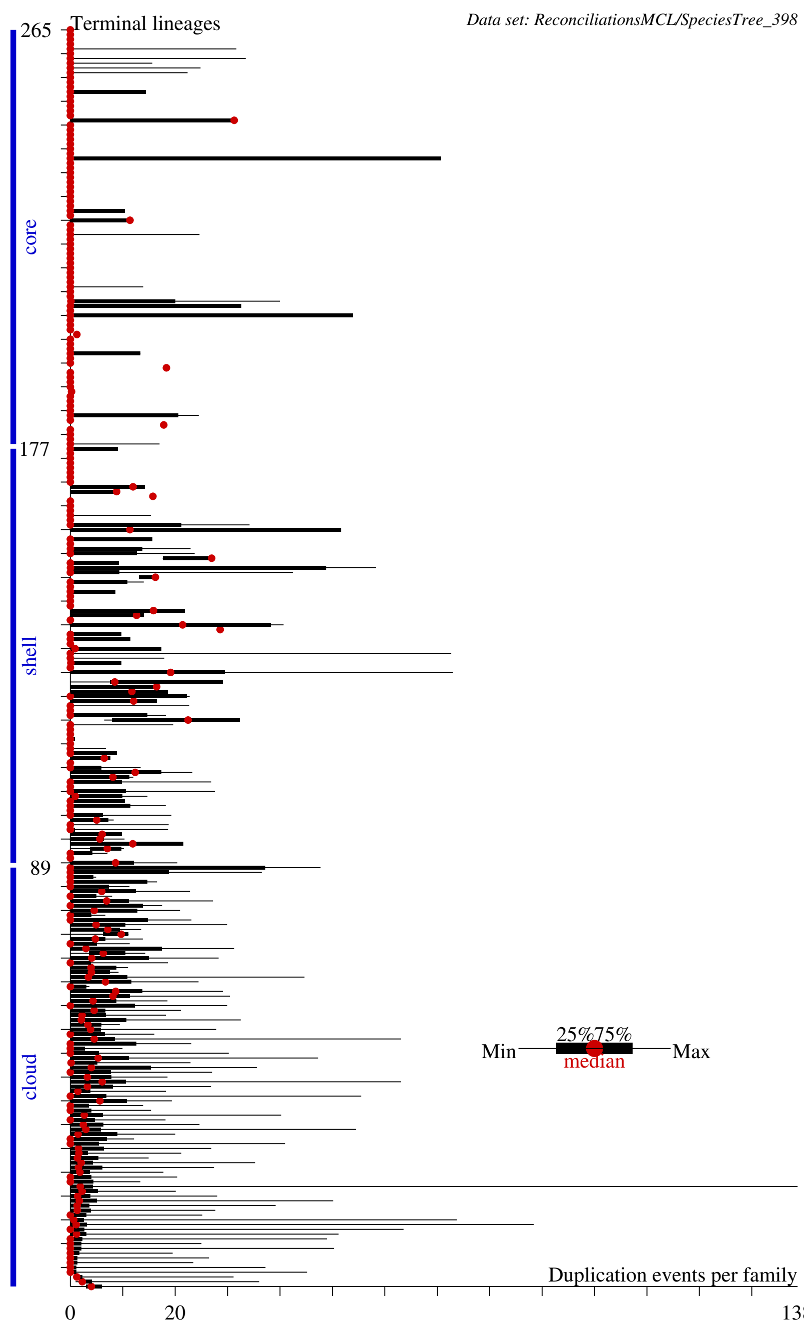

Fig. S38. Number of duplication events per family inferred by ALE reconciliation on 265 Bacteria over MCL families from (7). The horizontal box-and-whisker plot shows median, interquartile range and extreme values, by family frequency in extant genomes (along the Y axis).

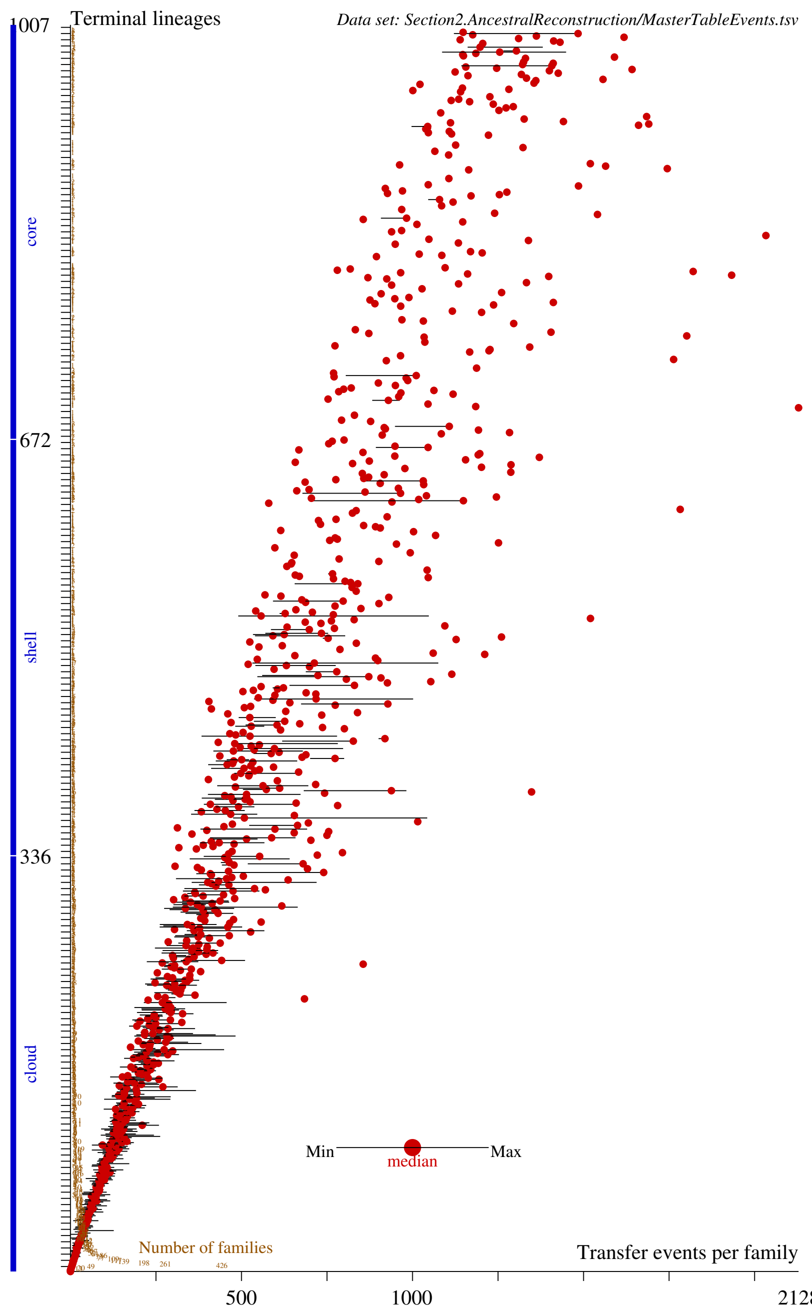

Fig. S39. Number of transfer events per family and number of families by frequency in ALE reconciliations on 1007 Bacteria over COG families from (11). The horizontal box-and-whisker plot shows median, interquartile range and extreme values, by family frequency in extant genomes (along the Y axis).

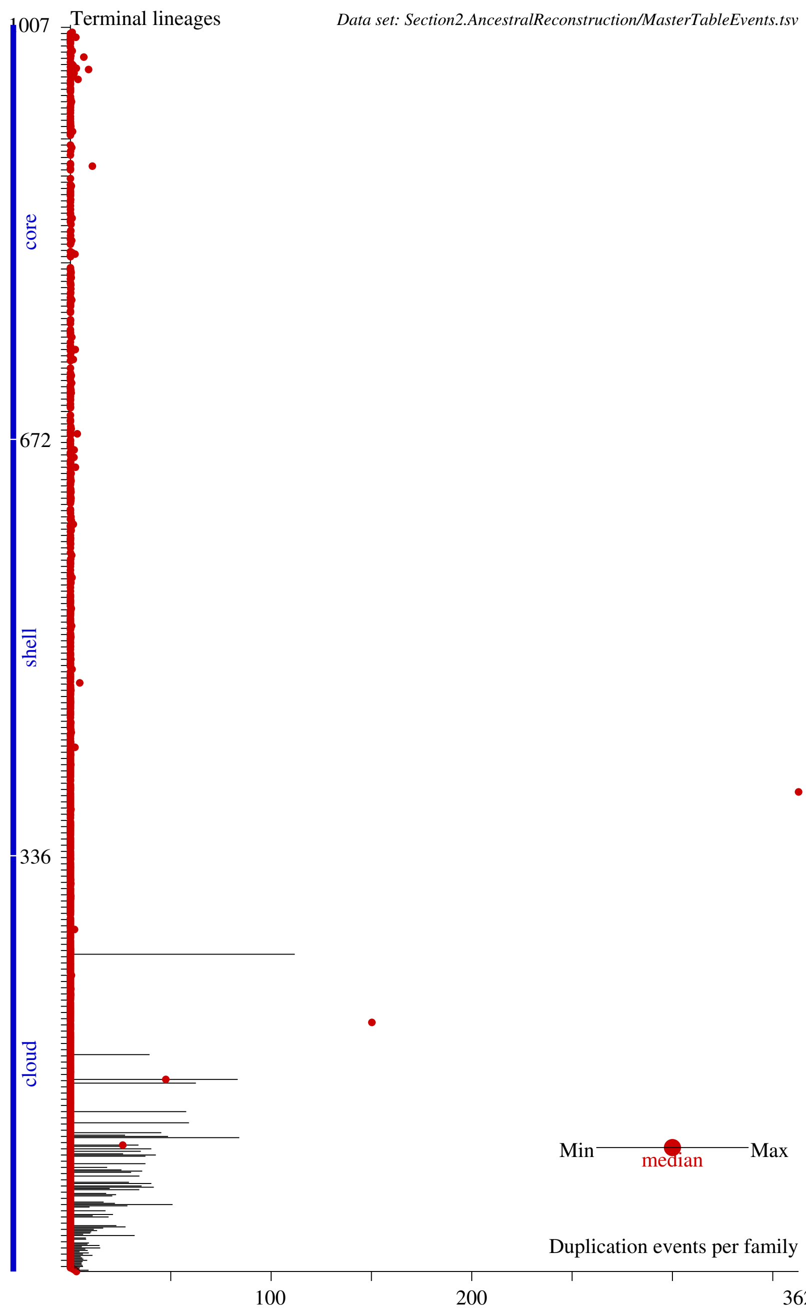

Fig. S40. Number of duplication events per family in ALE reconciliations on 1007 Bacteria over COG families from (11). The horizontal box-and-whisker plot shows median, interquartile range and extreme values, by family frequency in extant genomes (along the Y axis).

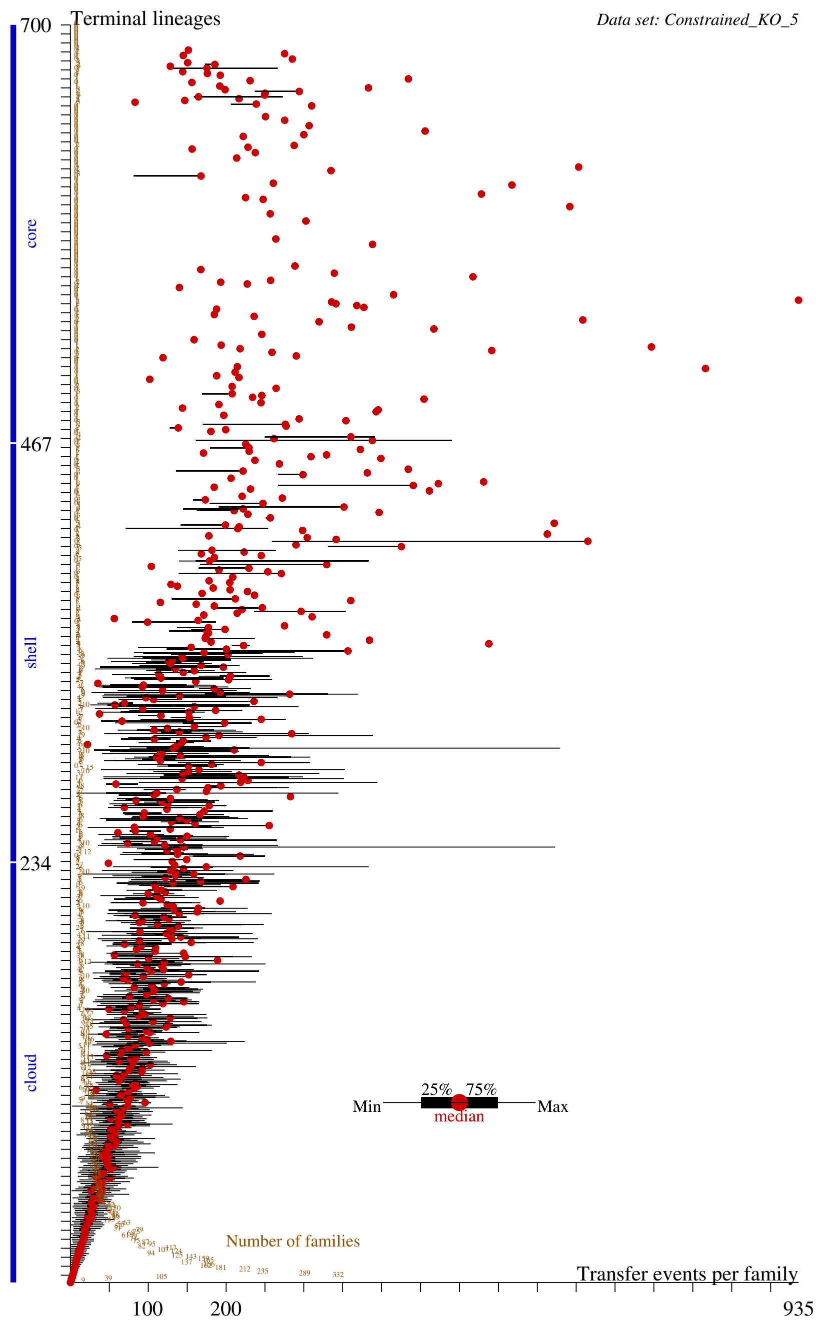

Fig. S41. Number of transfer events per family and number of families by frequency in ALE reconciliations on 700 genomes over KEGG families from (12). The horizontal box-and-whisker plot shows median, interquartile range and extreme values, by family frequency in extant genomes (along the Y axis).

B.8. Loss-dominated histories. Figures S43 and S44 shows the inheritance of families from the THM and Nanobdellati ancestor, in simulations and orthologs datasets T94, E114, D80, and ED194.

Figure S44 shows that on expectation, hundreds of families present only in 2–7 extant genomes (eg. a few Altarchaeota) are inherited from the Nanobdellati ancestor. For the THM ancestor over the E114 or ED194 reconstructions, a median number of 44 or 36 lineages (out of 94) inherit a copy, and the middle half of THM-originated families appear in the shell for that clade. Simulated data have similar interquartile statistics, but in true ortholog data, there are more core and fewer shell families that are traced back to the THM ancestor, echoing the difference of family frequencies on the original dataset (Figure S26). Furthermore, the number of cloud profiles that appears to be inherited is much larger in true family profiles than expected; such profiles have support from the distribution among Nanobdellati, often within the deepest branching phylum of (free-living) Altarchaeota. For the Nanobdellati ancestor over the D80 or ED194 reconstruction, about 3/4 of the inherited families belong to the cloud.

Family gain means the acquisition of at least one surviving copy ($0 < \tilde{\xi}_v$ and $0 = \tilde{\eta}_v$ at node v). *Family loss* means the loss of all surviving copies ($0 = \tilde{\eta}_v$ and $0 < \tilde{\xi}_u$ on edge uv).

Figure S45 shows the distribution of family gain and loss events in function of family frequency. Observe that 2–3 family gains are needed for most family profiles, except for the rarest and most common families where 1 suffices. 20–40 loss events generate a wide range of profiles (present in 1/6–5/6 of terminal taxa).

Figure S45 shows the distribution of family gain events in posterior and parsimony reconstructions over ED194. The log-linear plot suggests geometric decay: around 2/3 as many families are acquired in $n+1$ lineages as in n lineages for $n = 1, 2, \ldots, 7$. An exponential decay (with different rates) can also be observed in numerical parsimony reconstructions with an adjustable family gain penalty g. The parsimony reconstructions use a modified Wagner-parsimony penalization π of for the state transitions $\xi_u \to \xi_v$ on each edge uv: if $\xi_u = 0$, then $\pi = g + (\xi_v - 1)$ or else $\pi = \left|\xi_u \to \xi_v\right|$. We retrieved the minimum-parsimony reconstruction for $g = 1.0001, 3.0001, 5.0001$ by Sankoff's dynamic programming algorithm.

The stochastic BDI processes underlying the GLD model predict that the number of non-vertical events in a family follows a Poisson distribution since such events arrive with a constant rate γ_u on the edge leading to a node u. We observe fewer family gains in consequence of tree topology: one gain at the root suffices to hide subsequent transfer events creating pseudoparalogs, and multiple family gains are created mostly in multiple independent subtrees, with exponentially decaying probability by event count. As a consequence of the combinatorics of subtrees, the distribution of independent acquisitions for a family displays geometric decay, either by maximum-likelihood (accounting for XGD), or by parsimony (not counting XGD) reconstructions.

Figures S46–S48, S49–S51 and S52–S54 show the dynamics of family evolution on the E114, ED1904 and P75 datasets, when reconstructed from samples with observation bias $\Omega_{\min} = 4, 2, 1$. Observe how terminal lineages shift towards a gain-dominated history. For the three plots across the same taxon sets, posterior expectations were computed using the GLD model optimized with $\Omega_{\min} = 4$.

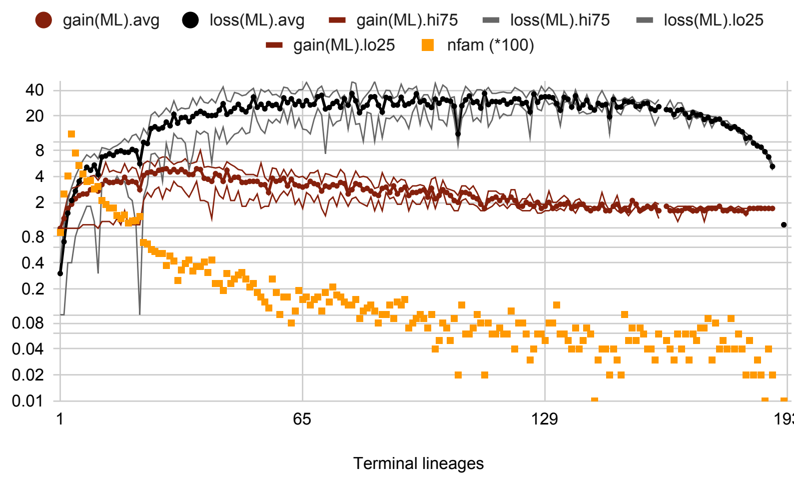

Fig. S42. Number of family events in function of family presence in terminal lineages of ED194. Averages (avg) and lower/upper quartiles (lo25 and hi25) are calculated across nfam families present in the given number of lineages. The peaks at 11 and 22 lineages are caused by core profiles in the classes of Thermococci and Halobacteria, respectively.

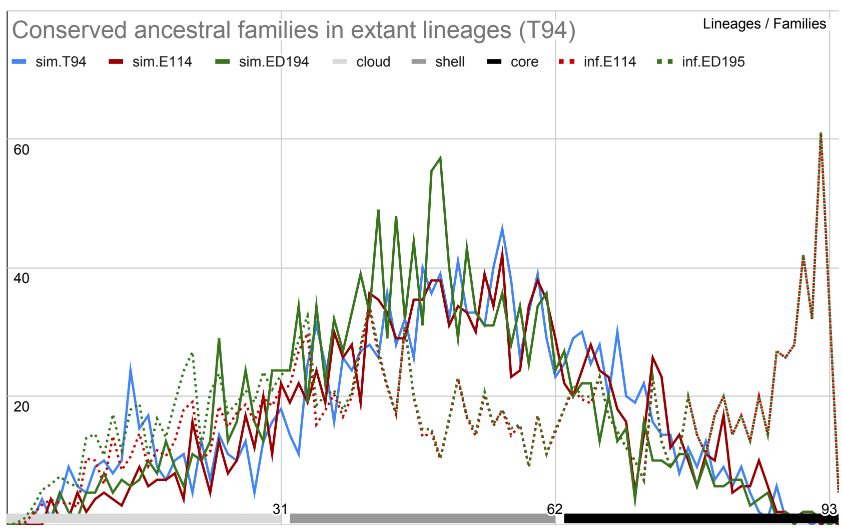

Fig. S43. Inheritance of ancestral families within THM (T94). Simulated datasets(sim.X) and inferred ancestral presence (inf.X) are shown over datasets X=T94, E114 and ED194 (with observation bias $\Omega_{\min} = 4$). X axis shows the number of extant THM lineages in which a family is present; Y axis shows the number of families inferred to be present at the THM ancestor.

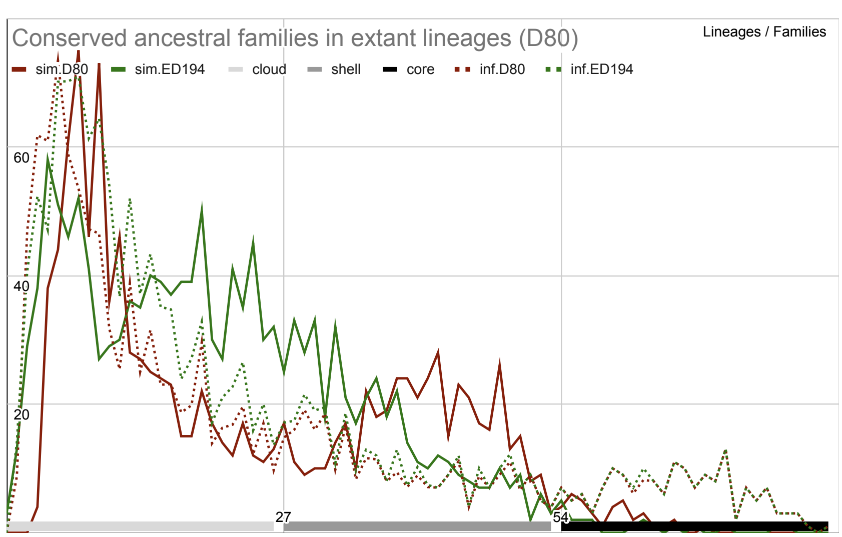

Fig. S44. Inheritance of ancestral families within Nanobdellati (D80). Simulated datasets(sim.XXX) and inferred ancestral presence (inf.X) are shown over datasets X=D80 and ED194 (with observation bias $\Omega_{\min} = 4$). X axis shows the number of extant Nanobdellati lineages in which a family is present; Y axis shows the number of families inferred to be present at the ancestor.

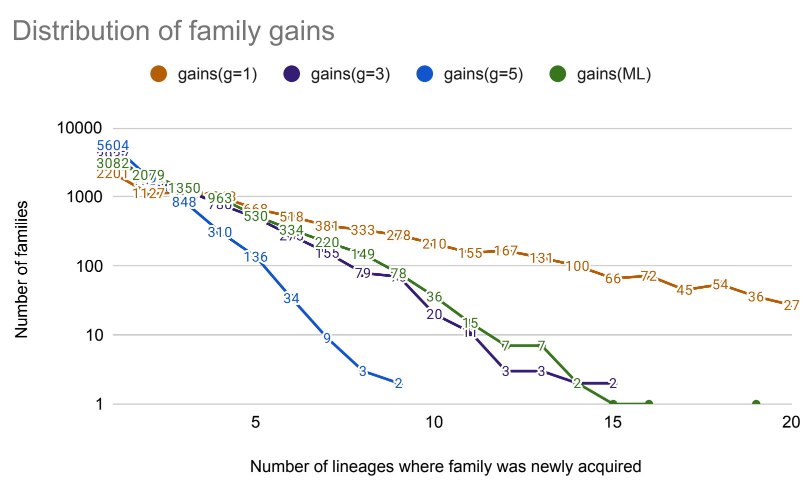

Fig. S45. Distribution of the number of gains per family over the ED194 dataset (minimum 4 copies). Numerical parsimony with family gain penalty g and posteriors from maximum-likelihood reconstruction

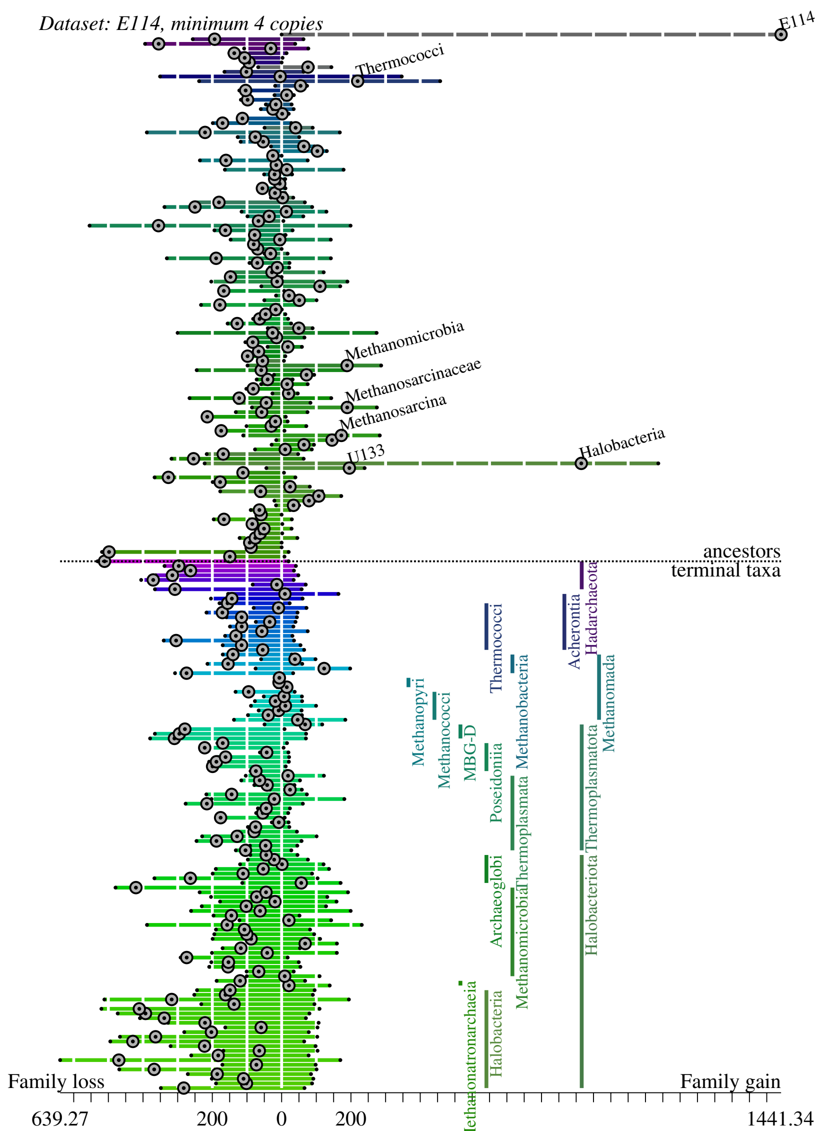

Fig. S46. Evolutionary dynamics of family evolution in Methanobacteriati lineages (E114 families with minimum 4 copies). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure ??. Dotted circles denote net change; largest increases are labeled.

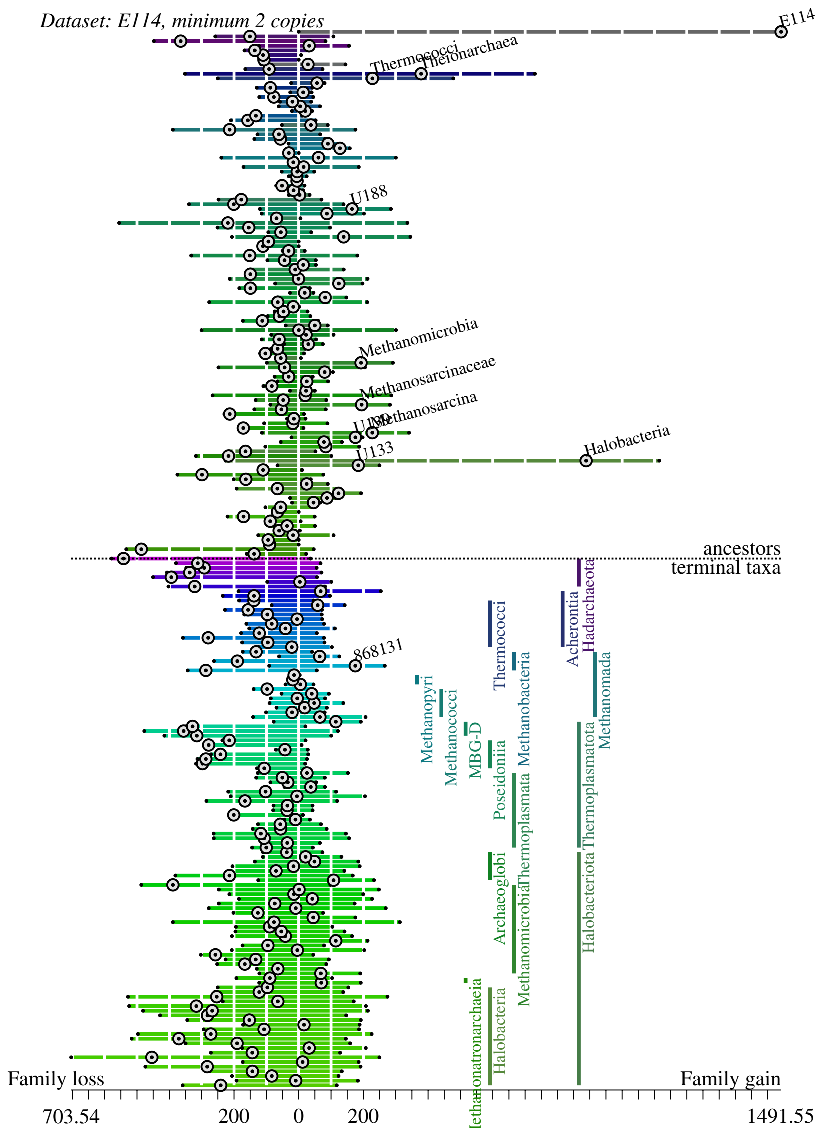

Fig. S47. Evolutionary dynamics of family evolution in Methanobacteriati lineages (E114 families with minimum 2 copies). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure ??. Dotted circles denote net change; largest increases are labeled

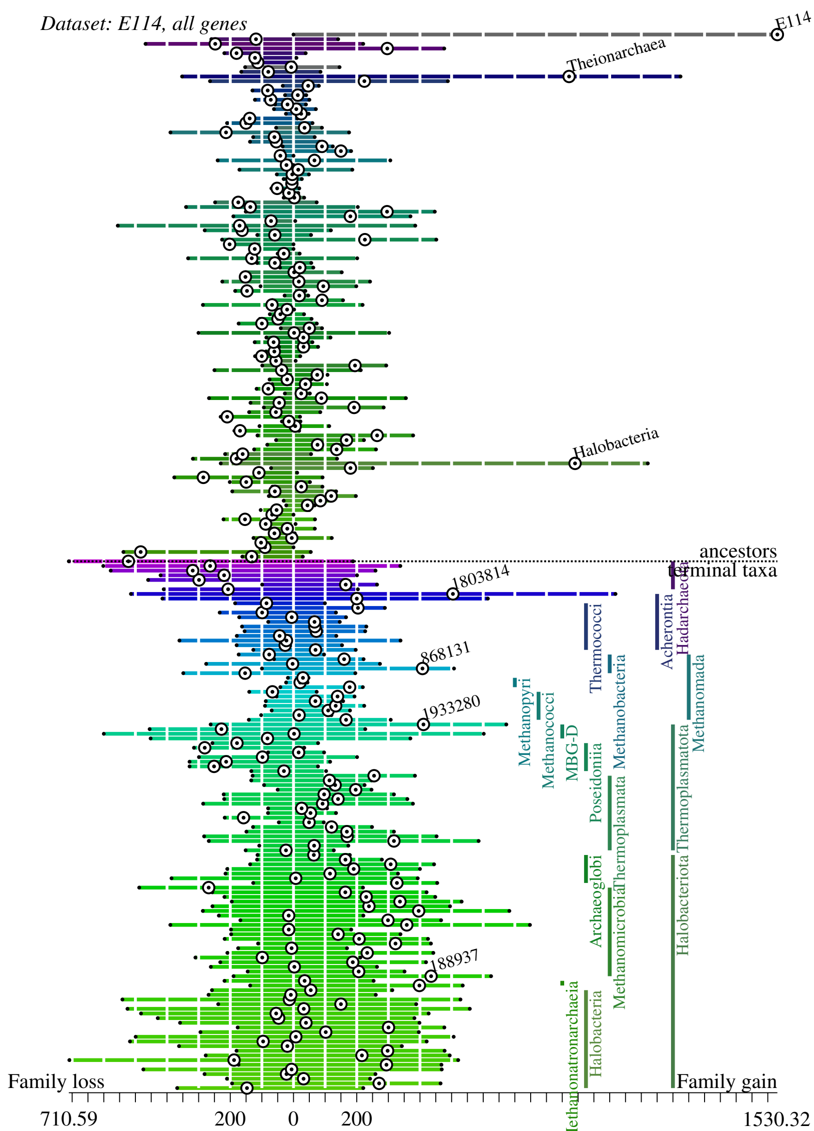

Fig. S48. Evolutionary dynamics of family evolution in Methanobacteriati lineages (E114, all genes). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure ??. Dotted circles denote net change; largest increases are labeled.

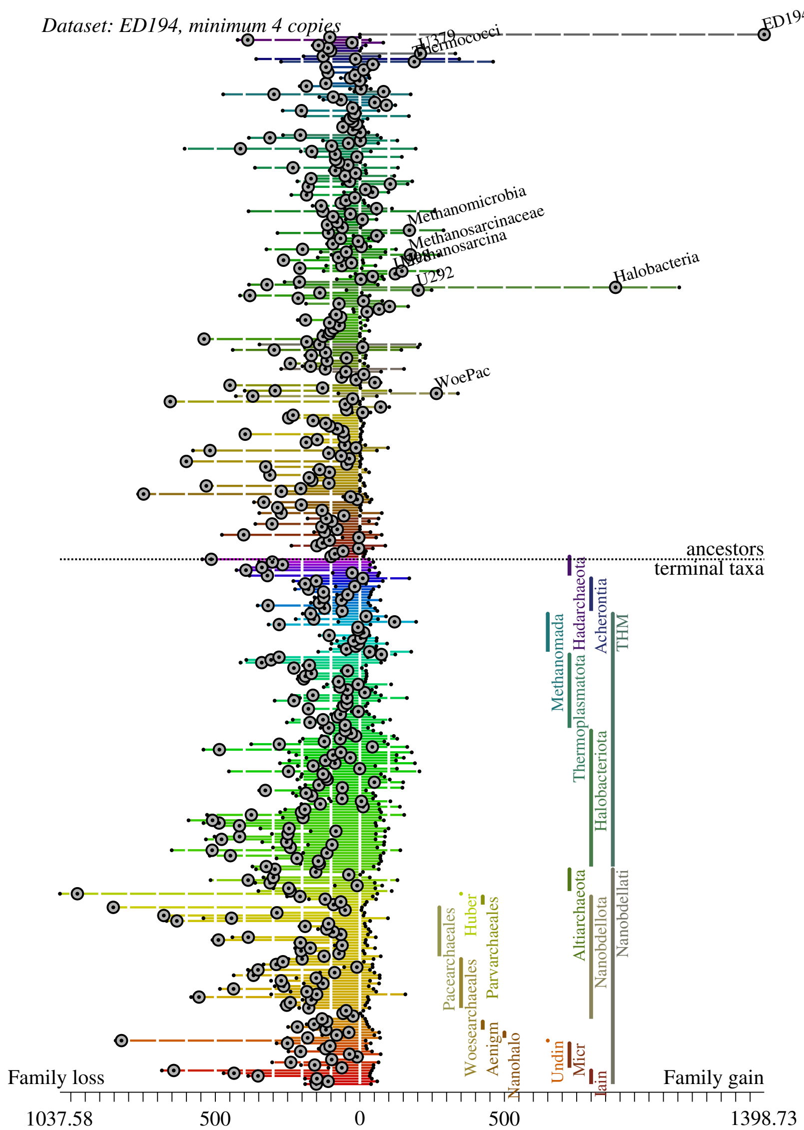

Fig. S49. Evolutionary dynamics of family evolution in Methanobacteriati+Nanobdellati lineages (ED194 families with minimum 4 copies). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure ??. Dotted circles denote net change; largest increases are labeled.

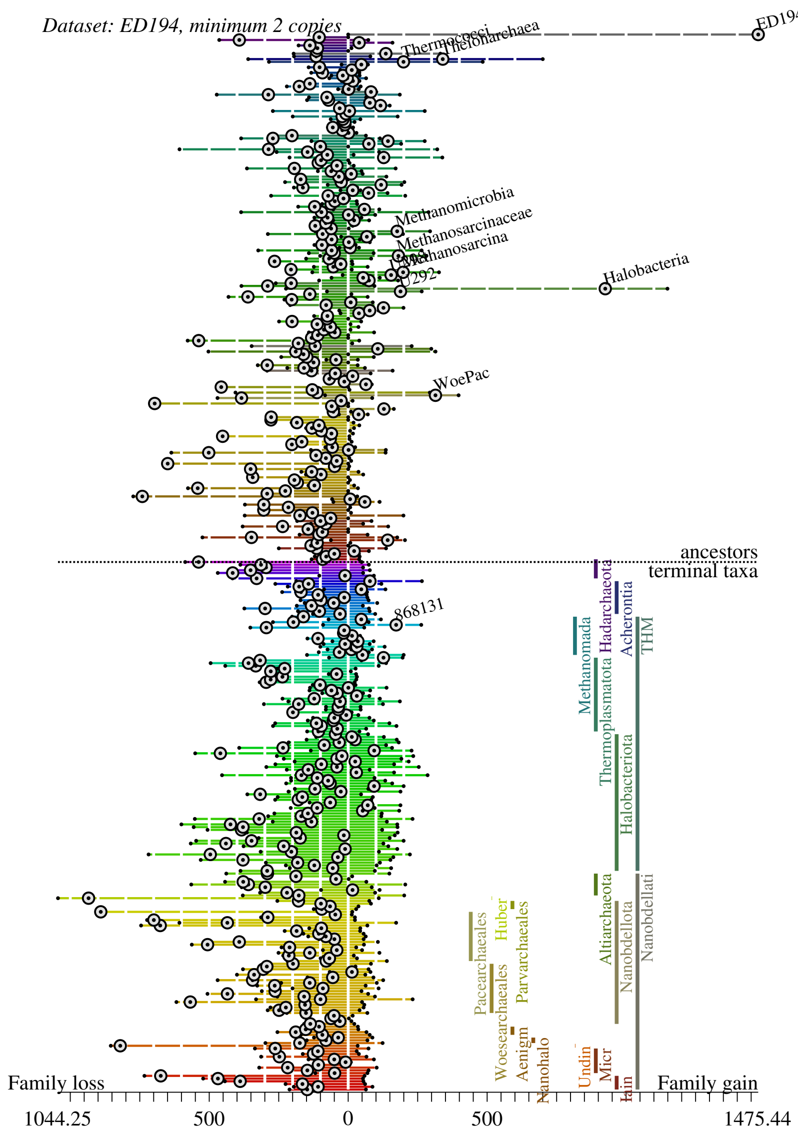

Fig. S50. Evolutionary dynamics of family evolution in Methanobacteriati+Nanobdellati lineages (ED194 families with minimum 2 copies). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure ??. Dotted circles denote net change; largest increases are labeled.

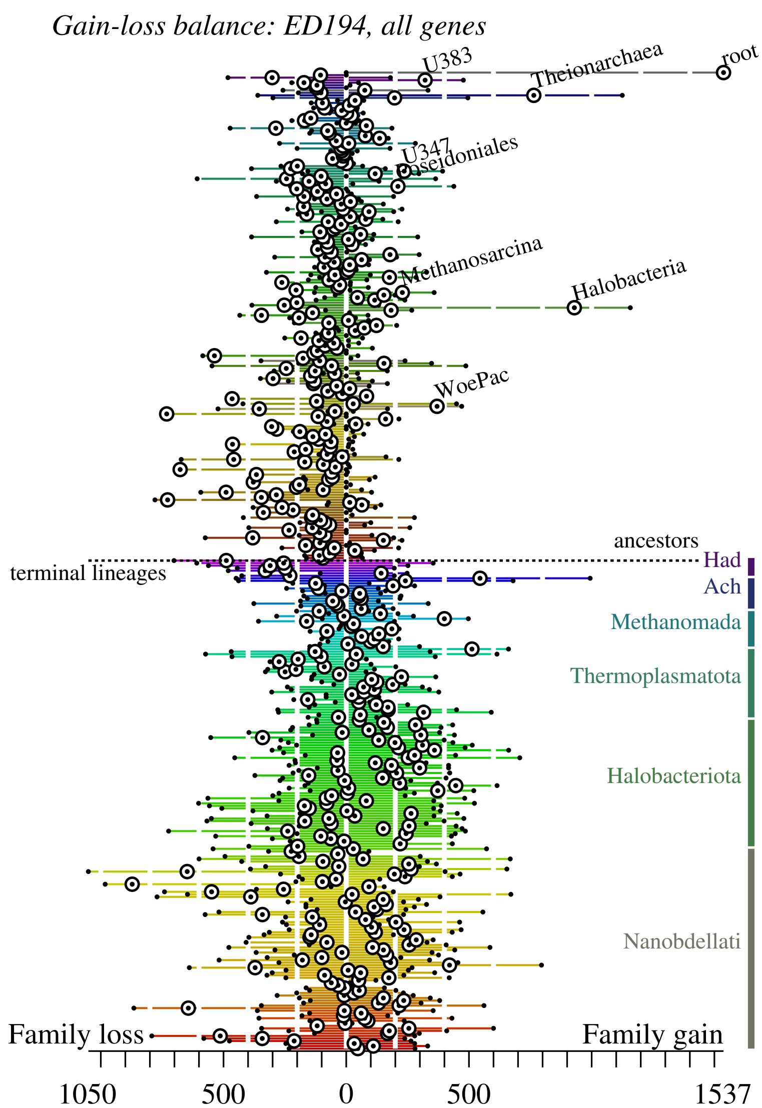

Fig. S51. Evolutionary dynamics of family evolution in Methanobacteriati+Nanobdellati lineages (ED194, all genes). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure ??. Dotted circles denote net change; largest increases are labeled.

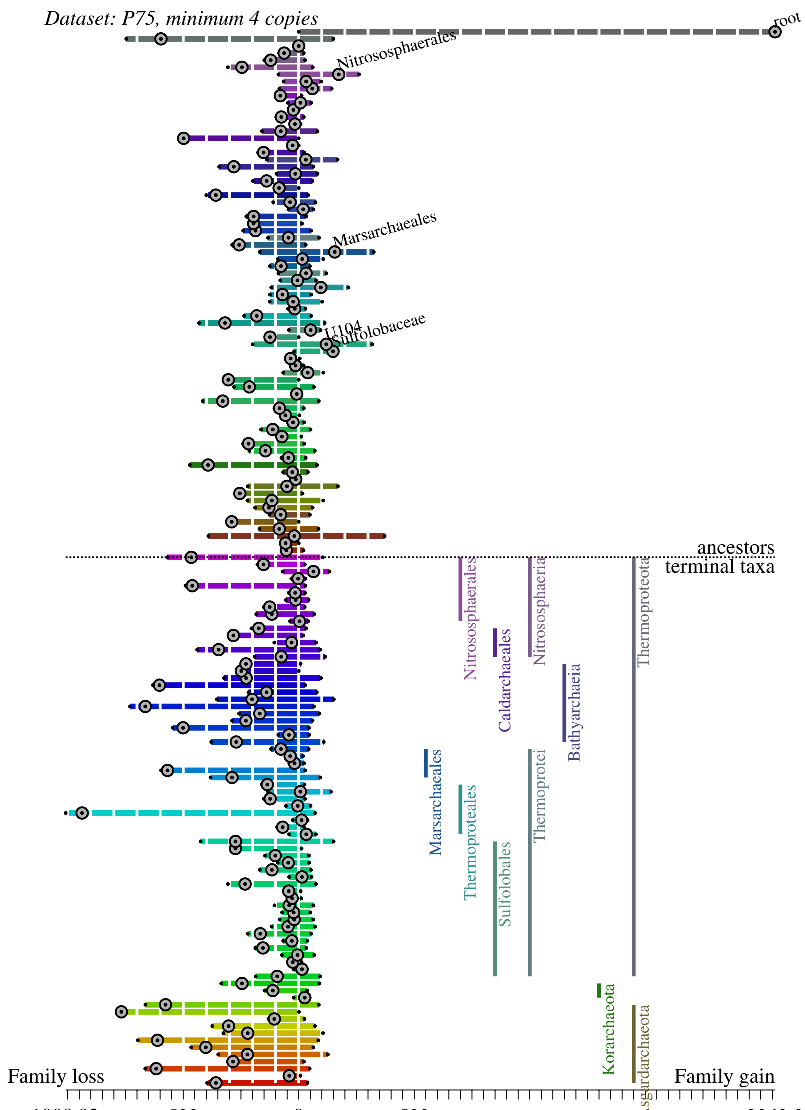

Fig. S52. Evolutionary dynamics of family evolution in Proteoarchaeota lineages (P75 families with minimum 4 copies). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure S23. Dotted circles denote net change; largest increases are labeled.

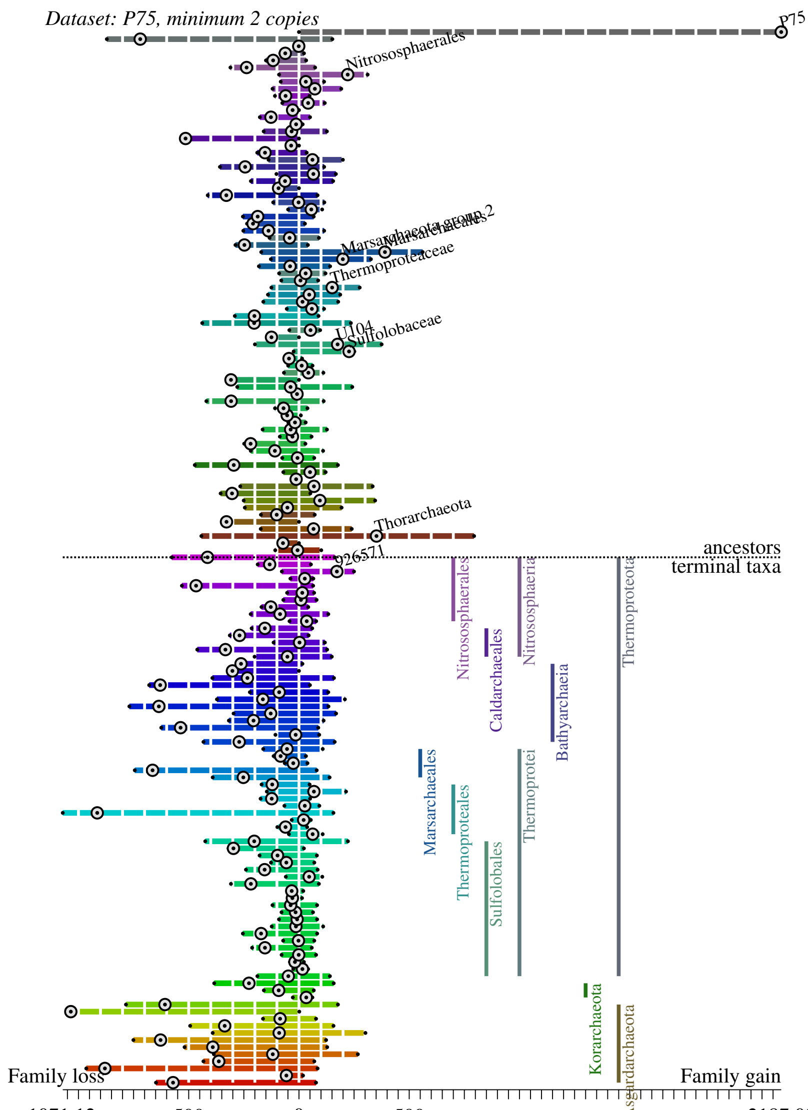

Fig. S53. Evolutionary dynamics of family evolution in Proteoarchaeota lineages (P75 families with minimum 2 copies). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure S23. Dotted circles denote net change; largest increases are labeled.

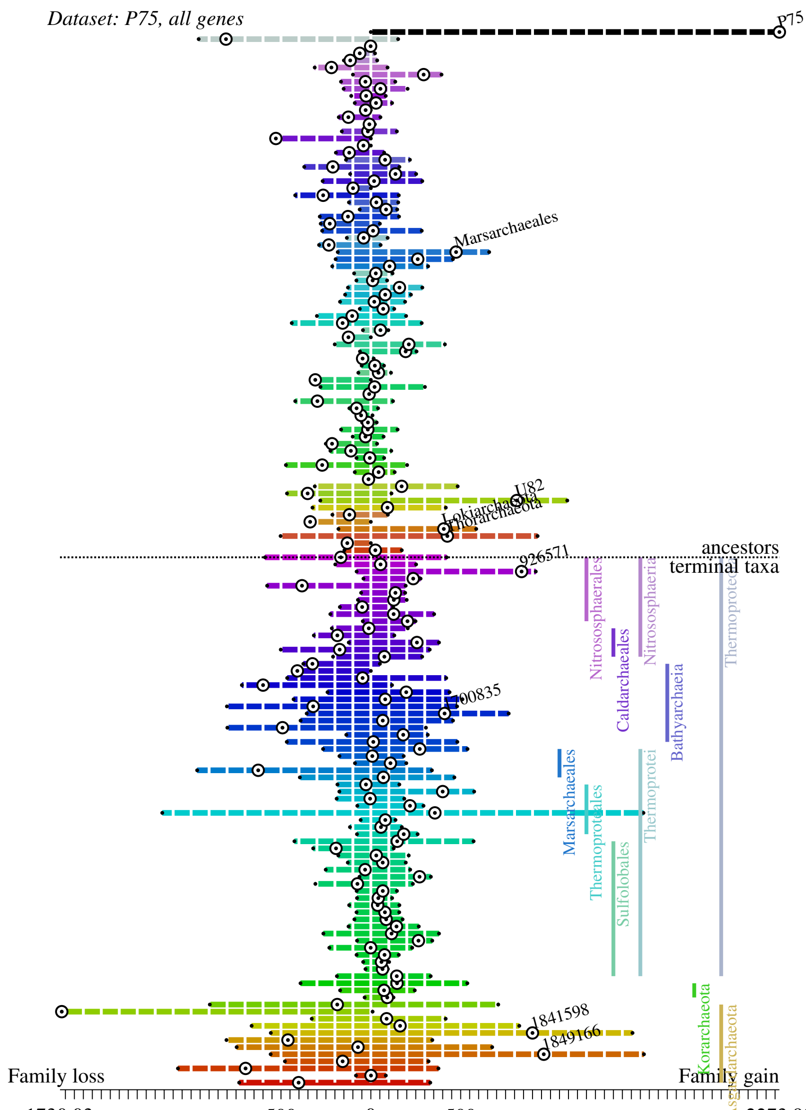

Fig. S54. Evolutionary dynamics of family evolution in Proteoarchaeota lineages (P75, all genes). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Figure S23. Dotted circles denote net change; largest increases are labeled.